Lecture 7. Time series: forecasting, dynamical systems and learned dynamics

Deep Learning for Scientific Discovery

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1 Time series in science

The regression methods introduced earlier in the course learn a relation between inputs and targets. A measured time series gives that task a temporal structure. A laser detector records irregular pulses, and a model must use the observations already available to predict later intensities. We also want to determine what the record reveals about the physical law that produces them.

Many systems of the natural sciences are dynamical systems, systems whose condition changes in time according to a fixed rule. At every moment their condition is described by a state, a set of numbers $\mathbf{x} = (x_1, \dots, x_d)$ that fixes everything needed to determine the future. For a pendulum the state is the angle and the angular velocity, for a planet its position and velocity, for a fluid the velocity and the temperature at every point of space. The number d of state variables is the number of initial values that must be specified before the future is determined. A position alone does not suffice for a pendulum, since a pendulum passing the lowest point may be moving to the left or to the right, and the velocity must be given as well.

The set of all possible states is the phase space, here $\mathbb{R}^d$ or a part of it, and every state is a point in it. For a pendulum the phase space is a plane with the angle on one axis and the angular velocity on the other, and a swinging pendulum traces a closed curve around the origin. The law of motion

$$\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}) \tag{1.1}$$

assigns to every point of phase space a velocity $\mathbf{F}(\mathbf{x})$, the direction and speed in which the state moves when it is at $\mathbf{x}$. The function $\mathbf{F}$ is called the vector field. It can be pictured as an arrow attached to every point of phase space, and the motion of the system as a point that always moves along the local arrow. As time passes the point traces a trajectory, also called an orbit, through phase space.

A deterministic law gives one arrow at each point, and a smooth field gives one trajectory through each point. Starting from a given state, the system can move in one way only. Two trajectories therefore never cross in phase space (Figure 1). At a crossing point the same state would continue along two different curves, which would mean that the present state does not fix the future. Determinism and the absence of crossings are the same statement. The same argument run backwards in time shows that the present state also fixes the whole past. A curve that appears to cross itself when plotted, such as a measured coordinate plotted against time or a projection of a trajectory onto two of its coordinates, is not a trajectory in the full phase space, and its crossings signal that some state variables are missing from the plot. This observation returns in Section 4, where it decides how much of the past is needed to forecast a measured record.

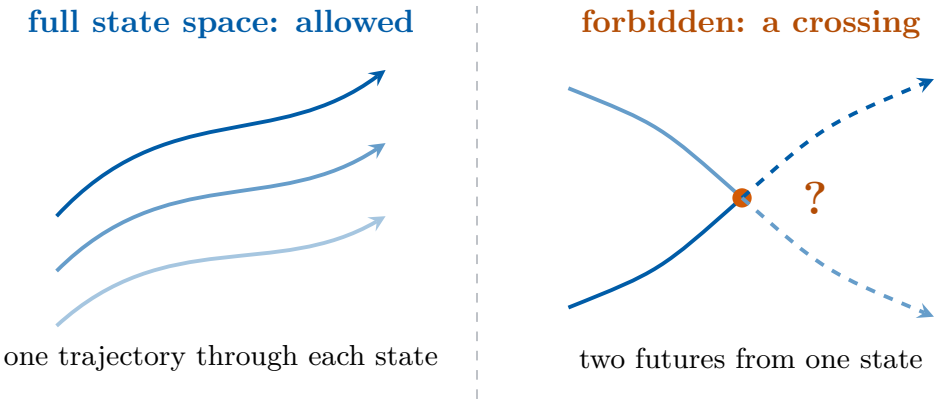


Figure 1: Left, trajectories of a deterministic law in the full phase space, one through each state. Right, a crossing, which would give one state two futures and is excluded for a smooth law $\mathbf{F}$.

Laplace imagined an intelligence with complete knowledge of the forces and the present state, able to calculate the entire past and future. This paraphrase of his argument expresses the predictive ideal of classical mechanics [1].

In the language of this chapter, the situation of all bodies is the state $\mathbf{x}$, the forces are the vector field $\mathbf{F}$, and the formula is the solution of equation (1.1). The statement is a consequence of the classical description of nature, in which a state is a point and the law of motion is a deterministic differential equation. It is not a statement about nature itself. In quantum mechanics the state determines only the probabilities of the outcomes of measurements, and the outcome of a single measurement is random even for a completely known state. The forecasting systems introduced here are classical, and for them this ideal describes the mathematical structure correctly. Two limitations nevertheless remain, and both are central to what follows. No observer has the intelligence that Laplace postulates, since the state is known only partly and only to finite precision, and for a chaotic system, as Section 3 shows, a small imprecision of the initial state grows until the forecast is lost.

Randomness in a classical system therefore has a precise meaning. It describes the information that the observer lacks and not an indeterminacy of the motion, and is called epistemic. The toss of a coin is the standard example. Its motion follows Newton's equations, and the outcome is fixed by the initial height, the upward velocity and the rate of spin at the moment the coin leaves the hand [2]. In the plane of the initial velocity and the spin rate, the initial conditions that give heads and those that give tails form alternating bands, which become narrower as the velocity and spin increase. A real toss is launched with an imprecision much larger than the width of a band, so the observer cannot tell into which band it falls, and the outcome is described by the probability one half. The coin is unpredictable because its initial state is not known well enough, and a machine that launched it with sufficient precision would produce the same outcome every time. Every record of this chapter will be examined from this point of view. An irregular record may come from a deterministic law with few variables, as the coin does, or from many unobserved influences that are best treated as random.

An experiment rarely sees the point $\mathbf{x}$ in phase space. It measures one quantity, or a few, at successive times separated by an interval Δt, and the measured value is a function of the state,

$$s_k = h(\mathbf{x}(k\Delta t)) + \eta_k, \qquad k = 0, 1, \ldots, N-1, \tag{1.2}$$

where the measurement function h maps the d numbers of the state to one number and η_k is the noise of the instrument. Figure 2 shows what this equation does. The state moves

continuously along its trajectory in the d-dimensional phase space. At the sampling times $t = k\Delta t$, and only then, the instrument evaluates one function of the current state, and the result is stored as the number s_k. Two reductions take place at once. Time becomes discrete, since the motion between the sampling times is not recorded, and the d coordinates of the state are reduced to one number, so that the measurement is a projection of the motion in phase space onto a single axis. The sequence $s_0, s_1, \ldots, s_{N-1}$ is a time series, the record of this projection at successive sampling times. Two different states can have the same projection, so a single measured value does not fix the state, and the record is a low-dimensional shadow of a motion in many dimensions. For the laser introduced in Section 1.1, the state consists of three variables, and the detector records only the intensity of the light.

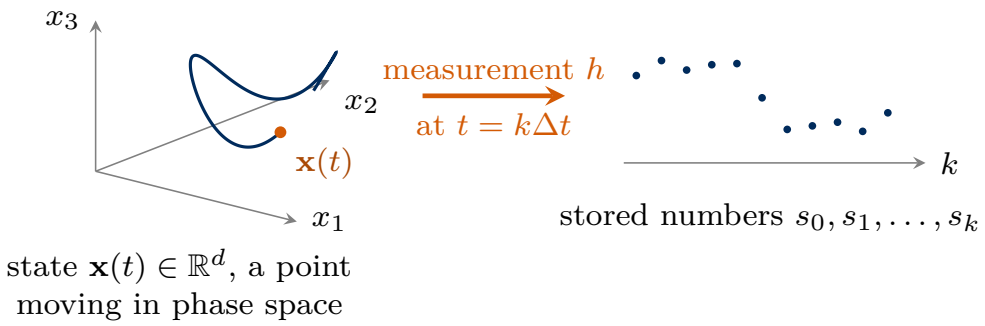


Figure 2: The state $\mathbf{x}(t)$ moves along a trajectory in phase space. At the times $k\Delta t$ the measurement h records one number, and the stored record $s_0, s_1, \ldots$ is all that remains of the motion.

Records of this kind are among the most common data of science. Weather stations record the temperature every day, oceanographers record the temperature of the sea surface every month, solar observatories count sunspots, electrodes record the electrical activity of the heart and of the brain, ecologists count animal populations year after year, and an experiment records the output of its detector at the rate of its electronics. In every case the measured values come from a system that changes according to some law, and in almost every case the record is a projection onto one or a few numbers of a state with many more.

Two questions are asked of such a record. The first is predictive. It asks for the value τ steps after time k, given the values up to time k. A forecast of tomorrow's temperature, of the next solar maximum or of the next pulse of a laser answers this question. The second question is explanatory. It asks for the law of motion that generates the record and for its parameters. A physicist who finds that a record is described by three coupled differential equations has learned something about the system, beyond the ability to forecast it.

Both questions meet three obstacles. The first is partial observation. The state of the system, the full set of variables that determines its future, is almost never measured. A thermometer records one number of an atmosphere described by millions of variables, and a detector records the intensity of a laser field but not the state of the molecules that amplify it. The second obstacle is an unknown or imprecise law. For a pendulum or a planetary orbit the equations of motion are known and their parameters can be measured. For a living cell, an ecosystem or the climate they are known only approximately, with missing terms and uncertain parameters. The third obstacle is that small errors can grow. In a chaotic system two nearly equal states separate exponentially in time, so even the exact law forecasts only over a limited horizon. In a stochastic system, driven by many influences that nobody records, the future is not fixed by the past at all, and the best possible forecast is an average over the possible futures.

Figure 3 contrasts the two situations. With the initial state and the law, the equations of

motion give the future, limited only by the growth of errors. With the record alone, a model has to learn a stand-in for the hidden dynamics from the compressed data, and its forecast is approximate.

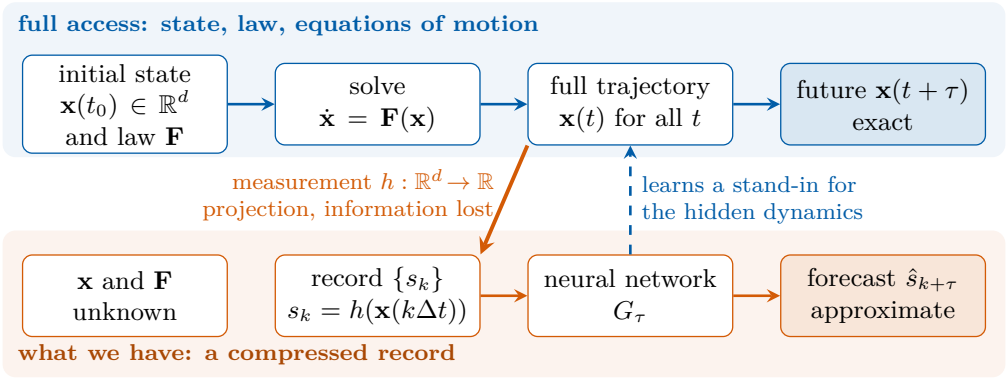


Figure 3: Top, full access to the state and the law gives the trajectory and the future state. Bottom, the measurement compresses the state into the record $\{s_k\}$, and a learned map G_τ forecasts $\hat{s}_{k+\tau}$ from it.

The approach of this lecture follows from these obstacles and proceeds in the order in which they arise. We first describe a system by its state and its law of motion, and a record by a noisy measurement of the state. We then quantify how fast small errors grow, which sets a limit to every forecast. Next we show that, for a deterministic system, a window of past values of a single measured variable can replace the hidden state, which turns a forecast into a regression problem, a map from a window of past values to a future value. We set up the training, the testing and the scoring of such a map and fit it with models of increasing flexibility, from linear autoregression through nonlinear features of the window to neural networks with learned features and memory. We apply the whole procedure to two measured records of different character, a daily temperature record and the sunspot number. Finally we turn to the second question and learn the law of motion itself, first a network for the vector field of a differential equation, then a network for an unknown term of a partly known law, and last a network for the solution of a known law.

The main measured example of the lecture is the intensity of a laser, introduced next. Its law is known in principle, but only one of its variables is measured. The Lorenz system, a three-variable model of convection with the same algebraic form as the laser equations, provides a simulated counterpart whose full state is available. Solutions of the Mackey–Glass delay equation, periodic for one parameter and chaotic for others, are benchmarks of different complexity. The temperature record and the sunspot number bring in measured records for which no law is established, and a damped pendulum is the example on which the law itself is learned.

1.1 A laser that pulses irregularly

A laser consists of an amplifying medium placed between two mirrors, which form the optical cavity, and of a source of energy, the pump, which keeps more molecules of the medium in an excited state than in the lower state. Light travelling back and forth in the cavity is amplified by stimulated emission, and a part of it leaves through one mirror, where a detector records its intensity. For most lasers and most pump strengths the output intensity settles to a constant value. In a far-infrared laser whose medium is ammonia gas (NH_3), studied experimentally in reference [3], it does not. The intensity rises in a sequence of pulses whose heights grow over

several pulses, then drops abruptly to a low level, and the growth starts again. Neither the number of pulses before a collapse nor the height of the largest pulse repeats. Figure 4 shows the arrangement, and Figure 5 shows 4000 consecutive samples of this record. It is known as data set A of the Santa Fe time-series competition [4], and we refer to it as the Santa Fe laser record.

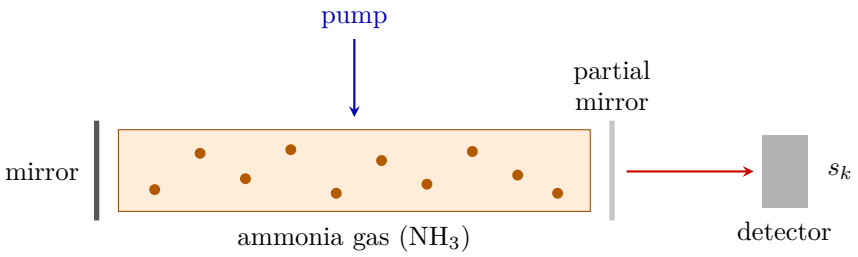


Figure 4: The far-infrared ammonia laser of reference [3]. The pump keeps the gas between the mirrors excited, light is amplified while it travels back and forth, and the part that leaves through the partial mirror is recorded by the detector as the intensity s_k.

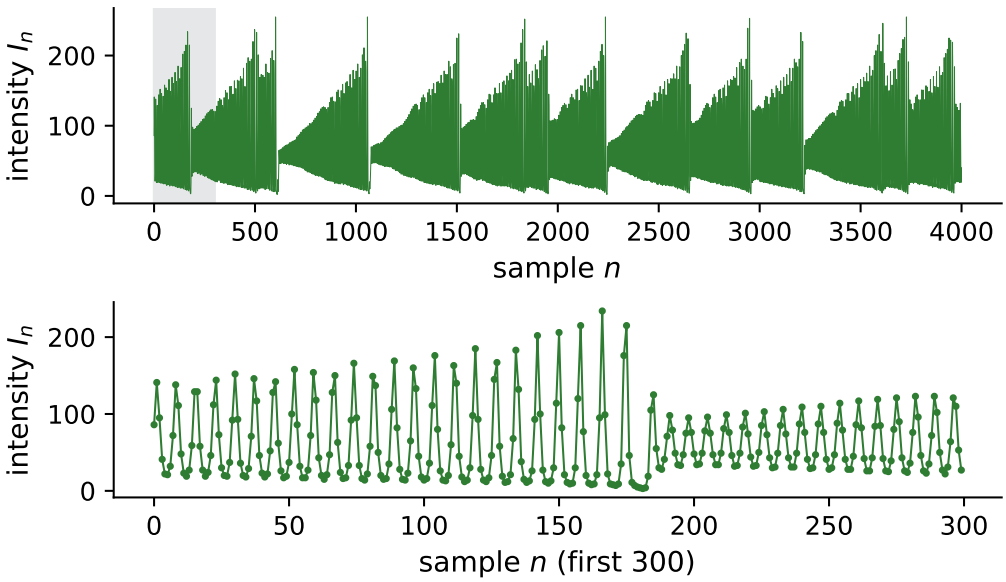


Figure 5: Santa Fe laser record, with the intensity I_n of the axis labels equal to s_k at sample $n = k$. Top, the 4000 samples used in this chapter. Bottom, the first 300 samples, which resolve the individual pulses and one collapse.

What is observed is a single sequence of numbers $s_0, s_1, \ldots, s_{N-1}$, the intensity at successive sampling times separated by a fixed interval Δt. The detector output was digitised with 8 bits, so every sample is an integer between 0 and 255, and the rounding to these levels is part of the measurement noise. The pulses recur roughly every 7.3 samples. We measure time in samples, $\Delta t = 1$.

For this record the predictive question asks for the value $s_{k+\tau}$, τ samples after the last known sample k. The sample k is called the forecast origin and $\tau \geq 1$ the horizon or lead. The forecast itself is written $\hat{s}_{k+\tau}$, where the hat marks an estimate. A useful forecast must anticipate the collapses, since they distinguish this record from a regular oscillation. The explanatory question asks for a law of motion that generates records of this kind and, where possible, for its parameters.

All three obstacles of the previous section are present here. The physics of a single-mode laser is known in principle. It involves three variables. The cavity field E is the light in the laser. The polarisation P is the electric dipole moment per unit volume that the field induces in the molecules. The population inversion D is the excess of molecules in the upper energy level over the lower one. The three variables obey the Maxwell–Bloch equations

$$\dot{E} = -\gamma_c(E - gP), \qquad \dot{P} = -\gamma_\perp(P - ED), \qquad \dot{D} = -\gamma_\parallel(D - D_0 + EP), \tag{1.3}$$

where the dot denotes the time derivative, γ_c, $\gamma_\perp$ and $\gamma_\parallel$ are the decay rates of the field, of the polarisation and of the inversion, g is a coupling constant and D_0 is the inversion maintained by the pump. The first obstacle, partial observation, appears because the detector measures only the intensity, which is proportional to E^2, and neither P nor D. At any moment P and D are hidden, and intensity also loses the sign of E. The equations are invariant under $(E, P, D) \mapsto (-E, -P, D)$, while E^2 is unchanged. These symmetry partners therefore produce the same entire intensity record. A reconstruction from intensity can at best distinguish states up to this symmetry; this can still suffice for predicting future intensity. The second obstacle, an imprecise law, appears because the rates and the pump level of the experiment are not known to the precision needed for a quantitative prediction. In most lasers the three rates differ by orders of magnitude, the fast variables follow the slow ones, and the motion is effectively one- or two-dimensional. In the far-infrared ammonia laser the three rates are of the same order, the field decays faster than the matter variables (a bad cavity), and all three variables remain active [3]. After a linear rescaling of the variables, equations (1.3) take the algebraic form of the Lorenz equations [5], a three-variable model whose solutions are chaotic for suitable parameters. This is the source of the third obstacle. For a chaotic system, small errors in the initial state grow exponentially, so that even the correct equations predict the state only over a limited time.

The difficulties of the laser record are those of every record produced by a system whose state is only partly observed.

2 States, flows and measured records

2.1 State and law of motion

The law of motion (1.1) is a system of first-order ordinary differential equations for the state $\mathbf{x} \in \mathbb{R}^d$,

$$\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}), \tag{2.1}$$

where the vector field $\mathbf{F} : \mathbb{R}^d \to \mathbb{R}^d$ assigns to every state its velocity in state space. Constant parameters, such as the decay rates in equations (1.3), are part of $\mathbf{F}$ and are not state variables. Equations of higher order are brought to this form by introducing derivatives as new variables. Newton's equation $m\ddot{q} = f(q)$ for one coordinate q becomes $\dot{q} = p/m$, $\dot{p} = f(q)$ with the momentum p, so one mechanical degree of freedom contributes two state variables. For the laser, $\mathbf{x} = (E, P, D)$ and $d = 3$. A law that does not depend explicitly on time, as in equation (2.1), is called autonomous.

The non-crossing of trajectories, used throughout this chapter, rests on the existence and uniqueness theorem for ordinary differential equations. If $\mathbf{F}$ is continuously differentiable, then through every initial state passes one and only one solution, at least for a finite time interval.

The vector field and the evolution over a finite time are two descriptions of the same dynamics. The field $\mathbf{F}$ gives the instantaneous velocity at each state, the local arrow. The evolution over

a time t is obtained by following the arrows from a given initial state, and it defines a map on phase space.

Definition 2.1 (Flow map)**.** The flow map $\varphi_t : \mathbb{R}^d \to \mathbb{R}^d$ sends an initial state $\mathbf{x}_0$ to the state reached after the time t along the solution of equation (2.1),

$$\varphi_t(\mathbf{x}_0) = \mathbf{x}(t; \mathbf{x}_0), \qquad \frac{d}{dt}\varphi_t(\mathbf{x}_0) = \mathbf{F}(\varphi_t(\mathbf{x}_0)). \tag{2.2}$$

The uniqueness of solutions gives the flow map three properties,

$$\varphi_0 = \mathrm{id}, \qquad \varphi_{t+s} = \varphi_t \circ \varphi_s, \qquad \varphi_{-t} = \varphi_t^{-1}. \tag{2.3}$$

Evolving for zero time leaves every state unchanged. Evolving for s and then for t reaches the same state as evolving for $t + s$, because the law does not change in time and through the intermediate state passes only one trajectory. Evolving for t and then for $-t$ returns to the initial state, so every flow map is invertible where the solutions exist backwards in time. Maps with these properties form a one-parameter group, and the vector field is its generator, $\mathbf{F}(\mathbf{x}) = \frac{d}{dt}\varphi_t(\mathbf{x})\big|_{t=0}$.

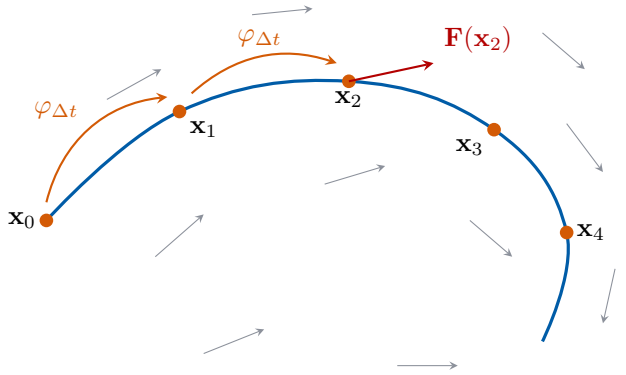


Figure 6: The vector field $\mathbf{F}$ attaches a velocity to every state (grey arrows), and a trajectory (blue) is tangent to the arrows at every point. Sampling at the interval Δt picks the states $\mathbf{x}_k = \mathbf{x}(k\Delta t)$ (orange), and the flow map $\varphi_{\Delta t}$ carries each of them to the next one.

A record sampled at the interval Δt sees the system only through the map $\varphi_{\Delta t}$ (Figure 6). With $\mathbf{x}_k = \mathbf{x}(k\Delta t)$, the composition rule (2.3) gives

$$\mathbf{x}_{k+1} = \varphi_{\Delta t}(\mathbf{x}_k), \qquad \mathbf{x}_{k+n} = \underbrace{\varphi_{\Delta t} \circ \cdots \circ \varphi_{\Delta t}}_{n}(\mathbf{x}_k) = \varphi_{n\Delta t}(\mathbf{x}_k), \tag{2.4}$$

so a continuous-time system observed at discrete times is a discrete-time map $\mathbf{x}_{k+1} = \mathbf{f}(\mathbf{x}_k)$ with $\mathbf{f} = \varphi_{\Delta t}$. The distinction between the velocity $\mathbf{F}$ and the one-step map $\varphi_{\Delta t}$ matters for every model of this chapter. A forecasting model that maps the present to a later value approximates a flow map, while the neural ODE of Section 10 approximates the vector field and obtains the flow map by integration.

2.2 Dissipation and attractors

A laser loses energy through the output mirror and through the decay of the polarisation and the inversion. When dissipation is accompanied by confinement, motion can approach a

subset A of phase space after a transient and remain close to it. An attractor is an invariant set attracting a neighbourhood of initial states. The simplest attractor is a fixed point, a state with $\mathbf{F}(\mathbf{x}^*) = 0$, such as a pendulum at rest. A limit cycle is a closed trajectory, as in a periodically pulsing laser. A strange attractor is a bounded set on which typical trajectories never close and which has a non-integer dimension. Dissipation can be stated as a property of the vector field, through the evolution of phase-space volumes.

Rate of change of phase-space volume. *Let a small set of initial states occupy the volume $V(t)$ at time t. Then*

$$\frac{dV}{dt} = \int_{A_t} \nabla \cdot \mathbf{F}\, d^d x, \qquad \nabla \cdot \mathbf{F} = \sum_{i=1}^{d} \frac{\partial F_i}{\partial x_i}, \tag{2.5}$$

where A_t is the region occupied at time t.

Derivation. A volume element $dV(0)$ is mapped into $dV(t) = \det M_t\, dV(0)$, where M_t with entries $(M_t)_{ij} = \partial x_i(t)/\partial x_j(0)$ is the Jacobian matrix of the flow map. Its determinant is positive because the derivative of a smooth invertible flow starts at the identity and cannot become singular. Differentiating $\dot{\mathbf{x}}(t) = \mathbf{F}(\mathbf{x}(t))$ with respect to the initial state gives, by the chain rule, $\dot{M}_t = J(\mathbf{x}(t))\, M_t$ with $M_0 = \mathbb{1}$, where $J_{ij} = \partial F_i/\partial x_j$ is the Jacobian of the vector field. Jacobi's formula for the derivative of a determinant, $\frac{d}{dt} \det M = \det M \operatorname{tr}(M^{-1}\dot{M})$, then yields $\frac{d}{dt} \det M_t = \det M_t \operatorname{tr} J = \det M_t\, \nabla \cdot \mathbf{F}$. Integrating $\frac{d}{dt} dV = \nabla \cdot \mathbf{F}\, dV$ over the region gives equation (2.5). □

Equation (2.5) contains a classical result of mechanics as a special case. For conservative mechanical systems, a Hamiltonian $H(\mathbf{q}, \mathbf{p})$ of the coordinates $\mathbf{q} = (q_1, \ldots, q_n)$ and the conjugate momenta $\mathbf{p} = (p_1, \ldots, p_n)$ generates the law of motion through Hamilton's equations,

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q_i}, \qquad i = 1, \ldots, n, \tag{2.6}$$

with the state $\mathbf{x} = (\mathbf{q}, \mathbf{p})$ and $d = 2n$. The frictionless pendulum, introduced in Section 2.3, is the simplest example, with one coordinate, the angle, one conjugate momentum, equal to angular velocity in the dimensionless units used below, and the energy (2.11) as Hamiltonian.

Liouville's theorem. *The flow of Hamilton's equations (2.6) conserves the volume of every region of phase space.*

Derivation. The divergence of the Hamiltonian vector field is

$$\nabla \cdot \mathbf{F} = \sum_{i=1}^{n} \Big(\frac{\partial \dot{q}_i}{\partial q_i} + \frac{\partial \dot{p}_i}{\partial p_i} \Big) = \sum_{i=1}^{n} \Big(\frac{\partial^2 H}{\partial q_i\, \partial p_i} - \frac{\partial^2 H}{\partial p_i\, \partial q_i} \Big) = 0, \tag{2.7}$$

since the mixed second derivatives of a smooth function are equal. Equation (2.5) then gives $dV/dt = 0$. □

A cloud of initial states of a frictionless system can therefore be deformed and stretched by the motion, but it keeps its volume, as an incompressible fluid does. Volume preservation excludes an attracting set of smaller volume with a basin of positive volume. Friction changes this, as the damped pendulum of the next subsection shows. A negative divergence contracts infinitesimal volumes. Uniform contraction maintained over time makes their volume tend to zero. An attracting bounded region also requires confinement of trajectories, which must be established separately. Within such a region, continued contraction can coexist with complicated motion.

2.3 The pendulum

The pendulum is the simplest system on which the notions introduced so far can be followed by explicit calculation, and it returns in Sections 10 and 11, where its law is learned from data. A bob of mass m hangs on a rigid massless rod of length l and swings in a vertical plane. Its position is given by the angle θ between the rod and the downward vertical. Gravity exerts the torque $-mgl\sin\theta$ about the pivot, which pulls the bob back towards the lowest point, and friction in the pivot and in the air exerts a torque that opposes the motion. For slow motion the friction torque is proportional to the angular velocity, $-b\dot\theta$ with a friction coefficient $b>0$. The moment of inertia of the bob is ml^2, and Newton's second law for rotation reads

$$ml^2\,\ddot\theta = -mgl\sin\theta - b\,\dot\theta. \tag{2.8}$$

Dividing by ml^2 and measuring time in units of $1/\omega_0$, where $\omega_0=\sqrt{g/l}$ is the angular frequency of small oscillations, removes all constants except one,

$$\ddot\theta = -\sin\theta - \gamma\,\dot\theta, \qquad \gamma = \frac{b}{ml^2\omega_0}, \tag{2.9}$$

where the dimensionless damping γ is the friction rate measured in units of the oscillation frequency. In these units a small oscillation without friction has the period 2π.

Equation (2.9) is of second order. The angle alone does not fix the future, since a pendulum passing the lowest point may move in either direction, and the angular velocity $\omega=\dot\theta$ becomes the second state variable. The state is $\mathbf{x}=(\theta,\omega)$, the phase space is the plane of angle and angular velocity, and the law of motion takes the first-order form of equation (2.1),

$$\dot\theta=\omega, \qquad \dot\omega=-\sin\theta-\gamma\,\omega, \qquad \text{so that} \qquad \mathbf{F}(\theta,\omega)=\begin{pmatrix}\omega\\ -\sin\theta-\gamma\,\omega\end{pmatrix}. \tag{2.10}$$

The first component says that the angle changes at the rate ω. The second contains the two physical torques, gravity and friction.

The energy shows what friction does. The kinetic energy of the bob and its potential energy, counted from the lowest point, give in the units of equation (2.9)

$$E(\theta,\omega)=\tfrac12\,\omega^2+1-\cos\theta. \tag{2.11}$$

Differentiating along a trajectory and inserting equations (2.10) gives

$$\frac{dE}{dt}=\omega\,\dot\omega+\sin\theta\,\dot\theta=\omega\,(-\sin\theta-\gamma\,\omega)+\sin\theta\,\omega=-\gamma\,\omega^2. \tag{2.12}$$

Without friction, $\gamma=0$, the energy is conserved, and every trajectory runs along a curve of constant E in the phase plane. For $E<2$ these curves are closed and the pendulum swings back and forth. For $E>2$ the bob passes over the top and the angle grows without bound. The curve $E=2$ separates the two kinds of motion. With friction the energy decreases whenever the pendulum moves, so every trajectory crosses the curves of constant energy inwards, and a swinging pendulum spirals towards rest.

The same conclusion follows from the divergence of the vector field (2.10),

$$\nabla\cdot\mathbf{F}=\frac{\partial\dot\theta}{\partial\theta}+\frac{\partial\dot\omega}{\partial\omega}=0+(-\gamma)=-\gamma. \tag{2.13}$$

By equation (2.5), an area $\mathcal{A}$ of initial states in the phase plane evolves as

$$\mathcal{A}(t)=\mathcal{A}(0)\,e^{-\gamma t}. \tag{2.14}$$

For $\gamma = 0$ the area is conserved, as Liouville's theorem requires, and for $\gamma > 0$ every region of initial states contracts towards a set of zero area.

The fixed points are the states at which the vector field vanishes. The condition $\mathbf{F}(\theta^*, \omega^*) = 0$ gives $\omega^* = 0$ from the first component and $\sin\theta^* = 0$ from the second, so the fixed points are the hanging position $(\theta^*, \omega^*) = (0, 0)$ and the upright position $(\theta^*, \omega^*) = (\pi, 0)$, each repeated every 2π in the angle. Their stability follows from the Jacobian of the vector field,

$$J(\theta, \omega) = \begin{pmatrix} \partial\dot{\theta}/\partial\theta & \partial\dot{\theta}/\partial\omega \\ \partial\dot{\omega}/\partial\theta & \partial\dot{\omega}/\partial\omega \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\cos\theta & -\gamma \end{pmatrix}. \tag{2.15}$$

A small deviation from a fixed point evolves with the Jacobian at that point, and it grows or decays as $e^{\mu t}$, where μ is an eigenvalue of J. The eigenvalues are the roots of the characteristic equation,

$$\det\left(J(\theta^*, 0) - \mu\mathbb{1}\right) = \mu^2 + \gamma\,\mu + \cos\theta^* = 0, \qquad \mu_\pm = -\frac{\gamma}{2} \pm \sqrt{\frac{\gamma^2}{4} - \cos\theta^*}. \tag{2.16}$$

At the hanging position $\cos\theta^* = 1$, and for weak damping, $0 < \gamma < 2$, the square root is imaginary,

$$\mu_\pm = -\frac{\gamma}{2} \pm i\sqrt{1 - \frac{\gamma^2}{4}}. \tag{2.17}$$

The imaginary part is the frequency of the oscillation and the negative real part its decay, so a deviation circles the hanging position with an amplitude that decreases as $e^{-\gamma t/2}$. The hanging position is a stable spiral. At $\gamma = 0$ its linearisation is a centre and the conserved energy prevents attraction. At $\gamma = 2$ the eigenvalues coincide and the linearised response is critically damped. For $\gamma > 2$ both eigenvalues are real and negative, giving a stable node. At the upright position $\cos\theta^* = -1$, and

$$\mu_\pm = -\frac{\gamma}{2} \pm \sqrt{1 + \frac{\gamma^2}{4}}. \tag{2.18}$$

The square root is larger than $\gamma/2$, so $\mu_+ > 0$ and $\mu_- < 0$. A deviation along one direction grows and along the other decays, and the upright position is a saddle. A pendulum balanced upside down falls at the slightest disturbance, while trajectories that arrive along the decaying direction approach the top ever more slowly. These incoming trajectories divide the phase plane into regions of trajectories that pass over the top a different number of times before they settle into a swing around one of the hanging positions.

In the examples of this chapter $\gamma = 0.200$. Equation (2.17) then gives $\mu_\pm = -0.100 \pm 0.995\,i$. The frequency is almost unchanged by the friction, and the small-amplitude envelope decreases by a factor e in the time $2/\gamma = 10.000$. The upright position retains a growing mode for every positive damping coefficient.

2.4 The Lorenz model of convection

The laser supplies a measured record with hidden coordinates. To test how much information a record retains, we also need a system whose full state and equations are available. The Lorenz model supplies that comparison. Its three equations have the same algebraic structure as the rescaled laser equations, although they describe the circulation and temperature of a fluid heated from below [6, 7].

A horizontal fluid layer of height H lies between plates at temperatures $T_0 + \Delta T$ below and T_0 above. This arrangement is called Rayleigh–Bénard convection. Heating makes fluid expand and lowers its density, so a warm parcel tends to rise through colder surroundings. Viscosity resists the motion, and thermal diffusion reduces temperature differences. At weak heating, the fluid remains at rest and carries heat by conduction. Increasing the heating can destabilise this state and produce circulating rolls.

The velocity $\mathbf{v}(\mathbf{r}, t_{\mathrm{phys}})$ and temperature $T(\mathbf{r}, t_{\mathrm{phys}})$ are fields, with values throughout the layer. The Boussinesq approximation assumes small relative density changes, $\alpha\Delta T \ll 1$, and retains them in the buoyancy force while using a constant reference density ρ_{m} elsewhere. With reduced pressure p, kinematic viscosity ν, thermal diffusivity χ, thermal expansion coefficient α, gravitational acceleration g and upward unit vector $\hat{\mathbf{z}}$, the equations are

$$\partial_{t_{\mathrm{phys}}}\mathbf{v} + (\mathbf{v}\cdot\nabla)\mathbf{v} = -\frac{\nabla p}{\rho_{\mathrm{m}}} + \nu\nabla^2\mathbf{v} + g\alpha(T - T_0)\hat{\mathbf{z}}, \tag{2.19}$$

$$\nabla\cdot\mathbf{v} = 0, \tag{2.20}$$

$$\partial_{t_{\mathrm{phys}}}T + (\mathbf{v}\cdot\nabla)T = \chi\nabla^2 T. \tag{2.21}$$

The reference hydrostatic pressure has been absorbed into p. In the momentum equation, the two terms on the left give the acceleration of a moving fluid parcel. The pressure gradient, viscous friction and buoyancy determine that acceleration. Incompressibility states that a parcel keeps its volume within this approximation. In the temperature equation, motion transports heat and diffusion smooths temperature differences. The advection terms couple the fields nonlinearly, and specifying their values throughout space requires an infinite-dimensional state.

The Prandtl number $\sigma = \nu/\chi$ compares viscous and thermal diffusion. The Rayleigh number measures buoyancy relative to both dissipative effects,

$$\mathrm{Ra} = \frac{g\alpha H^3\Delta T}{\nu\chi}. \tag{2.22}$$

The conduction profile $T_{\mathrm{cond}}(\zeta) = T_0 + \Delta T(1 - \zeta/H)$, with zero velocity, solves the field equations at every heating strength. Its stability depends on Ra. A parcel displaced upwards can remain warmer than its new surroundings and continue rising. Diffusion and viscosity suppress this feedback at sufficiently small Ra.

We reduce the fields to a few amplitudes by restricting the motion to a vertical plane, with horizontal coordinate ξ and height ζ. Write

$$\mathbf{v} = (-\partial_\zeta\Upsilon, 0, \partial_\xi\Upsilon), \qquad T = T_{\mathrm{cond}} + \delta T. \tag{2.23}$$

The stream function Υ enforces incompressibility because its mixed derivatives cancel. Taking the curl of the momentum equation eliminates the pressure. Define the advection operator $\mathcal{J}(\Upsilon, f) = \partial_\xi\Upsilon\,\partial_\zeta f - \partial_\zeta\Upsilon\,\partial_\xi f$. The resulting equations are

$$\begin{aligned} \partial_{t_{\mathrm{phys}}}\nabla^2\Upsilon + \mathcal{J}(\Upsilon, \nabla^2\Upsilon) &= \nu\nabla^4\Upsilon + g\alpha\partial_\xi\delta T, \\ \partial_{t_{\mathrm{phys}}}\delta T + \mathcal{J}(\Upsilon, \delta T) &= \chi\nabla^2\delta T + \frac{\Delta T}{H}\partial_\xi\Upsilon. \end{aligned} \tag{2.24}$$

The last term describes warm fluid carried upwards through the conduction gradient.

For horizontally periodic rolls and impermeable, free-slip plates held at fixed temperatures, the boundary conditions at $\zeta = 0, H$ are $v_\zeta = 0$, $\partial_\zeta v_\xi = 0$ and $\delta T = 0$. They impose zero normal flow, zero tangential viscous stress and a fixed plate temperature. Sine and cosine

modes can satisfy these conditions. Let $k_x = \pi a/H$, $k_z = \pi/H$ and $\Lambda^2 = k_x^2 + k_z^2$. We retain one stream-function mode and two temperature modes,

$$\begin{aligned} \Upsilon &= \mathsf{A}(t_{\text{phys}}) \sin(k_x \xi) \sin(k_z \zeta), \\ \delta T &= \mathsf{B}(t_{\text{phys}}) \cos(k_x \xi) \sin(k_z \zeta) - \mathsf{C}(t_{\text{phys}}) \sin(2 k_z \zeta). \end{aligned} \tag{2.25}$$

Each roll has width H/a. The amplitude A describes circulation, B a temperature difference between rising and sinking fluid, and C a change in the horizontally averaged temperature profile.

Projection means multiplying each field equation by a retained spatial mode and integrating over a horizontal period and the layer height. Orthogonality removes contributions from the other modes. Here $\nabla^2 \Upsilon = -\Lambda^2 \Upsilon$, so $\mathcal{J}(\Upsilon, \nabla^2 \Upsilon) = 0$. The retained vorticity equation therefore has no nonlinear self-advection term. In the temperature equation, the product of the circulation and temperature patterns generates both retained modes and a mode proportional to $\cos(k_x \xi) \sin(3 k_z \zeta)$, which is discarded. Projection gives

$$\begin{aligned} \frac{d\mathsf{A}}{dt_{\text{phys}}} &= -\nu \Lambda^2 \mathsf{A} + \frac{g \alpha k_x}{\Lambda^2} \mathsf{B}, \\ \frac{d\mathsf{B}}{dt_{\text{phys}}} &= \frac{k_x \Delta T}{H} \mathsf{A} - \chi \Lambda^2 \mathsf{B} - k_x k_z \mathsf{A} \mathsf{C}, \\ \frac{d\mathsf{C}}{dt_{\text{phys}}} &= \frac{k_x k_z}{2} \mathsf{A} \mathsf{B} - 4 \chi k_z^2 \mathsf{C}. \end{aligned} \tag{2.26}$$

These equations display the origin of the nonlinear terms before any change of units.

Introduce the dimensionless time $t = \chi \Lambda^2 t_{\text{phys}}$ and amplitudes

$$X = \frac{k_x k_z \mathsf{A}}{\sqrt{2} \chi \Lambda^2}, \qquad Y = \frac{g \alpha k_x^2 k_z \mathsf{B}}{\sqrt{2} \nu \chi K^6}, \qquad Z = \frac{g \alpha k_x^2 k_z \mathsf{C}}{\nu \chi K^6}. \tag{2.27}$$

The projected equations become the Lorenz model,

$$\dot{X} = \sigma(Y - X), \qquad \dot{Y} = X(\rho - Z) - Y, \qquad \dot{Z} = XY - \beta Z, \tag{2.28}$$

with

$$\rho = \frac{\text{Ra}}{\text{Ra}_c}, \qquad \text{Ra}_c = \frac{\pi^4 (1 + a^2)^3}{a^2}, \qquad \beta = \frac{4}{1 + a^2}. \tag{2.29}$$

The diffusion time used as one unit is $H^2/[\pi^2(1+a^2)\chi]$. Minimising Ra_c over a gives $a^2 = 1/2$, $\beta = 8/3$ and $\text{Ra}_c = 27\pi^4/4 = 657.511$. This threshold belongs to the free-slip model.

The term σY expresses buoyant driving of the circulation, and $-\sigma X$ its viscous decay. The term ρX creates a temperature contrast by transporting the conduction profile, while $-Y$ describes thermal diffusion. The circulation and temperature contrast alter the mean profile through XY, and $-\beta Z$ relaxes that alteration. The modified profile reduces the production of temperature contrast through $-XZ$. Figure 7 relates these amplitudes to the fluid layer.

The reduction uses five restrictions. Density variations enter only through buoyancy, with constant transport coefficients. The flow is two-dimensional. The horizontal wavelength is fixed. The plates are impermeable and free-slip at fixed temperatures. Only the three specified spatial modes are retained. Real rigid plates impose different velocity boundary conditions, and stronger convection can excite additional spatial modes. The three-mode system is therefore a controlled definition of our benchmark, while a quantitative prediction for a particular fluid requires checking these approximations. Its irregular motion at $\sigma = 10$,

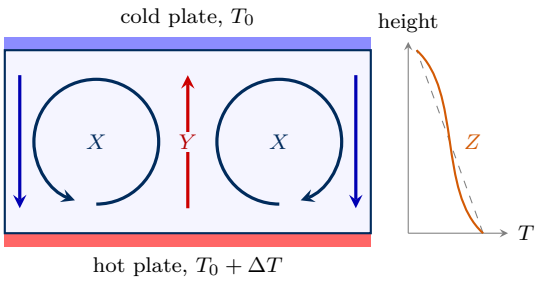


Figure 7: The convecting fluid layer of the Lorenz model. X is the speed of the two convection rolls, with its sign giving the sense of rotation, Y the temperature difference between the rising warm fluid and the sinking cold fluid, and Z the bending of the mean vertical temperature profile (solid) away from the straight profile of pure conduction (dashed).

$\rho = 28$ and $\beta = 8/3$ provides a small system in which to study predictability. We retain the amplitudes (X, Y, Z) in the derivation and use (x, y, z) in the numerical plots.

The fixed points of equations (2.28) follow from setting the right-hand sides to zero. The first equation gives $X = Y$, the third $Z = X^2/\beta$, and the second then $X(\rho - 1 - Z) = 0$. The origin, pure conduction, is one solution. The other two are the steady rolls

$$C_\pm = \Big(\pm\sqrt{\beta(\rho-1)},\ \pm\sqrt{\beta(\rho-1)},\ \rho - 1 \Big) = (\pm 8.485,\ \pm 8.485,\ 27), \tag{2.30}$$

two steady convection patterns of equal strength that turn in opposite senses, one for each sign of X. Their stability is decided by the Jacobian of the Lorenz field,

$$J(X,Y,Z) = \begin{pmatrix} -\sigma & \sigma & 0 \\ \rho - Z & -1 & -X \\ Y & X & -\beta \end{pmatrix}, \tag{2.31}$$

evaluated at the fixed point. A small deviation $\boldsymbol{\delta}$ from a fixed point obeys $\dot{\boldsymbol{\delta}} = J\boldsymbol{\delta}$, and its solution is a sum of terms proportional to $e^{\mu t}$, where μ runs over the eigenvalues of J, the roots of $\det(J - \mu\mathbb{1}) = 0$. A deviation along an eigenvector with negative real part of μ decays and one with positive real part grows, while an imaginary part makes the deviation rotate. If all real parts are negative the point attracts its neighbourhood, as a stable node for real eigenvalues or a stable spiral for complex ones. If some are negative and some positive the point is a saddle, which attracts along some directions and repels along others. A complex pair with positive real part gives an unstable spiral, along which the deviation circles the point with growing amplitude.

The loss of stability of the steady rolls follows from their characteristic polynomial. Substitution of $C_\pm$ into $\det(\mu\mathbb{1} - J) = 0$ gives

$$\mu^3 + (\sigma + \beta + 1)\mu^2 + \beta(\sigma + \rho)\mu + 2\sigma\beta(\rho - 1) = 0. \tag{2.32}$$

A complex pair crosses the imaginary axis when $\mu = i\Omega$ with $\Omega \neq 0$. Its imaginary and real parts require

$$\Omega^2 = \beta(\sigma + \rho), \qquad (\sigma + \beta + 1)\beta(\sigma + \rho) = 2\sigma\beta(\rho - 1). \tag{2.33}$$

Solving for the heating gives

$$\rho_c = \frac{\sigma(\sigma + \beta + 3)}{\sigma - \beta - 1}, \qquad \sigma > \beta + 1. \tag{2.34}$$

For a cubic $\mu^3 + a_1\mu^2 + a_2\mu + a_3$, the Routh–Hurwitz stability criterion is $a_1, a_2, a_3 > 0$ and $a_1 a_2 > a_3$. Here, for $\rho > 1$, the last inequality holds below ρ_c and becomes equality at the crossing. The remaining real eigenvalue at the crossing is $-(\sigma + \beta + 1)$. For the parameters used here $\rho_c = 24.737$ and $\Omega = 9.625$. These calculations locate the oscillatory instability. Chaotic motion above the crossing requires the further evidence from bounded trajectories and positive Lyapunov exponents developed below [7].

The eigenvalues are collected in Table 1. The origin has two negative eigenvalues and one positive one, so it is a saddle. Conduction is unstable, and a perturbation with a component along the unstable direction grows. At the steady rolls $C_\pm$ one real eigenvalue is strongly negative, which pulls trajectories onto a nearly flat sheet, and a complex pair has a small positive real part. Within the sheet the rolls are weakly unstable spirals. A trajectory circles one of them with slowly growing amplitude until it comes close enough to the origin to be thrown towards the other roll, where the same spiralling starts again. This mechanism produces the two lobes of Figure 8 and the irregular switching of $x(t)$ between positive and negative values. The steady rolls lose their stability when the heating exceeds $\rho_c = 24.74$, so at $\rho = 28$ no stable fixed point remains. The stored signal is $s_k = x(k\Delta t)$ with $\Delta t = 0.02$ time units.

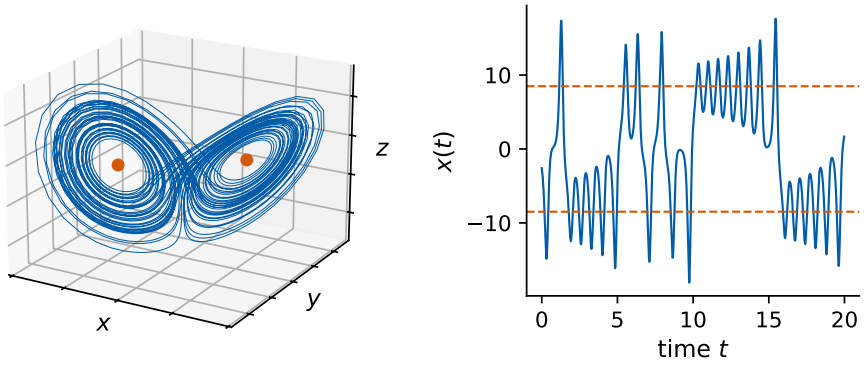


Figure 8: Lorenz system at $\rho = 28$. Left, a trajectory in phase space with the steady rolls $C_\pm$ (dots). Right, the coordinate $x(t)$, which switches irregularly between the two lobes.

2.5 A delayed feedback model

The convection model has a finite-dimensional state. A biological feedback loop can require a history because its response arrives after a maturation delay. The Mackey–Glass equation [8], models the regulation of blood-cell production,

$$\dot{x}(t) = \frac{0.2\, x(t - \tau_{\mathrm{MG}})}{1 + x(t - \tau_{\mathrm{MG}})^{10}} - 0.1\, x(t). \tag{2.35}$$

Production at time t depends on the concentration x a maturation delay τ_{MG} earlier, and removal is proportional to the present concentration. The future of a delay equation is fixed only by the whole history of x on an interval of length τ_{MG}, so its state space is infinite-dimensional. Numerically, the solution is periodic for $\tau_{\mathrm{MG}} = 15$ and chaotic for $\tau_{\mathrm{MG}} = 25$ and 35. The three cases are sampled once per time unit and provide a periodic reference and two chaotic signals of different complexity. The numerical approximation stores the delayed history on a grid, so its Lyapunov spectrum and dimension also depend on convergence of this discretisation [9].

2.6 Measurement, noise and randomness

The measured record (1.2) retains the part of the state to which the instrument responds, through the measurement function $h : \mathbb{R}^d \to \mathbb{R}$, and adds the measurement noise η_k. For the laser, $h(E, P, D) \propto E^2$ and the noise includes the 8-bit rounding. Equation (1.2) assumes that the measurement does not disturb the dynamics, which is appropriate for the classical systems of this chapter.

Randomness in a record has two different origins in classical physics. Measurement noise corrupts the record but does not affect the future of the system. Unresolved degrees of freedom, by contrast, act on the dynamics. A pollen grain in water is pushed by molecules whose positions are not recorded, and a description in terms of the grain alone must include a random force. Such randomness is epistemic in the sense of Section 1, a statement about missing information, and it decreases when more of the system is measured. For each record the question is whether a deterministic description with few variables is adequate or whether a stochastic description is required. An irregular appearance alone does not decide this question, since a chaotic deterministic signal such as the Lorenz coordinate and a stochastic signal can look equally irregular. Statistics such as the mean and the autocorrelation do not separate the two, and a geometric diagnostic is needed.

The statistical description used throughout treats a record as one realisation of a stochastic process, a family of random variables $s_1, s_2, \ldots$ indexed by time. The process is weakly stationary when its mean μ, its variance and its autocovariance $\gamma(m) = \langle (s_k - \mu)(s_{k+m} - \mu) \rangle$ do not depend on k, where $\langle \cdot \rangle$ denotes the expectation. The autocorrelation $\gamma(m)/\gamma(0)$ is estimated from a record of length N with mean $\bar{s}$ as

$$\rho(m) = \frac{\sum_{k=0}^{N-1-m} (s_k - \bar{s})(s_{k+m} - \bar{s})}{\sum_{k=0}^{N-1} (s_k - \bar{s})^2}, \qquad \rho(0) = 1. \tag{2.36}$$

It measures how well s_k predicts s_{k+m} by a linear function, since the best predictor of the form $\mu + b(s_k - \mu)$ has $b = \rho(m)$ and the mean squared error $\gamma(0)(1 - \rho(m)^2)$. A small autocorrelation therefore excludes a useful linear forecast from the current value alone and leaves nonlinear and multi-value forecasts open. The laser record is an example. Its autocorrelation is 0.526 at a separation of one sample and -0.202 at two samples so a forecast that repeats the latest value is of little use two samples ahead, although the pulses are evidently not random. A low-dimensional deterministic model is physically motivated for this laser. Its consistency with the measured record requires diagnostics beyond the autocorrelation.

The power spectrum describes the same record by frequencies. The discrete Fourier transform

$$X_j = \sum_{k=0}^{N-1} (s_k - \bar{s})\, e^{-2\pi i j k / N} \tag{2.37}$$

decomposes the record into frequencies $f_j = j/(N\Delta t)$. For an even-length real record, the one-sided periodogram on $j = 0, \ldots, N/2$ is

$$P_j = \frac{\Delta t}{N} w_j |X_j|^2, \qquad w_j = \begin{cases} 1, & j = 0 \text{ or } j = N/2, \\ 2, & 0 < j < N/2. \end{cases} \tag{2.38}$$

Interior bins combine the equal powers at positive and negative frequencies. With $\Delta f = 1/(N\Delta t)$, Parseval's identity gives $\sum_j P_j \Delta f = N^{-1} \sum_k (s_k - \bar{s})^2$, so the spectrum is a variance density per unit frequency. A peak at f_j marks a component that repeats with the period $1/f_j$, and the frequency resolution is $1/(N\Delta t)$, the inverse length of the record. An infinite periodic

signal has discrete spectral lines. A finite window broadens them and can spread power into adjacent bins. Phase variation can broaden peaks further, while uncorrelated noise has a flat population spectrum. The notebook averages windowed periodograms by the Welch method to reduce fluctuations in the spectral estimate. By the Wiener–Khinchin theorem the power spectrum is the Fourier transform of the autocovariance, so it carries the same information as the autocorrelation, arranged by frequency. The pulses of the laser recur roughly every 7.3 samples, the period of the strongest peak of its power spectrum.

3 Sensitivity to initial conditions

A known deterministic law assigns one future to each initial state, while measurements specify that state only to finite precision. The usefulness of a calculated trajectory therefore depends on how the flow amplifies the remaining uncertainty. Local stability at a fixed point extends to growth rates along an entire trajectory.

3.1 Growth of small errors

Suppose that the law $\mathbf{F}$ is known without error and that the initial state is known up to a small error $\boldsymbol{\delta}_0$. The forecast then follows the trajectory $\tilde{\mathbf{x}}(t)$ that starts at $\tilde{\mathbf{x}}(0) = \mathbf{x}(0) + \boldsymbol{\delta}_0$ instead of the true trajectory $\mathbf{x}(t)$, and the question is how the separation $\boldsymbol{\delta}(t) = \tilde{\mathbf{x}}(t) - \mathbf{x}(t)$ between the two trajectories evolves. Both trajectories obey the same law, so the separation obeys $\dot{\boldsymbol{\delta}} = \mathbf{F}(\mathbf{x} + \boldsymbol{\delta}) - \mathbf{F}(\mathbf{x})$. As long as the separation is small, a Taylor expansion of $\mathbf{F}$ to first order in $\boldsymbol{\delta}$ gives the tangent equation

$$\dot{\boldsymbol{\delta}} = J(\mathbf{x}(t))\,\boldsymbol{\delta}, \qquad J_{ij} = \frac{\partial F_i}{\partial x_j}, \tag{3.1}$$

where the Jacobian J of the vector field is evaluated along the true trajectory. The tangent equation is linear in $\boldsymbol{\delta}$, although the law itself is nonlinear, and its coefficients change in time as the true state moves through phase space.

A linear equation has a linear solution. The error at time t is a linear function of the initial error,

$$\boldsymbol{\delta}(t) = M_t\,\boldsymbol{\delta}_0, \qquad (M_t)_{ij} = \frac{\partial x_i(t)}{\partial x_j(0)}, \tag{3.2}$$

where the matrix element $(M_t)_{ij}$ measures how much the coordinate x_i at time t changes per unit change of the coordinate x_j at time 0. M_t is the Jacobian of the flow map φ_t, the matrix already met in the derivation of equation (2.5). Inserting equation (3.2) into the tangent equation (3.1) for every initial error $\boldsymbol{\delta}_0$ shows that M_t itself obeys

$$\dot{M}_t = J(\mathbf{x}(t))\,M_t, \qquad M_0 = \mathbb{1}, \tag{3.3}$$

so it is computed by integrating d^2 linear equations together with the trajectory.

The matrix M_t maps errors, and its effect on their length is the quantity of interest. The squared length of the error at time t is

$$|\boldsymbol{\delta}(t)|^2 = \boldsymbol{\delta}(t)^{\top}\boldsymbol{\delta}(t) = \boldsymbol{\delta}_0^{\top}\,\tilde{M}_t\,\boldsymbol{\delta}_0, \qquad \tilde{M}_t = M_t^{\top}M_t. \tag{3.4}$$

The matrix $\tilde{M}_t$ is symmetric and positive definite for this invertible finite-time flow, so it can be diagonalised by an orthonormal basis of eigenvectors,

$$\tilde{M}_t\,\mathbf{v}_i = \sigma_i(t)^2\,\mathbf{v}_i, \qquad \mathbf{v}_i^{\top}\mathbf{v}_j = \delta_{ij}, \qquad \sigma_1(t) \geq \sigma_2(t) \geq \cdots \geq \sigma_d(t) > 0. \tag{3.5}$$

The numbers $\sigma_i(t)$ are the singular values of M_t. Expanding the initial error in this basis, $\boldsymbol{\delta}_0 = \sum_i c_i \mathbf{v}_i$, and inserting it into equation (3.4) gives

$$|\boldsymbol{\delta}(t)|^2 = \sum_{i=1}^{d} c_i^2 \, \sigma_i(t)^2. \tag{3.6}$$

An initial error $\varepsilon \mathbf{v}_i$ along the i-th eigenvector grows to the length $\varepsilon\sigma_i(t)$, and a small sphere of initial errors of radius ε becomes an ellipsoid with semi-axes $\varepsilon\sigma_1(t), \dots, \varepsilon\sigma_d(t)$. For a generic initial error all coefficients c_i are nonzero, and once the σ_i have separated, the sum (3.6) is dominated by its first term, so $|\boldsymbol{\delta}(t)| \approx |c_1|\sigma_1(t)$. The growth of a typical error is set by the largest singular value.

On a chaotic attractor the singular values grow or decay exponentially, at rates that become well defined over long times.

Definition 3.1 (Lyapunov exponents)**.** The Lyapunov exponents of a trajectory are the asymptotic growth rates of the semi-axes,

$$\lambda_i = \lim_{t\to\infty} \frac{1}{t} \ln \sigma_i(t), \qquad i = 1, \dots, d. \tag{3.7}$$

The multiplicative ergodic theorem of Oseledets [10] guarantees these limits for almost every initial state with respect to an invariant probability measure, under integrability conditions on the tangent propagator and its inverse. If that measure is ergodic, the exponents are the same almost everywhere with respect to it. Different ergodic components can have different spectra. A positive λ_i means exponential growth of errors in the corresponding direction, a negative one exponential decay. A bounded, non-periodic motion with $\lambda_1 > 0$ is called chaotic. For a bounded trajectory of an autonomous flow that does not approach a fixed point, one exponent is zero, in the direction of motion. The tangent perturbation $\boldsymbol{\delta}_0 = \epsilon\, \mathbf{F}(\mathbf{x}(0))$ evolves as $\boldsymbol{\delta}(t) = \epsilon\, \mathbf{F}(\mathbf{x}(t))$, the first-order effect of a time shift. Its exponent is zero when the speed has no exponential growth or decay. Boundedness alone is insufficient: a trajectory approaching a stable equilibrium can have only negative exponents, as for $\dot{x} = -x$.

The sum of the exponents is fixed by the vector field alone. The determinant of M_t is the factor by which the flow changes small volumes, and the derivation of equation (2.5) showed that it obeys $\frac{d}{dt} \det M_t = (\nabla \cdot \mathbf{F})(\mathbf{x}(t)) \det M_t$ with $\det M_0 = 1$, a scalar linear equation with the solution

$$\det M_t = \exp\Big(\int_0^t (\nabla \cdot \mathbf{F})\big(\mathbf{x}(t')\big)\, dt' \Big). \tag{3.8}$$

The determinant is positive because the exponential is positive. Equivalently, a smooth finite-time flow has a nonsingular derivative and starts at the identity, so its orientation cannot change continuously. The same determinant follows from the singular values, since $\det \tilde{M}_t = (\det M_t)^2 = \prod_i \sigma_i(t)^2$, so $|\det M_t| = \sigma_1(t) \cdots \sigma_d(t)$. Taking the logarithm, dividing by t and letting $t \to \infty$ gives

$$\sum_{i=1}^{d} \lambda_i = \lim_{t\to\infty} \frac{1}{t} \ln |\det M_t| = \lim_{t\to\infty} \frac{1}{t} \int_0^t \nabla \cdot \mathbf{F}\big(\mathbf{x}(t')\big)\, dt' = \langle \nabla \cdot \mathbf{F} \rangle, \tag{3.9}$$

the time average of the divergence along the trajectory. The sum of the Lyapunov exponents is the average rate at which the flow changes phase-space volumes. For a Hamiltonian system the divergence vanishes by Liouville's theorem, and the exponents sum to zero. The symplectic structure of Hamilton's equations implies more, namely that the exponents come in pairs $\pm\lambda$, so that every expanding direction is matched by a contracting one. A negative sum means

that infinitesimal volumes shrink on average along the trajectory. For a bounded attracting motion this is compatible with growth of errors in individual directions. Volume contraction alone does not establish the existence of a bounded attractor.

The pairing in a Hamiltonian system follows from preservation of the symplectic form. In canonical coordinates write

$$\mathcal{S} = \begin{pmatrix} 0 & \mathbb{1}_n \\ -\mathbb{1}_n & 0 \end{pmatrix}, \qquad \dot{\mathbf{x}} = \mathcal{S}\nabla H, \qquad M_t^{\top}\mathcal{S}M_t = \mathcal{S}. \tag{3.10}$$

To obtain the last identity, differentiate its left-hand side, insert $\dot{M}_t = \mathcal{S}H''M_t$, and use the symmetry of the Hessian H''. The two terms cancel, so the expression keeps its initial value $\mathcal{S}$. With $P_t = M_t^{\top}M_t$, this also gives $P_t\mathcal{S}P_t = \mathcal{S}$. If $P_t\mathbf{v} = \sigma^2\mathbf{v}$, then

$$P_t(\mathcal{S}\mathbf{v}) = \sigma^{-2}(\mathcal{S}\mathbf{v}). \tag{3.11}$$

The singular values therefore occur as reciprocal pairs. Taking logarithms and dividing by time gives the exponent pairs $\lambda, -\lambda$ [11]. This restriction is stronger than a zero sum. Dissipative flows generally lack this pairing.

Computing the full spectrum requires preserving the contracting perturbations as well as the expanding ones. If several tangent vectors are propagated for a long time without adjustment, their components along the fastest expanding direction dominate. Numerical round-off then makes them nearly parallel, and information about the other directions is lost. We prevent that loss by repeatedly orthonormalising the vectors [11].

Let Q_k contain an orthonormal set of perturbation vectors at time t_k. Integrate the tangent equation over a short interval δt to obtain $A_k = D\varphi_{\delta t}(\mathbf{x}(t_k))Q_k$. Its QR factorisation is

$$A_k = Q_{k+1}R_k, \qquad Q_{k+1}^{\top}Q_{k+1} = \mathbb{1}, \qquad \widehat{\lambda}_i = \frac{1}{K\delta t}\sum_{k=0}^{K-1} \ln|(R_k)_{ii}|. \tag{3.12}$$

The upper-triangular matrix R_k records the growth factors removed when constructing the new orthonormal basis. We continue integrating from Q_{k+1} and accumulate their logarithms. After a transient in both the trajectory and tangent basis, long averages give the exponents under the same ergodic assumptions as equation (3.7). The signals notebook uses this procedure for Lorenz and for a discretised Mackey–Glass history. A finite-time QR estimate can differ from the finite-time singular-value rates of the whole propagator, even though their long-time limits agree under the usual hypotheses.

The Lorenz spectrum in Table 1 has one positive exponent, a middle exponent consistent with zero, and one strongly negative exponent. Its sum agrees with the constant divergence.

Expansion persists because the flow also brings elongated pieces back into a bounded region. Around the Lorenz rolls, nearby trajectories separate and turn before entering the other lobe. This repeated stretching and folding creates fine layers while strong transverse contraction reduces their thickness. Flow uniqueness still prevents trajectories from crossing in the full state space. Overlaps in a plotted projection arise because a coordinate has been suppressed. The positive exponent, bounded motion and layered geometry describe different parts of this mechanism.

3.2 The prediction horizon

An initial error of size δ_0 along a generic direction grows approximately as $\delta_0 e^{\lambda_1 t}$, because the component along the most expanding direction soon dominates. A forecast remains useful

Table 1: The Lorenz system at $\sigma = 10$, $\beta = 8/3$, $\rho = 28$. Times are in the time units of equations (2.28).

quantity	value
eigenvalues at the origin	-22.83, -2.67, 11.83 (saddle)
eigenvalues at $C_\pm$	-13.85, $0.094 \pm 10.195\,i$ (unstable spirals)
heating at which $C_\pm$ lose stability, ρ_c	24.74
Lyapunov exponents $\lambda_1, \lambda_2, \lambda_3$	0.907, 0.0001, -14.573
$\lambda_1 + \lambda_2 + \lambda_3$ and divergence $-(\sigma + 1 + \beta)$	-13.667 and -13.667
Lyapunov time $T_\lambda = 1/\lambda_1$	1.103 (55.141 samples at $\Delta t = 0.02$)
predicted horizon for $\delta_0 = 10^{-8}$, $\Delta = 1$	20.3
measured horizon, mean of $\ln \lvert\boldsymbol{\delta}\rvert$ over 200 pairs	20.4
Kaplan–Yorke dimension D_{KY}	2.06

until the error reaches a tolerance Δ, which happens at

$$T_{\mathrm{pred}} = \frac{1}{\lambda_1} \ln \frac{\Delta}{\delta_0}. \tag{3.13}$$

The time $T_\lambda = 1/\lambda_1$ is the Lyapunov time, over which errors grow by the factor e. The twin experiment of Figure 9 tests equation (3.13) on the Lorenz system. Two trajectories start $\delta_0 = 10^{-8}$ apart, a precision far beyond any experiment, and the forecast is declared lost when they are a distance $\Delta = 1$ apart, small compared with the size of the attractor. The equation predicts $\ln(10^8)\,T_\lambda$, about twenty time units, and the average over many pairs agrees with this prediction (Table 1). On the logarithmic scale of the right panel the distance grows along a straight line of slope λ_1 with fluctuations, because the local stretching rate varies over the attractor and only its long-time average equals λ_1. The growth stops when the distance reaches the size of the attractor, where the linearisation no longer applies and the two trajectories are as far apart as two unrelated states.

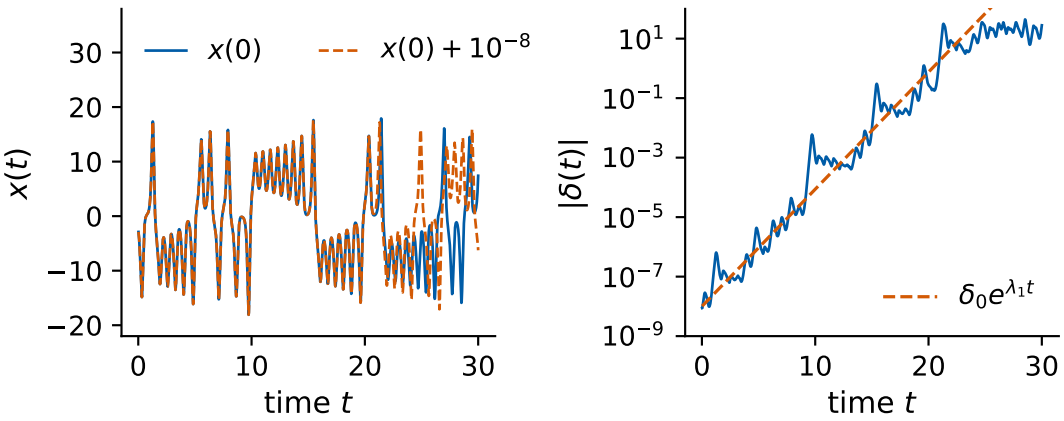


Figure 9: Lorenz trajectories from two initial states 10^{-8} apart. Left, $x(t)$ of both. Right, their distance $|\boldsymbol{\delta}(t)|$ on a logarithmic scale with the line $\delta_0 e^{\lambda_1 t}$.

The logarithm in equation (3.13) has a practical consequence. Reducing the initial error by a factor of 1000, which in an experiment may require a new instrument, extends the horizon by only $\ln 1000\,T_\lambda \approx 6.9\,T_\lambda$. For any forecasting method, the natural unit of the horizon is therefore the Lyapunov time, and the quality of a forecast must be stated as a function of the horizon. Equation (3.13) applies to forecasts of individual trajectories. It does not bound the prediction of averages over the attractor, such as the mean and variance of x, whose long-time

prediction depends on the stability of the invariant statistics and their sampling uncertainty, and fresh measurements reset the error.

3.3 Attractor dimension and the Kaplan–Yorke estimate

The largest Lyapunov exponent measures the rate at which nearby predictions separate. Reconstructing a state from observations also depends on the geometry of the set of possible states. A periodic orbit lies on a curve even when its equations use many coordinates. A chaotic attractor can occupy a set of zero volume with structure at progressively smaller scales. To relate that geometry to a forecasting window, we need a definition of dimension that also applies to such sets.

Cover a bounded set with boxes of side ε and let N_ε be the smallest number required. A smooth curve of finite nonzero length needs a number proportional to ε^{-1} at small scales, while a smooth surface needs a number proportional to ε^{-2}. The box-counting dimension extracts this exponent,

$$D_B = \lim_{\varepsilon \to 0} \frac{\ln N_\varepsilon}{\ln(1/\varepsilon)}, \tag{3.14}$$

when the limit exists. If the ratio has no limit, its lower and upper limits define two box dimensions. Coordinates are measured in fixed reference units, so the logarithm compares dimensionless scales.

Lyapunov exponents offer a different route to a dimension estimate. Order them as $\lambda_1 \geq \cdots \geq \lambda_d$ and write

$$S_0 = 0, \qquad S_j = \sum_{i=1}^{j} \lambda_i. \tag{3.15}$$

The product of the j largest singular values of the tangent map is the maximal expansion factor of an infinitesimal j-dimensional volume. Its asymptotic logarithmic growth rate is S_j. In the Lorenz flow, lengths and two-dimensional areas can expand while three-dimensional volumes contract. This suggests placing an effective dimension between the last non-contracting volume and the next contracting one.

For a noninteger value $D = j + s$ with $0 \leq s \leq 1$, we interpolate the logarithmic growth rates by assigning the factor $\sigma_1 \cdots \sigma_j \sigma_{j+1}^s$ to the tangent map. Its asymptotic rate is

$$S_j + s\lambda_{j+1}. \tag{3.16}$$

Its zero occurs at the Kaplan–Yorke estimate [12],

$$j = \max\{k \in \{0, \ldots, d\} : S_k \geq 0\}, \qquad D_{\mathrm{KY}} = j + \frac{S_j}{|\lambda_{j+1}|} \quad (j < d). \tag{3.17}$$

When $j = d$, the convention is $D_{\mathrm{KY}} = d$. When all exponents are negative, $j = 0$ and the result is zero. A stable limit cycle has one zero exponent along its motion and negative transverse exponents, giving $D_{\mathrm{KY}} = 1$. The interpolation defines a Lyapunov dimension. Identifying it with a dimension obtained by covering or measuring an attractor requires additional dynamical assumptions.

For the Lorenz benchmark, the spectrum in Table 1 gives $S_1 > 0$, $S_2 = 0.907 > 0$ and $S_3 = -13.667 < 0$. Thus

$$D_{\mathrm{KY}} = 2 + \frac{\lambda_1 + \lambda_2}{|\lambda_3|} = 2.06. \tag{3.18}$$

The contracting exponent has a magnitude about 16.072 times the largest exponent, leaving a dimension only slightly above two. This is consistent with the thin layered structure visible in the attractor.

The Mackey–Glass signals show that dimension and the rate of error growth are separate properties. Their state is a history, a function on an interval of length τ_{MG}, so the phase space is infinite-dimensional. Discretising the history and applying the same QR procedure gives $D_{\mathrm{KY}} \approx 3.158$ for $\tau_{\mathrm{MG}} = 25$ and 3.406 for $\tau_{\mathrm{MG}} = 35$, while the periodic signal at $\tau_{\mathrm{MG}} = 15$ has $D_{\mathrm{KY}} = 1$. The motion of an infinite-dimensional system thus settles on an attractor of a few dimensions, and the longer delay has the larger dimension although its largest exponent is smaller. The dimension decides how many delay coordinates a reconstruction needs, Section 4, while the largest exponent sets the prediction horizon (3.13).

Estimated from a scalar record, the Kaplan–Yorke dimension is uncertain, because neighbourhood fits resolve the expansion along the attractor much better than the contraction across it. The reconstruction theorem of the next section is stated with the box dimension (3.14), and D_{KY} serves as an estimate of it.

4 Reconstructing the state from one measured variable

4.1 One value does not determine the future

Section 3 assumed that the state is known. The laser record provides one number per sample, and one coordinate does not determine the state. In the Lorenz record we find $x = 4.62$ at sample 266, after which x increases by 0.63 in one step, and $x = 4.64$ at sample 325, after which it decreases by 1.24 (Figure 10). Two nearly equal values have different futures because the hidden coordinates y and z differ. No function of the current value alone can therefore forecast the next value. The same ambiguity explains why the rapid decay of the autocorrelation of the laser record, noted in Section 2.6, says little about the predictability of a deterministic record.

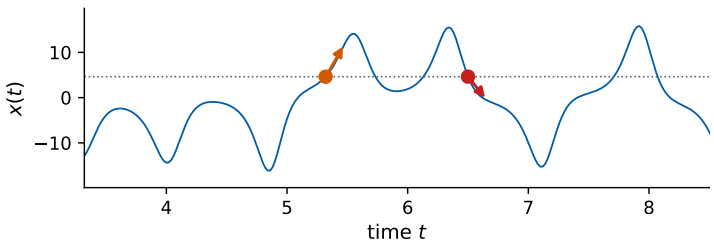


Figure 10: A section of the Lorenz coordinate $x(t)$. At the two marked times x has nearly the same value, once on a rising and once on a falling flank.

Past values carry the missing information. Through the dynamics, the value of x a short time ago depends on y and z now, and a sequence of past values constrains the hidden coordinates. For a harmonic oscillator this is elementary, since the velocity, which a single position does not reveal, follows from two successive positions as $v_k \approx (x_k - x_{k-1})/\Delta t$. The question is whether a finite window of past measurements determines the full state in general.

4.2 Delay vectors and the embedding theorem

The aim is to find a number m of past values and a spacing κ between them such that the window of these values fixes the present state $\mathbf{x}_k$, and with it every future value of the record. Figure 11 shows the chain of objects involved. The Lorenz state has three coordinates, but the measured record contains only one of them. One measurement per step supplies only one number. The past values supply the missing ones, because the dynamics has written the hidden coordinates into the later values of the measured one. In the Lorenz system the value of y now determines how fast x changes, so the course of x over the preceding samples carries information about y and z.

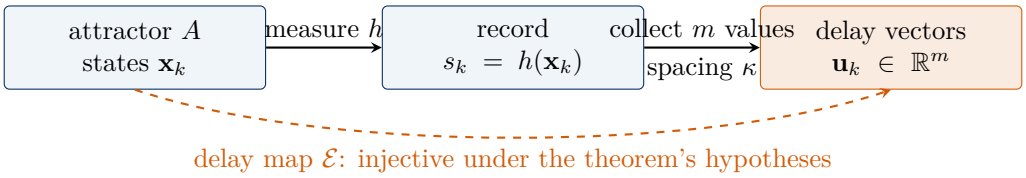


Figure 11: From the hidden state to the delay vector. The measurement h turns each state into one stored number, and collecting m stored numbers spaced by κ samples gives the delay vector. The composite map from the state to the delay vector is the delay map $\mathcal{E}$ of equation (4.6).

The construction collects several past values of the record into one vector.

Definition 4.1 (Delay vector)**.** For a scalar record s_k, an integer delay $\kappa \geq 1$ and a number of coordinates m, the delay vector at time k is

$$\mathbf{u}_k = (s_k,\, s_{k-\kappa},\, \ldots,\, s_{k-(m-1)\kappa}) \in \mathbb{R}^m. \tag{4.1}$$

Each entry of the delay vector is a measurement of an earlier state, and every earlier state is determined by the present one through the flow map. For noise-free data, the measurement equation (1.2) and the properties (2.3) of the flow map give for the j-th entry

$$s_{k-j\kappa} = h(\mathbf{x}_{k-j\kappa}) = h\big(\varphi_{-j\kappa\Delta t}(\mathbf{x}_k)\big), \qquad j = 0, 1, \ldots, m-1, \tag{4.2}$$

since the state $j\kappa$ samples earlier is obtained from the present state by evolving backwards over the time $j\kappa\Delta t$. All entries are therefore functions of the single present state $\mathbf{x}_k$, and the delay vector is the value of one map of phase space into $\mathbb{R}^m$.

The delay vector and the state are different objects. For the Lorenz system with the measured coordinate x and $m = 3$,

$$\mathbf{x}_k = (x(k\Delta t),\, y(k\Delta t),\, z(k\Delta t)), \qquad \mathbf{u}_k = (x(k\Delta t),\, x((k-\kappa)\Delta t),\, x((k-2\kappa)\Delta t)). \tag{4.3}$$

The state contains three different quantities at one time, the delay vector one quantity at three times. They are not equal, but they can carry the same information. For the harmonic oscillator, whose state is the position and the velocity, the velocity is approximately recovered from two successive positions,

$$v_k \approx \frac{x_k - x_{k-1}}{\Delta t}. \tag{4.4}$$

For a harmonic oscillator of angular frequency Ω_0, the exact relation at the delay $\kappa\Delta t$ follows from its sinusoidal solution,

$$x(t - \kappa\Delta t) = x(t)\cos(\Omega_0\kappa\Delta t) - \frac{v(t)}{\Omega_0}\sin(\Omega_0\kappa\Delta t). \tag{4.5}$$

Solving for $v(t)$ shows that the two positions determine the state when $\sin(\Omega_0 \kappa \Delta t) \neq 0$. Delays equal to an integer number of half-periods lose the velocity information. A very small delay retains it in principle but amplifies measurement noise when the position difference is divided by $\kappa \Delta t$. Figure 12 shows the same for both systems. The closed orbit of the oscillator and the two lobes of the Lorenz attractor are recognisable in the delay coordinates. Visual similarity motivates a reconstruction, whose mathematical conditions follow below.

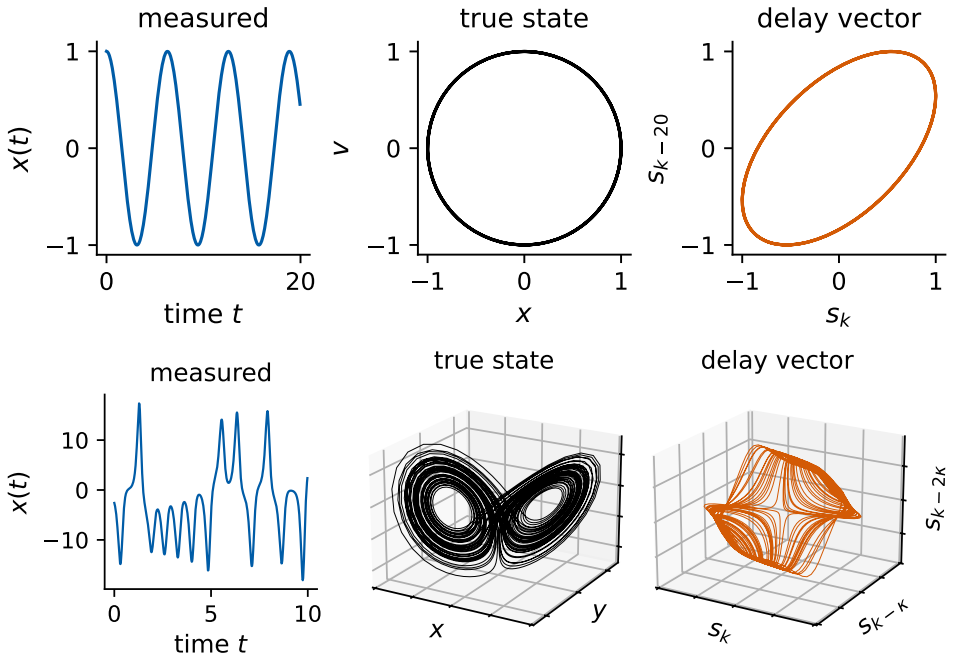


Figure 12: Phase space from one coordinate. Top, the harmonic oscillator. Bottom, the Lorenz system. Left, the measured coordinate $x(t)$. Middle, the true state, (x, v) and (x, y, z). Right, the delay vectors built from x alone, with the delays marked on the axes, which reproduce the closed orbit and the two lobes in deformed coordinates.

Definition 4.2 (Delay map)**.** Let $A \subset \mathbb{R}^d$ be the attractor on which the states lie. The delay map $\mathcal{E} : A \to \mathbb{R}^m$ assigns to every state the delay vector that a noise-free record would produce,

$$\mathcal{E}(\mathbf{x}) = \Big(h(\mathbf{x}),\, h(\varphi_{-\kappa\Delta t}(\mathbf{x})),\, \ldots,\, h(\varphi_{-(m-1)\kappa\Delta t}(\mathbf{x}))\Big), \qquad \mathbf{u}_k = \mathcal{E}(\mathbf{x}_k). \tag{4.6}$$

Its image $\mathcal{E}(A) \subset \mathbb{R}^m$ is the reconstructed attractor, the cloud of delay vectors.

The delay map involves the unknown measurement function and the unknown flow, and it is never evaluated. The data supply its values $\mathbf{u}_k$, and the mathematical question concerns its properties. The delay vectors can replace the state when $\mathcal{E}$ loses no information about it. A topological embedding is one-to-one and continuous with continuous inverse on its image. In particular, different states give different delay vectors,

$$\mathbf{x} \neq \mathbf{x}' \quad \Longrightarrow \quad \mathcal{E}(\mathbf{x}) \neq \mathcal{E}(\mathbf{x}'), \qquad \mathbf{x}, \mathbf{x}' \in A, \tag{4.7}$$

so that the inverse map $\mathcal{E}^{-1} : \mathcal{E}(A) \to A$ exists. Since A is compact and $\mathcal{E}$ is continuous, this inverse is continuous: nearby delay vectors correspond to nearby states. For a smooth manifold, a smooth embedding additionally preserves the local differential structure and has a smooth inverse onto its image. Injectivity on a fractal set alone does not provide this stronger property. In either case the state is a function of the delay vector,

$$\mathbf{x}_k = \mathcal{E}^{-1}(\mathbf{u}_k). \tag{4.8}$$

A single measured value violates condition (4.7), as the two nearly equal values of x with different futures in Figure 10 show. The question is how many coordinates are needed for it to hold.

The sufficient number of coordinates depends on which geometric description is available. A compact smooth state manifold has an integer dimension d, while an attractor inside it can have a smaller box dimension. Equation (3.14) defines that box dimension. A smooth closed curve has dimension one, and a fractal attracting set can have a noninteger value. Integer dimension alone excludes no chaotic dynamics.

Takens' delay embedding theorem. *Let the dynamics evolve on a compact smooth manifold of dimension d. For generic smooth vector fields and generic smooth measurement functions, the delay map (4.6) with $m \geq 2d+1$ coordinates is an embedding [14]. For an attractor A of box-counting dimension D_B, which need not be a manifold, almost every smooth measurement function gives a delay map that is one-to-one on A if $m > 2D_B$, under additional conditions on periodic orbits [15].*

For clarity, a sufficient set of periodic-orbit assumptions can be stated for the sampled backward map $f = \varphi_{-\kappa\Delta t}$. Assume it is a smooth invertible map with smooth inverse on a neighbourhood of the compact set A. For each integer $1 \leq p \leq m$, let $A_p = \{\mathbf{x} \in A : f^p(\mathbf{x}) = \mathbf{x}\}$. Require $\dim_B A_p < p/2$ and distinct eigenvalues of Df^p at these points. These are the hypotheses of the fractal delay theorem used here [15]. Upper box dimension is used if the limit defining D_B does not exist. The conclusion holds for a prevalent set of observations, meaning that almost every coefficient choice in suitable finite-dimensional smooth perturbations of any observation succeeds. It supplies no guarantee for a specified instrument without checking its structure.

The word generic excludes exceptional cases that can be destroyed by an arbitrarily small change, for example a constant measurement function, which cannot distinguish any states, or a delay that equals the period of an orbit, which makes all coordinates equal on that orbit. The bound $m > 2D_B$ is sufficient, and its origin is geometric. The dimension count for pairs of points suggests that intersections may persist when $2D \geq m$, whereas enough independent coordinates can separate them when $m > 2D$. This is a motivation for the theorem, and its proof must also control periodic orbits and the restrictions of delayed observations. A closed curve ($D = 1$) drawn in the plane ($m = 2$) can cross itself, while in three coordinates a small perturbation removes any crossing. Figure 13 shows this for the periodic signal

$$s(t) = \cos t + 0.6\cos(2t + 0.7), \tag{4.9}$$

whose phase runs around a closed curve with $D_B = 1$. In two delay coordinates the curve crosses itself, and at the crossing two different states, marked in orange, have the same delay vector. There $\mathcal{E}^{-1}$ does not exist, and the delay vector does not determine whether the signal will continue along one branch or the other. In three coordinates, $m = 3 > 2D_B$, the two branches pass each other at different heights, every delay vector belongs to one state, and the future is unique.

The box-counting dimension is difficult to estimate reliably from a finite record. The Kaplan–Yorke dimension of Section 3 offers a heuristic guide to complexity, but substituting it for D_B does not preserve the theorem's guarantee. The diagnostics of Section 4.3 guide a practical choice of m.

When $\mathcal{E}$ is an embedding, the reconstructed attractor $\mathcal{E}(A)$ is a faithful copy of A. Distances and angles are distorted, but the topology is preserved. The two lobes of the Lorenz attractor remain two lobes, holes remain holes, and neighbouring states remain neighbouring delay

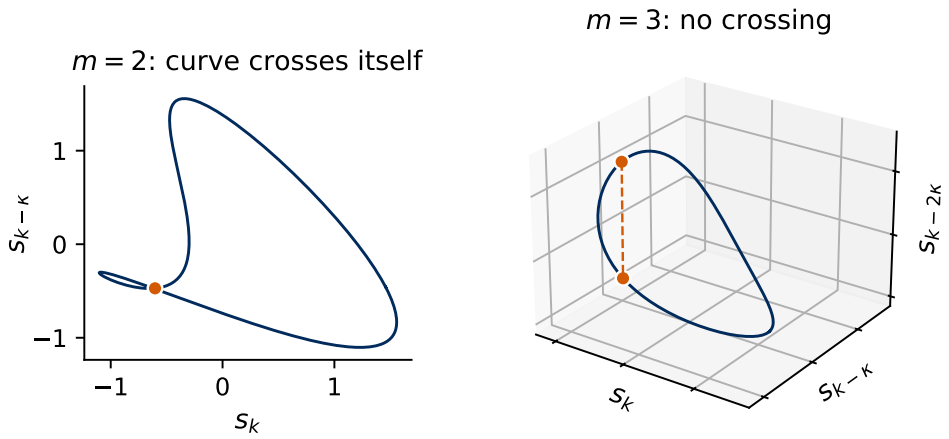


Figure 13: The periodic signal (4.9) in delay coordinates. Left, $m = 2$, where the closed curve crosses itself and two different states (orange) share one delay vector. Right, $m = 3$, where the crossing is removed and the two states have different delay vectors.

vectors. The dynamics itself carries over to the copy. Starting from a delay vector, the inverse map gives the state, the flow map advances it by one sample, and the delay map returns to delay coordinates,

$$\mathbf{u}_{k+1} = \mathcal{E}(\mathbf{x}_{k+1}) = \mathcal{E}(\varphi_{\Delta t}(\mathbf{x}_k)) = (\mathcal{E} \circ \varphi_{\Delta t} \circ \mathcal{E}^{-1})(\mathbf{u}_k). \tag{4.10}$$

The map $\mathcal{E} \circ \varphi_{\Delta t} \circ \mathcal{E}^{-1}$ is the same dynamics written in other coordinates. Lyapunov exponents are preserved by a smooth coordinate change with bounded derivatives and inverse derivatives along the motion. The injective fractal reconstruction theorem alone does not guarantee such a change in every direction. In particular, a record confined to an attractor need not reveal contraction transverse to it: motion on a stable limit cycle does not determine its radial contraction rate. Full physical spectra and D_{KY} should therefore not be assumed recoverable from every delay reconstruction.

The theorem yields the result on which all forecasting methods of this chapter rest. The future measurement can be written as a function of the present delay vector, by following the chain of maps in Figure 11 backwards to the state, forwards in time and back to the record. The measurement τ samples ahead is the measurement function applied to the future state, the future state is the present state advanced by the flow map, and the present state is the inverse delay map applied to the present delay vector,

$$s_{k+\tau} = h(\mathbf{x}_{k+\tau}) = h(\varphi_{\tau\Delta t}(\mathbf{x}_k)) = h\Big(\varphi_{\tau\Delta t}(\mathcal{E}^{-1}(\mathbf{u}_k))\Big) = G_\tau(\mathbf{u}_k), \qquad G_\tau = h \circ \varphi_{\tau\Delta t} \circ \mathcal{E}^{-1}, \tag{4.11}$$

up to measurement noise. Figure 18 shows the three steps of this chain. The forecast map G_τ exists and depends only on a finite window of the record; it is smooth when the reconstruction and its inverse are smooth. It is unknown, because h, φ and $\mathcal{E}$ are unknown, but it can be estimated from the pairs $(\mathbf{u}_k, s_{k+\tau})$ that the record itself provides. Forecasting a deterministic system from a scalar record is therefore a regression problem with a finite-dimensional input. The theorem does not supply the form of G_τ, which is nonlinear even for simple systems, and it does not address noise, which limits how finely nearby delay vectors can be distinguished.

4.3 Choosing the delay and the number of coordinates

Under its genericity and periodic-orbit hypotheses, the theorem gives a reconstruction with enough coordinates, but it says nothing about which delay works best with a finite and noisy

record, and its bound on the number of coordinates is often much larger than needed. Both numbers are therefore chosen from the record itself, the delay κ first and the number of coordinates m second.

A very short delay makes successive coordinates nearly equal, so the delay vectors lie close to the diagonal and noise dominates their differences. For dynamics whose correlations decay with time, a long delay can make successive coordinates weakly dependent and the reconstructed layers difficult to resolve in a finite sample. The delay should make the two coordinates as different as possible while keeping them related. The mutual information (MI) measures how much knowing one value tells about another,

$$\mathrm{MI}(\kappa) = \sum_{i,j} p_{ij}(\kappa) \ln \frac{p_{ij}(\kappa)}{p_i^{(1)}\, p_j^{(2)}}, \tag{4.12}$$

where the range of the record is divided into bins and the probabilities are estimated from a histogram of the pairs $(s_{k-\kappa}, s_k)$ of the record. If $n_{ij}(\kappa)$ of the $N-\kappa$ pairs fall in bin i for the earlier and bin j for the later value, then

$$p_{ij}(\kappa) = \frac{n_{ij}(\kappa)}{N-\kappa}, \qquad p_i^{(1)} = \sum_j p_{ij}(\kappa), \qquad p_j^{(2)} = \sum_i p_{ij}(\kappa), \tag{4.13}$$

where the marginal probabilities $p_i^{(1)}$ and $p_j^{(2)}$ are the fractions of earlier values in bin i and of later values in bin j. The MI is measured in nats when the natural logarithm is used. It vanishes when the two values are independent, since then $p_{ij} = p_i^{(1)} p_j^{(2)}$ and every logarithm is zero, and, unlike the autocorrelation, it detects nonlinear dependence.

The information interpretation follows by defining the binned entropy $H(U) = -\sum_i p_i^{(1)} \ln p_i^{(1)}$ for the earlier measurement U, and similarly $H(V)$ and $H(U,V)$ for the later measurement and the joint histogram. Expanding the logarithm in equation (4.12) gives

$$\mathrm{MI}(U,V) = H(U) + H(V) - H(U,V) = H(V) - H(V \mid U). \tag{4.14}$$

Here $H(V \mid U) = -\sum_{i,j} p_{ij} \ln(p_{ij}/p_i)$ is the entropy of the conditional distribution, averaged over the earlier measurement. Thus MI measures the reduction in uncertainty about V after observing U. For the true bin probabilities, Gibbs' inequality gives nonnegativity. To see this, put $q_{ij} = p_i p_j$ and apply $-\ln z \geq 1 - z$ to $z = q_{ij}/p_{ij}$ on cells with $p_{ij} > 0$. Summing gives $\sum p_{ij} \ln(p_{ij}/q_{ij}) \geq 0$, with equality at independence. Empty cells contribute zero by continuity. A finite histogram is a different object from the population distribution. Sampling fluctuations generally give a positive MI even for independent data, and changing the bin width changes both bias and resolution. The first minimum is consequently a practical choice to check for stability with record length and binning [16].

The procedure for the delay [17] uses 16 bins of equal width. The estimator above uses all $N-\kappa$ pairs at each lag. The benchmark diagnostics instead fix a maximum lag $\kappa_{\max} = 100$ and use $N - \kappa_{\max}$ pairs at every lag, pairing s_i with $s_{i+\kappa}$ for $i = 0, \ldots, N - \kappa_{\max} - 1$. Their joint histogram gives $p_{ij}(\kappa)$, normalised by this fixed pair count, and both marginal probabilities are recomputed from that histogram at each lag. Equation (4.12) then gives one value $\mathrm{MI}(\kappa)$ for each delay, and the curve $\mathrm{MI}(\kappa)$ is plotted against κ. The delay is taken at the first local minimum of this curve. This choice is a heuristic balance between redundant and weakly related coordinates. It does not in general maximise the information useful for forecasting. When the curve decreases without a clear minimum, a common alternative is the delay at which it has fallen to a fixed fraction, such as $1/e$, of its value at $\kappa = 0$.

The estimate of MI(κ) can be followed on the Lorenz coordinate in Figure 14. For a chosen delay, every pair $(s_{k-\kappa}, s_k)$ of the record is a point in the plane, and the grid of 16×16 cells divides the plane into the bins (i, j). Counting the points in each cell and dividing by the number of pairs gives $p_{ij}(\kappa)$, shown as a shaded image below the points. At $\kappa = 1$ the points lie on the diagonal, since x hardly changes within one sample, so knowing s_{k-1} almost fixes s_k and the mutual information is large. A second coordinate of this kind repeats the first. At $\kappa = 10$ the points spread over a curved band. The two values still depend on each other, so the reconstruction stays smooth, while each adds information that the other lacks. At $\kappa = 60$ the histogram has spread further and the mutual information is small, although the two values are not yet independent. The whole curve MI(κ), at the bottom of the figure, falls steeply over the first few samples and has its first minimum at $\kappa = 10$.

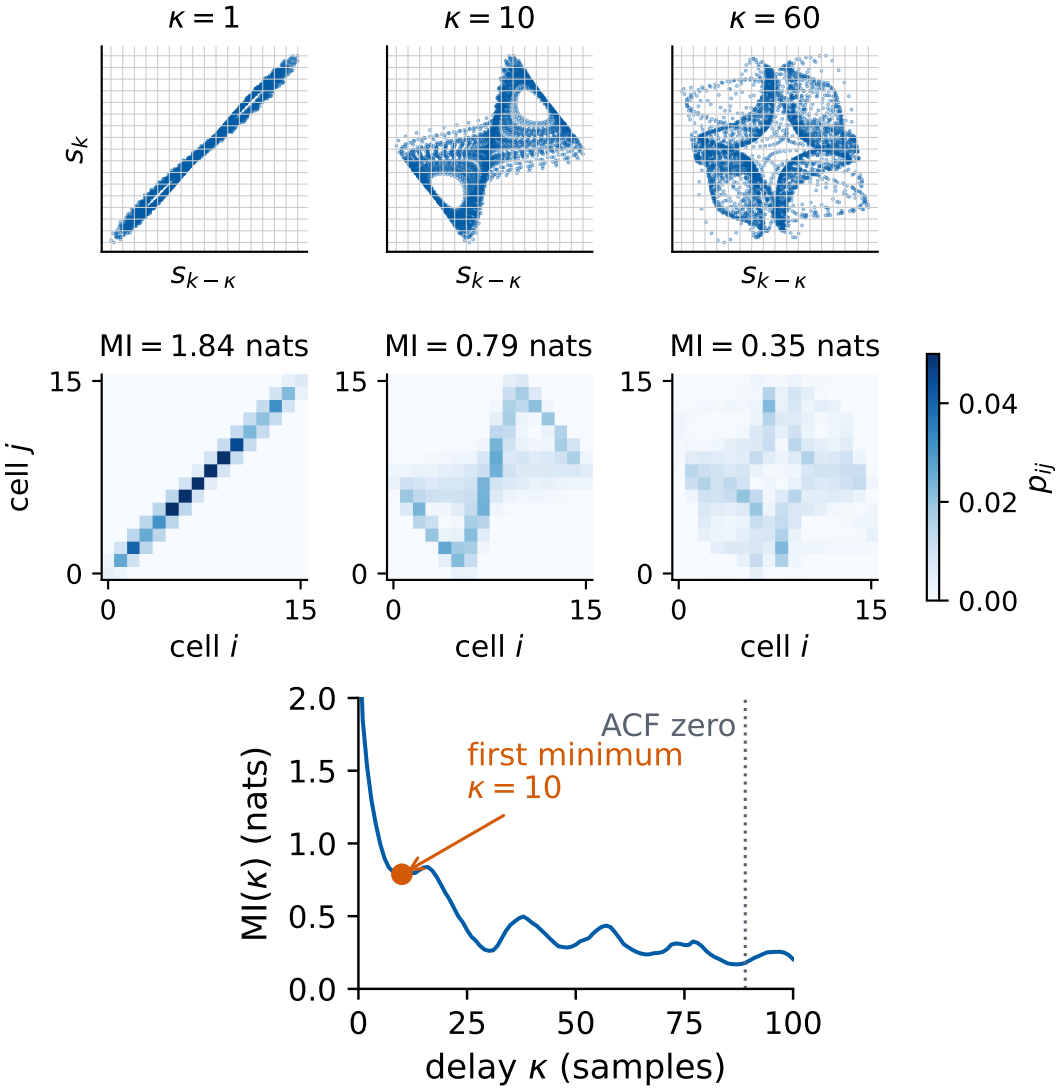


Figure 14: Mutual information of the Lorenz coordinate $s_k = x(k\Delta t)$, $\Delta t = 0.02$. Top, upper row, the pairs $(s_{k-\kappa}, s_k)$ for $\kappa = 1$, 10 and 60 with the grid of 16×16 cells, and lower row, the fraction p_{ij} of pairs in each cell, with the value of MI(κ). Bottom, MI(κ) against the delay, with its first minimum at $\kappa = 10$ and the first zero of the autocorrelation at $\kappa = 89$ (dotted).

Figure 15 shows what the choice of the delay does to the reconstructed picture. With $\kappa = 10$ the delay vectors $(s_k, s_{k-\kappa}, s_{k-2\kappa})$ form two lobes joined by the switching orbits, a smoothly deformed image of the true attractor. With $\kappa = 89$, the first zero of the autocorrelation, the reconstructed layers become difficult to distinguish at the resolution of the finite record. A

vanishing autocorrelation removes linear correlation and can leave nonlinear dependence. The autocorrelation measures only the linear dependence, and its first zero comes far too late for this signal.

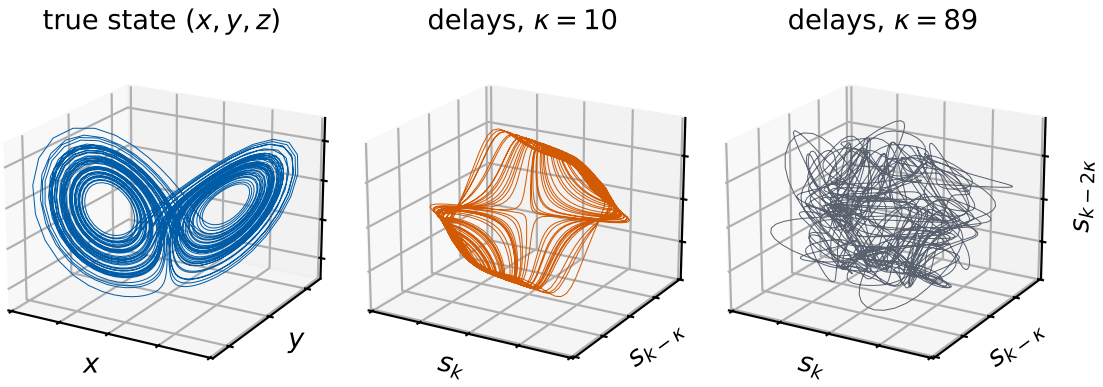


Figure 15: Lorenz system. Left, the true state (x, y, z). Middle, delay vectors $(s_k, s_{k-\kappa}, s_{k-2\kappa})$ of the measured $s_k = x(k\Delta t)$ with $\kappa = 10$, the first minimum of MI(κ). Right, the same with $\kappa = 89$, the first zero of the autocorrelation.

The number of coordinates is chosen next, with κ fixed. For an attractor with known box-counting dimension D_B, the sufficient threshold under the theorem's hypotheses is

$$m > 2D_B, \qquad \text{equivalently} \qquad m \geq \lfloor 2D_B \rfloor + 1. \tag{4.15}$$

The condition is sufficient for generic measurements, and a particular measurement may need fewer coordinates. For a smooth D-dimensional manifold a smooth embedding requires at least D coordinates, and its topology can require more. Neither this dimension obstruction nor the sufficient bound should be turned into a rigorous bracket using the estimated D_{KY}. For Lorenz, $D_{\mathrm{KY}} \simeq 2.06$ suggests testing a few coordinates; it does not by itself prove that five delay coordinates embed the measured attractor.

The sufficient condition (4.15) does not settle the practical value of m, and for a measured record D_{KY} itself must be estimated from delay vectors, which already require a choice of m. The method of false nearest neighbours [18] avoids this circle and looks for the self-intersections directly. Points that are close in m coordinates because they are close on the attractor stay close when a further coordinate is added. Points that are close only because the reconstruction has folded two distant parts of the attractor onto each other are separated by the new coordinate. Figure 16 shows such a false pair for the Lorenz coordinate. In two delay coordinates the orbit passes through the same region from two different lobes, and two points of different lobes lie close together. The third coordinate separates them by nearly a hundred times their distance in the plane. The two points are different states with different futures, so a forecast that relies on the neighbour would be wrong.

The procedure runs over $m = 1, 2, 3, \ldots$ with κ fixed by the mutual information and repeats the same steps for each m.

1. All delay vectors with m coordinates are formed,

$$\mathbf{u}_k^{(m)} = \big(s_k,\, s_{k-\kappa},\, \ldots,\, s_{k-(m-1)\kappa}\big). \tag{4.16}$$

2. For every k the nearest neighbour $\mathbf{u}_j^{(m)}$ is found, the delay vector at the smallest distance

$$r_k = \|\mathbf{u}_k^{(m)} - \mathbf{u}_j^{(m)}\| = \Big(\sum_{i=0}^{m-1} \big(s_{k-i\kappa} - s_{j-i\kappa}\big)^2 \Big)^{1/2}, \tag{4.17}$$

searched only among $|k-j|>50$, so that j is not simply a neighbouring sample on the same pass of the trajectory.

3. One coordinate is added, $m \to m+1$. The pair acquires the entries $s_{k-m\kappa}$ and $s_{j-m\kappa}$ and with them the new difference

$$\Delta_k = \left|s_{k-m\kappa} - s_{j-m\kappa}\right|. \tag{4.18}$$

4. The neighbour is false by the ratio test if the new coordinate pulls the pair apart far beyond its distance,

$$\Delta_k > 15\, r_k. \tag{4.19}$$

5. The fraction $\mathrm{FNN}(m)$ of indices k with a false neighbour is computed, and m is increased until it falls below 5%. The first such m is an empirical embedding-dimension estimate at this threshold; it does not prove injectivity.

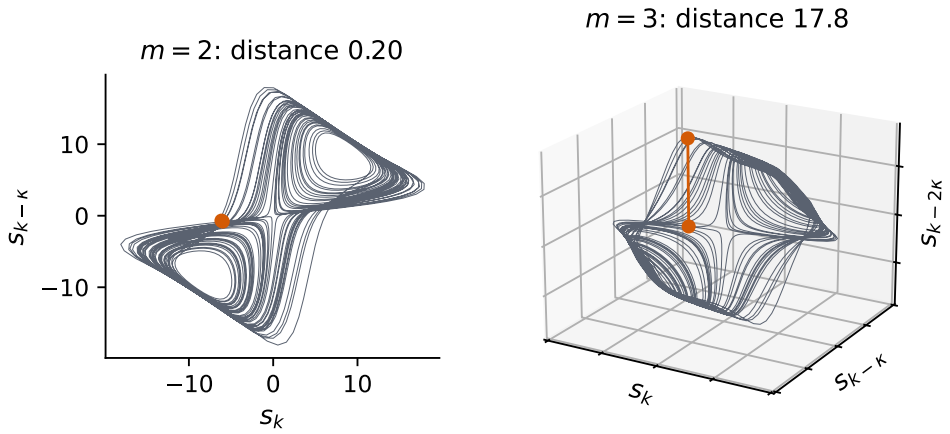


Figure 16: A false nearest neighbour of the Lorenz coordinate. Left, two points (orange) that are close in the plane $(s_k, s_{k-\kappa})$. Right, the same two points in three delay coordinates, where they belong to different lobes of the attractor.

For noise the ratio test alone is misleading, since in high dimension all nearest-neighbour distances become comparable and the ratio becomes small. An absolute test,

$$\sqrt{r_k^2 + \Delta_k^2} > 2\sigma_s, \tag{4.20}$$

with σ_s the standard deviation of the record, declares false the pairs that are not close at all. The combined fraction, in which a pair counts as false if either test fails, then distinguishes the two kinds of record by its level and its slope in m. For the Lorenz benchmark it settles at a few percent. Points in sparsely visited regions have distant nearest neighbours, so a finite deterministic record can retain a nonzero combined fraction. The independent Gaussian control instead has a high fraction that grows over the tested dimensions.

Figure 17 applies both criteria to the benchmark signals, and Table 2 lists the delays and dimensions they select. Together they suggest a window span (4.21) to test in forecasting. The results are read against what is known about each system. The laser ratio criterion first falls below its threshold at three coordinates. These are not the three Maxwell–Bloch variables, since the intensity cannot distinguish the sign-related states of Section 1.1, and a delay vector can only identify the pair of states with the same future intensity. The Lorenz coordinate passes the criterion at four coordinates, a diagnostic for this record that sets no

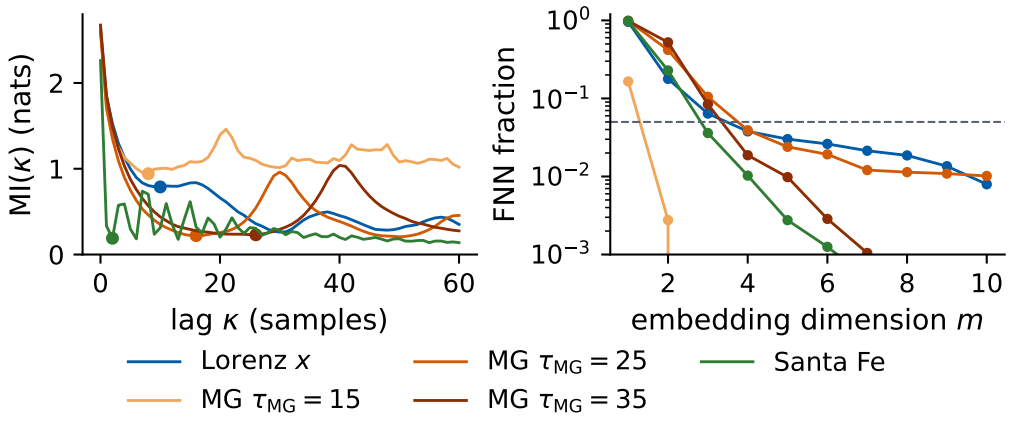


Figure 17: Left, mutual information MI(κ) of the five benchmark signals as a function of the delay. Right, fraction of false nearest neighbours by the ratio test alone as a function of the number of coordinates, with the 5% threshold dashed.

lower bound on the dimension. The periodic Mackey–Glass signal is satisfied with two. Its attractor is a closed curve, and a closed curve drawn in the plane crosses itself only at a few isolated points, which spoil the neighbourhoods of very few delay vectors. The chaotic Mackey–Glass signals need four coordinates, although the state of a delay equation is infinite-dimensional. The motion settles on an attractor of low dimension, and only the dimension of the attractor matters for the reconstruction.

A record without any deterministic structure tests the method from the other side. For a sequence of independent Gaussian values the ratio test alone reports almost no false neighbours at high m, which would wrongly suggest a successful embedding. In many dimensions all distances between random points become nearly equal, so the nearest neighbour is no closer than any other point, and adding a coordinate cannot increase its distance by a large factor. The absolute test exposes this case, since these neighbours are not close at all. The combined fraction then stays high and grows with m, while for the Lorenz record it levels off at a few percent. A combined fraction that remains high and grows with m gives no support for a well-resolved low-dimensional reconstruction at the available record length and noise level. Measurement noise or undersampling can produce this pattern even when the source has a low-dimensional deterministic law. The diagnostic depends on noise level, record length and thresholds, and it supports a conclusion only together with forecasting performance.

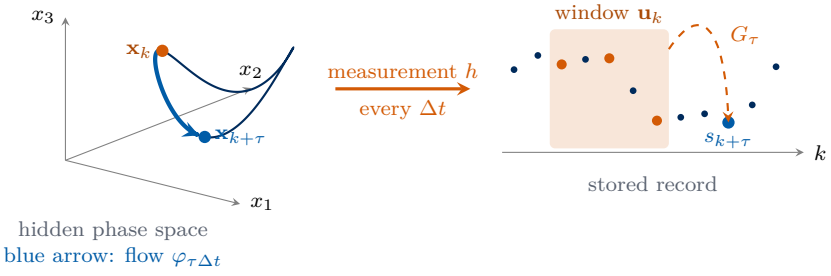


Figure 18: The forecast map. The hidden state $\mathbf{x}_k$ is moved by the flow of the vector field to $\mathbf{x}_{k+\tau}$ (blue arrow), and the measurement turns the states into the stored record. The forecast map G_τ of equation (4.11) goes directly from the window $\mathbf{u}_k$ of the record to the future value $s_{k+\tau}$ (dashed), without passing through the hidden phase space.

The window that a forecasting model must read spans the first and the last entry of the delay

Table 2: Delay embedding of the benchmark records, 4000 samples each. κ is the first minimum of the mutual information, m the smallest number of coordinates for which the ratio test of false nearest neighbours falls below 5%, and the last two columns give the combined fraction of false neighbours. For the Gaussian record the ratio test first reaches zero at $m = 7$ and remains zero at $m = 8$. This finite-sample outcome gives no valid deterministic embedding. Temporal exclusion is 50 samples for Lorenz and Mackey–Glass and 20 for the Gaussian control, which uses delay one.

record	κ	m	combined, $m = 4$	combined, $m = 8$
Lorenz x	10	4	0.040	0.040
Mackey–Glass, $\tau_{\mathrm{MG}} = 15$	8	2	–	–
Mackey–Glass, $\tau_{\mathrm{MG}} = 25$	16	4	–	–
Mackey–Glass, $\tau_{\mathrm{MG}} = 35$	26	4	–	–
laser	2	3	–	–
independent Gaussian values	–	–	0.174	0.275

vector, so its length in samples is

$$L = (m-1)\kappa + 1. \tag{4.21}$$

Definition 4.3 (Forecast map)**.** Let A be an invariant state set, φ_t its deterministic flow and $h : A \to \mathbb{R}$ the measurement function. Suppose the noise-free delay map $\mathcal{E} : A \to \mathcal{E}(A) \subset \mathbb{R}^m$ is an embedding. For a forecast horizon $\tau \in \{1, 2, \ldots\}$ measured in samples, define

$$G_\tau : \mathcal{E}(A) \longrightarrow \mathbb{R}, \qquad \boxed{G_\tau = h \circ \varphi_{\tau\Delta t} \circ \mathcal{E}^{-1}}. \tag{4.22}$$

Thus $G_\tau(\mathbf{u}_k) = s_{k+\tau}$. Reading the composition from right to left, $\mathcal{E}^{-1}$ recovers the state, $\varphi_{\tau\Delta t}$ advances it by τ sampling intervals, and h returns the future measurement. A smooth embedding and smooth dynamics give a smooth forecast map on the reconstructed state space.

The definition specifies G_τ on admissible delay vectors $\mathcal{E}(A)$. A forecasting model extends this relation to the inputs encountered in noisy data. An observational symmetry can also permit prediction of the measured quantity without an injective reconstruction of the full physical state, provided states with the same delay vector have the same future measurement. Learning to forecast means estimating the resulting input–output relation from the record.

5 Forecasting as regression on a window of the past

5.1 Windows as supervised examples

Equation (4.11) turns forecasting into supervised learning. For every origin k the input is the window $\mathbf{u}_k = (s_k, s_{k-1}, \ldots, s_{k-L+1})$ of the last L values, a delay vector with $\kappa = 1$ and $m = L$, and the target is $s_{k+\tau}$. A model $\hat{s}_{k+\tau} = G_\tau(\mathbf{u}_k; \boldsymbol{\Theta})$ with parameters $\boldsymbol{\Theta}$ is fitted by minimising the mean squared error over the available pairs. The squared error is the appropriate loss when the forecast is meant to estimate the conditional mean of $s_{k+\tau}$ given the window, since the conditional mean minimises the expected squared error. A dense window with L larger than the embedding window $(m-1)\kappa + 1$ contains the delay vector and leaves the selection of the relevant values to the fit.

Three choices distinguish the forecasting strategies, as sketched in Figure 19. A one-step model predicts s_{k+1}. A direct model has a separate map G_τ for every horizon, either as separate models or as one model with one output per horizon. A recursive model iterates a one-step model, appending each prediction to the window as if it were a measurement. The recursive forecast needs only one model, but each step feeds the error of the previous one into the input, and the model is evaluated on inputs that contain its own errors, which it never saw during fitting. A recursive forecast over many steps from a single origin is called a free run. For an illustrative one-step rule $\hat{s}_{k+1} = 0.5 + 0.8 s_k - 0.2 s_{k-1}$ and the last measured values $s_k = 6$, $s_{k-1} = 4$, the recursion gives

$$\hat{s}_{k+1} = 0.5 + 0.8 \cdot 6 - 0.2 \cdot 4 = 4.5, \qquad \hat{s}_{k+2} = 0.5 + 0.8 \cdot \hat{s}_{k+1} - 0.2 \cdot 6 = 2.9. \tag{5.1}$$

The second step uses the forecast $\hat{s}_{k+1}$ in place of the unknown measurement. A direct two-step model is fitted to the targets s_{k+2} and in general gives a different value.

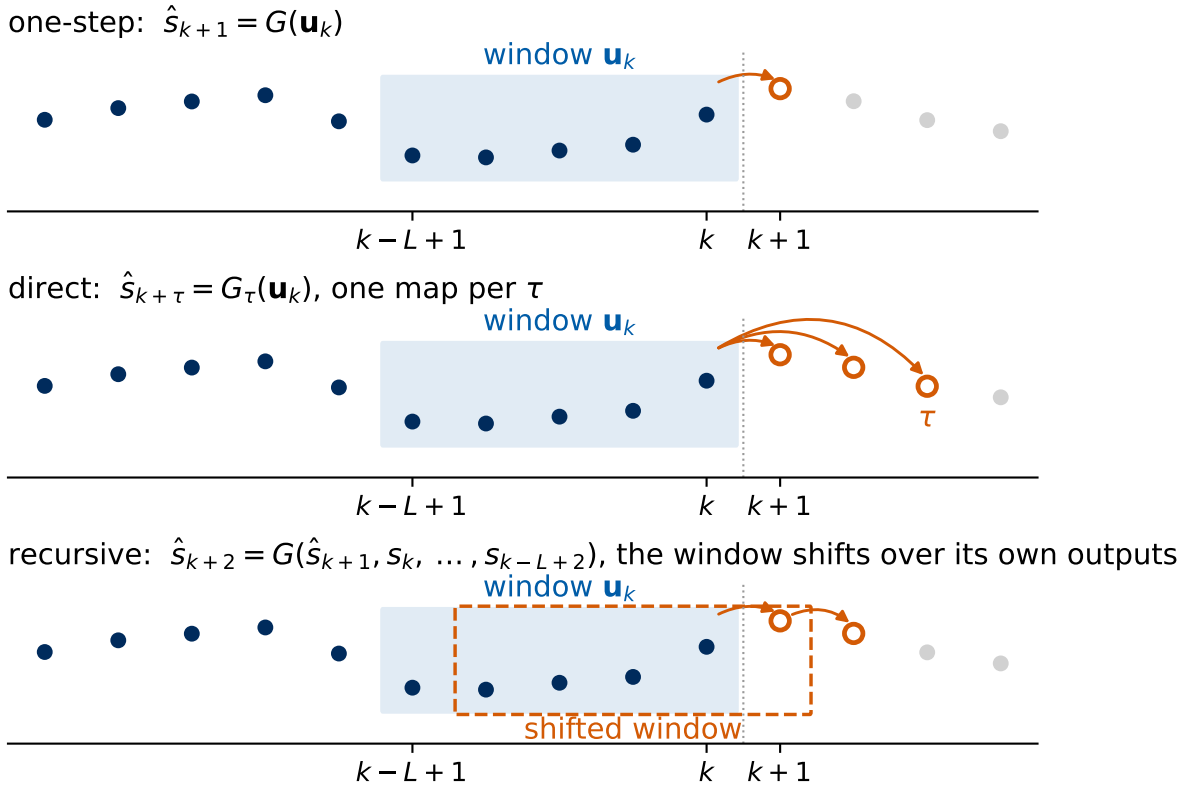


Figure 19: One-step, direct and recursive forecasts from the window $\mathbf{u}_k$ of the last L values. The recursive forecast shifts the window over its own predictions (orange).

5.2 Training and testing in time order

The pairs $(\mathbf{u}_k, s_{k+\tau})$ are not independent samples. Neighbouring windows share $L-1$ values, and a random split into training and test pairs would place nearly identical inputs on both sides, so that the test score would measure interpolation within the record. A forecast is a statement about the future, and its evaluation must respect the order of time. The record is therefore divided into consecutive blocks, the earliest for fitting the parameters, the next for selecting settings such as L or a penalty strength (validation), and the latest for the final test. The constraint applies to the targets as well as to the inputs. A fitting pair must have

its target inside the fitting block. Any preprocessing estimated from data, such as a rescaling, is estimated on the fitting data and then frozen.

All benchmark comparisons in this chapter follow one protocol, shared by all models. Each signal has $N = 4000$ samples, rescaled to $[0, 1]$ with the minimum and maximum of the first 3000. The rescaling thus uses the fitting and validation blocks together, a minor departure from the rule above that does not touch the test block. Samples 0 to 2499 form the fitting block, samples 2500 to 2999 the validation block and samples 3000 to 3999 the test block (Figure 20). The neural windows have $L = 64$ values and the horizons run from 1 to 50. The same earliest origin is used for all models, while AR and NVAR select their feature lengths, MA uses filtered innovations, and the ESN carries a continuously driven state. Each block loses some origins at its edges, because the targets of an origin up to $\tau = 50$ must lie inside the block, while its window may reach back into the preceding block, which leaves 2387, 451 and 951 origins in the three blocks.

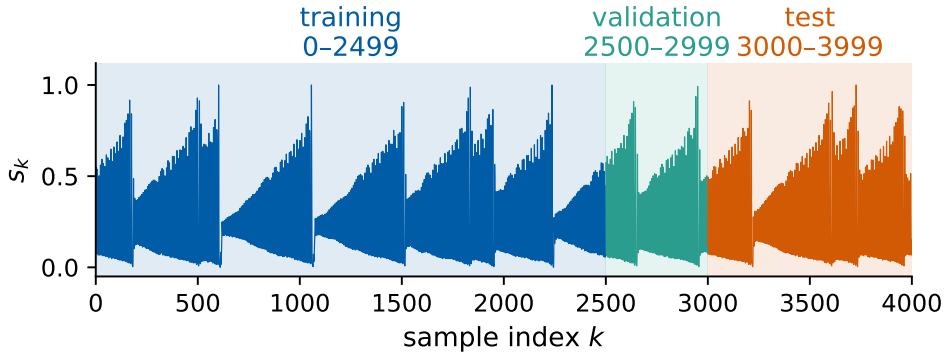


Figure 20: Chronological split of the laser record into fitting, validation and test blocks.

For a small example of the split, take $L = 3$, horizons 1 and 2, a fitting block ending at sample 5 and a test block starting at sample 8. Origin $k = 3$ gives the fitting input (s_3, s_2, s_1) and targets (s_4, s_5). Origin $k = 4$ is excluded from fitting because its second target s_6 is outside the fitting block. The first test origin can be $k = 7$, whose input (s_7, s_6, s_5) is already known and its targets (s_8, s_9) are in the test block. Shuffling the permitted fitting pairs into minibatches does not change their split assignment. Randomly assigning overlapping pairs to fitting and testing would.

5.3 Scoring a forecast

The simplest forecast that uses the record is persistence, $\hat{s}_{k+\tau} = s_k$. It needs no fitting, and every model must be compared with it, because a slowly varying signal makes persistence accurate without any knowledge of the dynamics. For each horizon the forecasts $\hat{y}_i = \hat{s}_{k_i+\tau}$ and targets $y_i = s_{k_i+\tau}$ over the test origins k_i are collected and scored by their correlation.

Definition 5.1 (Forecast capacity)**.** The capacity at horizon τ is the squared Pearson correlation between forecasts and targets over the test origins,

$$C(\tau) = r(\tau)^2, \qquad r(\tau) = \frac{\sum_i (\hat{y}_i - \bar{\hat{y}})(y_i - \bar{y})}{\sqrt{\sum_i (\hat{y}_i - \bar{\hat{y}})^2 \sum_i (y_i - \bar{y})^2}}, \tag{5.2}$$

with bars denoting averages over the origins. The total capacity is $C_{\mathrm{tot}} = \sum_{\tau=1}^{50} C(\tau)$, at most 50 for the benchmark protocol.

$C(\tau)$ is the fraction of the variance of the targets that a linear rescaling of the forecasts explains. It equals 1 when the forecasts are an exact affine function of the targets and 0 when they are uncorrelated. It is insensitive to offset and scale. Forecasts $\hat{y} = 2y + 5$ have $C = 1$ although every value is wrong, and so do forecasts $\hat{y} = 1 - y$, because squaring removes the sign of r. The capacity therefore measures how well the forecast follows the variations of the signal, which is the property that distinguishes the models compared in this chapter, and it is the score used throughout. Where the level and the amplitude of the forecast matter, as for the temperature in Section 9.1 and for the free runs of Section 10, the root-mean-square error is reported alongside it. A constant forecast has an undefined correlation and is assigned $C = 0$ by convention. The name capacity follows the reservoir-computing literature, where the same squared correlation measures how well a dynamical system reproduces a function of its past inputs [19].

Persistence behaves very differently on the benchmark signals (Table 7). On the laser its squared correlation is about a quarter even one step ahead. The pulses are only a few samples long, and the intensity has partly fallen one sample after a maximum. On the slowly varying periodic Mackey–Glass signal persistence scores much higher. These are correlation scores, and a literal repetition of the last value can have a substantial amplitude error even when C is large. Persistence supplies a baseline that a useful learned forecast should improve upon.

Persistence and an optimal linear forecast can be compared without fitting a neural network. For a stationary process with mean μ, variance v and lag-τ correlation $\rho(\tau)$, minimise $\mathbb{E}[(s_{k+\tau} - \mu - b(s_k - \mu))^2]$ over b. Expanding the square and differentiating gives

$$b_* = \rho(\tau), \qquad \mathrm{MSE}_{\mathrm{lin}} = v[1 - \rho(\tau)^2], \qquad \mathrm{MSE}_{\mathrm{pers}} = 2v[1 - \rho(\tau)]. \tag{5.3}$$

Persistence fixes $b = 1$, while the optimal coefficient allows regression towards the mean. Their excess error is $v[1 - \rho(\tau)]^2$. The two forecasts differ only by a scale factor and an offset, so both have the capacity $C(\tau) = \rho(\tau)^2$, while persistence has the larger mean squared error. The capacity ignores this difference in amplitude.

5.4 Features and a linear readout

All forecasting models of the following sections share one structure, which organises their comparison. A model first turns the past of the record into a vector of M numbers, the features, and then forms the forecast as a linear combination of the features,

$$(s_1, \ldots, s_k) \xrightarrow{\Phi} \boldsymbol{\phi}_k \in \mathbb{R}^M \xrightarrow{\text{linear}} \hat{y}_k = \mathbf{w}^\top \boldsymbol{\phi}_k, \tag{5.4}$$

where Φ is the feature map, $\boldsymbol{\phi}_k$ the feature vector at time k, $\mathbf{w} \in \mathbb{R}^M$ the readout weights and $\hat{y}_k$ the forecast of the target, for example $s_{k+\tau}$. Figure 21 shows the scheme. At every step the feature vector is one point in $\mathbb{R}^M$, and as the record advances these points trace a path in the feature space. The readout projects the path onto the direction $\mathbf{w}$.

The models differ in the feature map and in what is trained. The features can be the last L values themselves, a linear feature map, or nonlinear functions of them such as products of values or tanh units. They can also be a state that is updated recursively, $\mathbf{h}_k = U(\mathbf{h}_{k-1}, s_k)$, as in a recurrent network. In the autoregressive models of Section 6 and in nonlinear vector autoregression, Section 7.3, the feature map is fixed in advance and only the readout $\mathbf{w}$ is fitted, which is a linear least-squares problem; a positive ridge penalty makes its solution unique. In the neural networks of Section 8 the feature map has parameters of its own, and the feature map and the readout are trained together by gradient descent. The question that runs through the next three sections is which features a given record requires.

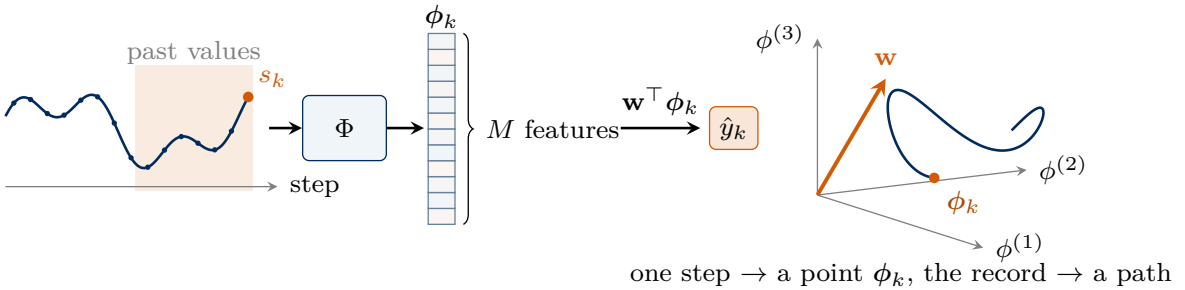


Figure 21: Features and a linear readout. The feature map Φ turns the past values of the record into a vector $\boldsymbol{\phi}_k$ of M features, and the forecast is the projection $\mathbf{w}^\top \boldsymbol{\phi}_k$. Right, each step gives one point in the feature space, and the record traces a path in it.

6 Linear models

6.1 Autoregression and least squares

The simplest choice for the forecast map is a linear function of the window. Before using it, we ask what kind of system a linear model can describe. For a scalar sinusoid $s_k = \cos(\omega k + \phi)$, the identity $\cos(a+b) + \cos(a-b) = 2\cos a \cos b$ with $a = \omega(k-1) + \phi$ and $b = \omega$ gives $s_k = 2\cos\omega\, s_{k-1} - s_{k-2}$, an exact linear recurrence of order two. A sum of n sinusoids obeys a linear recurrence of order $2n$, whose characteristic polynomial is the product of the quadratics of the individual frequencies. In the notebook on signals, a recurrence of order four fitted to a quasi-periodic sum of two sines with incommensurate frequencies continues it to rounding precision, although the signal never repeats. A finite sum of sinusoidal components therefore has an exact finite-order linear recurrence, however irregular it looks; this argument does not give a finite-order recurrence for every nonsinusoidal continuous-time oscillation.

Definition 6.1 (Autoregressive model)**.** The autoregressive model of order p, AR(p), forecasts the next value as a weighted sum of the last p values,

$$\hat{s}_{k+1} = c + \sum_{i=0}^{p-1} \phi_i \, s_{k-i}, \tag{6.1}$$

with an offset c and weights ϕ_i [21]. The direct AR forecast at horizon τ uses its own weights $w_{\tau,i}$ and offset c_τ for the target $s_{k+\tau}$.

The parameters are fitted by least squares. The windows of the fitting block form the rows of a matrix X, with a leading column of ones for the offset, the targets form a vector $\mathbf{y}$, and $\mathbf{a} = (c, \phi_0, \dots, \phi_{p-1})^\top$. The loss $\mathcal{L}(\mathbf{a}) = \|\mathbf{y} - X\mathbf{a}\|^2$ is a convex quadratic function of $\mathbf{a}$, so its minimum is where the gradient vanishes,

$$\nabla_{\mathbf{a}} \mathcal{L} = -2X^\top(\mathbf{y} - X\mathbf{a}) = 0 \quad \Longrightarrow \quad X^\top X\, \mathbf{a} = X^\top \mathbf{y}. \tag{6.2}$$

These normal equations are a single linear system, solved without iteration. For a centred stationary record the entries of $X^\top X/n$ tend to the autocovariances $\gamma(i-j) = \langle s_k s_{k+i-j} \rangle$ and those of $X^\top \mathbf{y}/n$ to $\gamma(i+1)$. Equation (6.2) then becomes the Yule–Walker equations, and the fitted model depends on the data only through the autocovariance function, the second-order statistics of the record. A linear model sees nothing of a record beyond its autocovariance, a limitation that the following sections remove. In practice a penalty $\lambda\|\boldsymbol{\phi}\|^2$ is added to stabilise the fit when the window is long, and the order and the penalty are chosen on the validation block (Table 3).

A recursive AR forecast is the iteration of a linear map on the window. With $\mathbf{u}_k$ as state and the offset removed by centring, $\mathbf{u}_{k+1} = A\mathbf{u}_k$, where the companion matrix A has the weights ϕ_i in its first row and shifts the window in the remaining rows. The iterated forecast is $A^n\mathbf{u}_k$, which decays to the mean if all eigenvalues of A lie inside the unit circle, equivalently if all roots of the polynomial $1 - \sum_i \phi_i z^{i+1}$ lie outside it. For AR(20) fitted to the Lorenz coordinate, all roots lie outside the unit circle, the closest at a modulus of 1.0541. The model is stable, and its free run relaxes to the mean. A stable linear map decays to a fixed point, an unstable one grows without bound, and a map with eigenvalues on the unit circle oscillates periodically or quasi-periodically. None of these produces a bounded aperiodic motion that switches between two lobes. Chaos requires nonlinearity.

6.2 Moving-average and ARMA models

A second linear construction forecasts from the recent errors of the model instead of the recent values.

Definition 6.2 (Moving-average and ARMA processes)**.** Let ξ_k be independent random innovations with zero mean and variance σ^2. The moving-average process of order q, MA(q), is

$$s_k = \mu + \xi_k + \sum_{j=1}^{q} \theta_j \xi_{k-j}, \tag{6.3}$$

with a mean level μ and weights θ_j. The ARMA(p, q) process adds an autoregressive part, $s_k = c + \sum_{i=1}^{p} \phi_{i-1} s_{k-i} + \xi_k + \sum_{j=1}^{q} \theta_j \xi_{k-j}$.

The name refers to a moving average of past innovations, and an MA process is distinct from a running mean of past measurements. The innovations are not observed. Given the parameters, they are estimated recursively by the one-step errors $e_k = s_k - \hat{s}_k$ of the model, and the one-step forecast is the conditional mean

$$\hat{s}_{k+1} = \mu + \sum_{j=1}^{q} \theta_j e_{k+1-j}, \tag{6.4}$$

since the new innovation ξ_{k+1} has mean zero. Because every e_k depends on the parameters through all earlier forecasts, the fit is a nonlinear optimisation. The parameters maximise the likelihood of the record under the assumption of Gaussian innovations, which is equivalent to minimising the sum of the squared one-step errors when the starting values are fixed.

Horizon of a moving-average forecast. *The MA(q) forecast of $s_{k+\tau}$ made at origin k equals the mean μ for every $\tau > q$.*

Derivation. From the definition, $s_{k+\tau} = \mu + \xi_{k+\tau} + \sum_{j=1}^{q} \theta_j \xi_{k+\tau-j}$. At origin k the innovations up to ξ_k are estimated by the errors up to e_k, and the future innovations are replaced by their expected value zero, so $\hat{s}_{k+\tau} = \mu + \sum_{j=1}^{q} \theta_j e_{k+\tau-j}$ with $e_{k+\tau-j}$ set to zero unless $k+\tau-j \le k$, that is $j \ge \tau$. For $\tau > q$ no index $j \le q$ satisfies this, so all correction terms vanish. □

The result is confirmed on the laser (Table 3). The MA(8) forecast is reasonable one step ahead, degrades over the next few steps and is constant, with zero capacity, at $\tau = 10$. A deterministic signal carries its information in its values, which MA models discard after q steps. MA models describe records in which a random shock acts for a fixed number of steps and is then forgotten, which is not how a dynamical system with a state behaves. ARMA forecasts on the benchmark signals stay close to those of AR.

6.3 Linear forecasts of the laser

Fitted directly for each horizon, AR on the laser triples the total capacity of persistence, and its capacity falls steadily with the horizon (Table 3). The first row of Figure 22 shows where it fails. Each point is one test origin, with the target on the horizontal axis and the forecast on the vertical axis. For small and medium intensities the points lie near the diagonal, while for the largest targets they bend below it. The model underestimates the high pulses and misses the collapses that follow them. On the periodic Mackey–Glass signal, a superposition of a few harmonics, the same direct AR model reaches nearly the maximal total capacity, in agreement with the recurrence argument above, while on the chaotic Mackey–Glass signals it falls well short of it (Table 7).

Table 3: Forecasts of the laser record on its test block. $C(\tau)$ is the capacity at horizon τ, C_{tot} its sum over $\tau = 1, \dots, 50$ (maximum 50), with the range over three random initialisations of the networks in brackets. The free run counts the steps from the origin $k_0 = 2999$ during which the error stays below 0.2 of the training range. Settings were chosen on the validation block.

model	settings	$C(1)$	$C(2)$	$C(5)$	$C(10)$	C_{tot}	free run
persistence	–	0.27	–	–	–	5.5	–
AR	$p = 64$, penalty 1	0.85	0.83	0.77	0.60	16.5	38
MA	$q = 8$	0.77	0.74	0.50	0.00	4.7	–
NVAR	$L = 32$, $\lambda = 10^{-2}$	0.98	0.95	0.93	0.87	34.7	36
MLP	64 hidden units	–	–	–	–	35.3 [35.3, 36.0]	9
RNN	$n_h = 32$	0.96	0.93	0.90	0.90	42.5 [41.6, 42.5]	23
LSTM	$n_h = 32$	0.94	0.93	0.93	0.92	44.3 [44.0, 44.9]	110

An affine autoregression has the form $F(\mathbf{u}) = c + \mathbf{a}^{\top}\mathbf{u}$, so $F(2\mathbf{u}) = 2F(\mathbf{u}) - c$. Proportional response holds for the centred variables when the intercept has been removed. The laser responds to a large pulse by a collapse, a response that is not proportional to the amplitude. The forecast map G_τ of equation (4.11) is a curved surface over the space of windows, and a linear model fits a plane to it. The next section asks how a model can remain linear in its parameters, and therefore keep the single least-squares solution, while becoming nonlinear in the data.

7 Nonlinear features and the Koopman operator

The laser forecasts leave systematic errors near large pulses, where a linear dependence on past intensities is too restrictive. Products and other functions of past values can describe this curvature while keeping the fitted readout linear in its coefficients. The Koopman operator explains the relation between nonlinear state evolution and linear evolution in a space of functions.

7.1 The sampled flow map

The laws of motion of this chapter are polynomials of low degree. The Lorenz field (2.28) and the Maxwell–Bloch field (1.3) are quadratic in the state. What a sampled record sees is not the field but the one-step flow map $\varphi_{\Delta t}$ of equations (2.2) and (2.4), which for a short interval

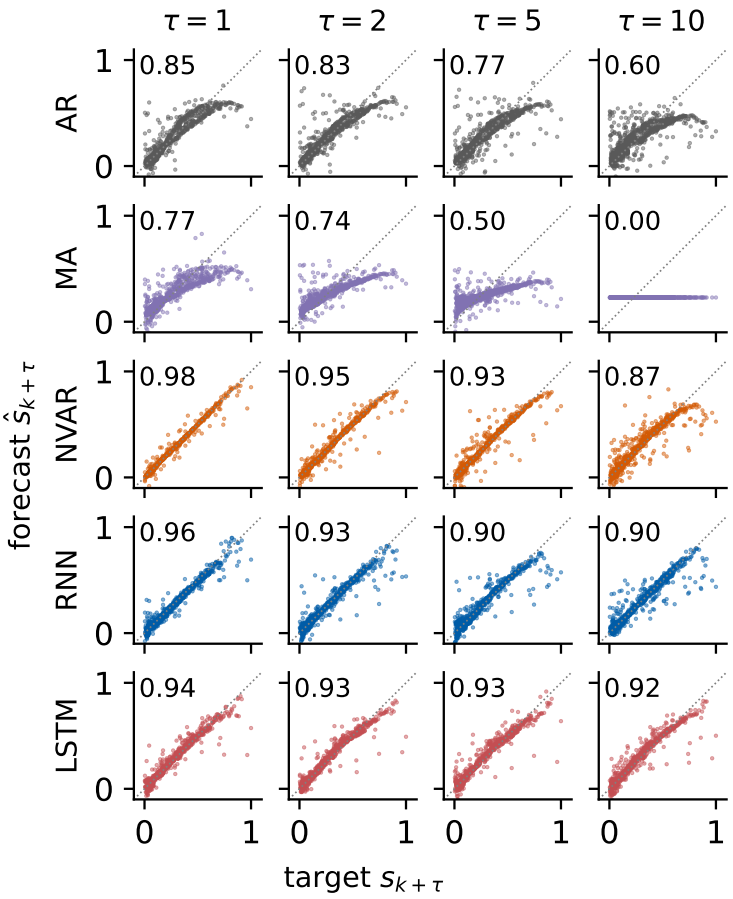


Figure 22: Forecast against target on the laser test block at $\tau = 1$, 2, 5 and 10 (columns) for AR, MA, NVAR, RNN and LSTM (rows), with the capacity C printed in each panel. These are the forecasts quoted in the text.

follows from integrating the field,

$$\varphi_{\Delta t}(\mathbf{x}) = \mathbf{x} + \int_0^{\Delta t} \mathbf{F}(\mathbf{x}(t'))\, dt' \approx \mathbf{x} + \Delta t\, \mathbf{F}(\mathbf{x}), \tag{7.1}$$

with the approximation of the explicit Euler step, accurate to first order in Δt. Figure 23 contrasts the two objects. The velocity $\mathbf{F}(\mathbf{x}_k)$ is the tangent to the trajectory at the present state, the direction and speed of the motion at this instant. The flow map gives the position after the time Δt, the sum of all velocities along the way, and only for a short interval is it close to one step along the tangent.

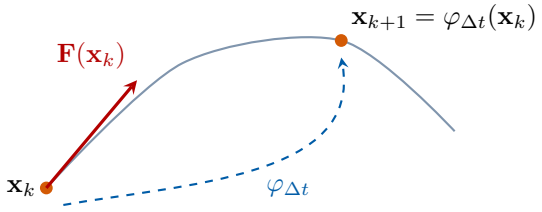


Figure 23: The velocity and the flow map. Red, the velocity $\mathbf{F}(\mathbf{x}_k)$, tangent to the trajectory at the present state. Blue, the flow map $\varphi_{\Delta t}$, which carries the state along the trajectory to its position one sampling interval later.

For a quadratic field and a short interval, the one-step map is a quadratic polynomial of the state up to terms of order Δt^2. When the measured vector is the full state, the one-step map is therefore close to a quadratic polynomial of the measured variables, a statement that follows from the field alone. When only a scalar window is measured, the hidden coordinates enter as unknown functions of the window, and the expectation that quadratic functions of the window suffice is a heuristic, which Section 7.3 tests. The Koopman operator makes this idea systematic and explains why a model that is linear in its parameters can describe a nonlinear system.

7.2 Linear evolution of observables

Linear and nonlinear evolution differ in one property, superposition. For a linear law, $\mathbf{x}_{k+1} = A\mathbf{x}_k$, the evolution of a sum of two initial states is the sum of their evolutions, and everything about the dynamics follows from the eigenvalues and eigenvectors of one matrix. The solution is a sum of modes, each multiplied at every step by its eigenvalue, and forecasting is a matrix power. A nonlinear law has no such decomposition. The evolution of a sum is not the sum of the evolutions, the behaviour near one state says little about the behaviour near another, and chaos, which Section 6 showed to be impossible for a linear map, becomes possible. The Koopman operator recovers linearity for any law, at a price, by changing what is evolved. Instead of the state, it evolves functions of the state.

An observable is any function $g : \mathbb{R}^d \to \mathbb{R}$ of the state, for example a coordinate, the energy, a product $x_1 x_3$ or the measurement h.

Definition 7.1 (Koopman operator)**.** The Koopman operator K_t acts on observables by composition with the flow map [22],

$$(K_t g)(\mathbf{x}) = g(\varphi_t(\mathbf{x})). \tag{7.2}$$

We write $K = K_{\Delta t}$ for one sampling interval.

$K_t g$ is the observable whose value now equals the value of g a time t later. The operator inherits from the properties (2.3) of the flow map the rules

$$K_0 = \mathrm{id}, \qquad K_t K_s = K_{t+s}, \qquad g(\mathbf{x}_{k+n}) = (K^n g)(\mathbf{x}_k), \tag{7.3}$$

so that advancing an observable by n samples is the n-th power of one operator. The operator is linear for every vector field, linear or not, since

$$K(\alpha g_1 + \beta g_2) = (\alpha g_1 + \beta g_2) \circ \varphi_{\Delta t} = \alpha\, g_1 \circ \varphi_{\Delta t} + \beta\, g_2 \circ \varphi_{\Delta t} = \alpha K g_1 + \beta K g_2. \tag{7.4}$$

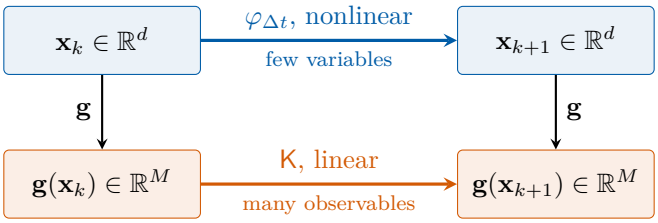


Figure 24: Two routes from the state at step k to the observables at step $k+1$. Along the top, the nonlinear flow map acts on the d state variables. Along the bottom, the observables are evaluated first and advanced by a linear map. When the dictionary $\mathbf{g}$ is closed, both routes give the same result for every state.

Figure 24 shows the trade. The state can be advanced by the nonlinear map $\varphi_{\Delta t}$ in few variables, or the observables can be advanced by a linear operator. The nonlinearity has not disappeared. It has been moved into the space on which K acts, the space of all functions on phase space, which is infinite-dimensional. Superposition holds for observables, while the states evolve nonlinearly. This has a formal analogy with the Heisenberg picture of quantum mechanics, where observables evolve linearly while the state is fixed. The Koopman construction here acts on classical functions of phase space and makes no quantum assumption.

The measured record itself is an instance of this picture (Figure 25). The measurement function h is an observable, and for noise-free data the Koopman operator applied to it gives the next measurement,

$$(Kh)(\mathbf{x}_k) = h(\varphi_{\Delta t}(\mathbf{x}_k)) = h(\mathbf{x}_{k+1}) = s_{k+1}, \qquad s_{k+n} = (K^n h)(\mathbf{x}_k). \tag{7.5}$$

A time series is the operator K acting again and again on one observable. Forecasting therefore amounts to finding observables, computed from the record, on which the evolution is close to linear. The models of this chapter are three ways of choosing them. AR uses the past values $s_k, s_{k-1}, \ldots$, NVAR the past values and their products, and the neural networks learn the observables from the data.

The operator K acts on infinitely many functions, and a practical method keeps a finite set of observables $\mathbf{g} = (g_1, \ldots, g_M)$, a dictionary. The set is useful if K maps it nearly into itself, so that an $M \times M$ matrix K advances it,

$$\mathbf{g}(\mathbf{x}_{k+1}) \approx \mathsf{K}\,\mathbf{g}(\mathbf{x}_k) \qquad \Longrightarrow \qquad \mathbf{g}(\mathbf{x}_{k+n}) \approx \mathsf{K}^n\,\mathbf{g}(\mathbf{x}_k), \tag{7.6}$$

and forecasting becomes repeated multiplication by K. The least-squares choice minimises $\sum_k \|\mathbf{g}(\mathbf{x}_{k+1}) - \mathsf{K}\,\mathbf{g}(\mathbf{x}_k)\|^2$ over the recorded states. With the matrices G and G_+ whose columns are $\mathbf{g}(\mathbf{x}_k)$ and $\mathbf{g}(\mathbf{x}_{k+1})$, the solution is $\mathsf{K} = G_+ G^+$, where G^+ is the Moore–Penrose pseudoinverse, which gives the least-squares solution when G is not square. The dictionary is closed if K maps every function in its span into its span. Then the relation is exact, the

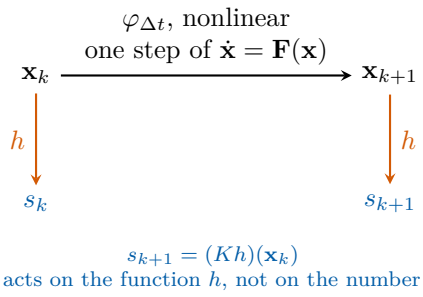


Figure 25: The measured record in the Koopman picture. The hidden state is advanced by the nonlinear flow map (top), and the measurement h turns each state into a stored value (bottom). The Koopman operator acts on observables: $h \mapsto Kh$ is linear, and evaluating the new function at the original state gives $s_{k+1} = (Kh)(\mathbf{x}_k)$. This does not define a scalar map from s_k alone to s_{k+1}.

diagram of Figure 24 commutes for every state, and the forecast over n steps is the matrix power $\mathbf{g}(\mathbf{x}_{k+n}) = \mathsf{K}^n \mathbf{g}(\mathbf{x}_k)$.

A two-coordinate example makes the change of representation explicit. Its nonlinear update is $x_{k+1} = ax_k$, $y_{k+1} = by_k + cx_k^2$, with fixed parameters a, b and c. In the original coordinates it can be written as

$$\begin{pmatrix} x \\ y \end{pmatrix}_{k+1} = \begin{pmatrix} a & 0 \\ c\,x_k & b \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}_k . \tag{7.7}$$

The lower-left entry in equation (7.7) depends on the input x_k. Consequently this matrix expression still represents a nonlinear map. Its successive updates use state-dependent matrices, whose products generally cannot be replaced by powers of one fixed matrix.

The Koopman construction applies this update to observables. Choose $g_1 = x$, $g_2 = y$ and $g_3 = x^2$. Composition with the nonlinear map gives $Kg_1 = ag_1$, $Kg_2 = bg_2 + cg_3$ and $Kg_3 = (ax)^2 = a^2 g_3$. All three images belong to the span of the same observables. The enlarged vector therefore evolves as

$$\underbrace{\begin{pmatrix} x \\ y \\ x^2 \end{pmatrix}_{k+1}}_{\mathbf{g}(\mathbf{x}_{k+1})} = \underbrace{\begin{pmatrix} a & 0 & 0 \\ 0 & b & c \\ 0 & 0 & a^2 \end{pmatrix}}_{\mathsf{K}} \underbrace{\begin{pmatrix} x \\ y \\ x^2 \end{pmatrix}_k}_{\mathbf{g}(\mathbf{x}_k)} . \tag{7.8}$$

The matrix in equation (7.8) is fixed for every input state. The nonlinear dependence x^2 is now a component of the observable vector, so a linear operator on three observables reproduces the nonlinear evolution of two state coordinates. The linearity identity (7.4) guarantees a linear action on observables. A finite matrix representation also requires closure of their span, which we have checked explicitly in this example.

For the parameters $a = 0.9$, $b = 0.5$, $c = 1$, the matrix has eigenvalues 0.9, 0.5 and 0.81, and its powers reproduce direct iteration of the nonlinear map to rounding precision. The eigenvalue $a^2 = 0.81$ belongs to the observable x^2, a decay rate available in the enlarged description. Fitting a matrix to (x, y) alone, as in dynamic mode decomposition (DMD) of the raw state, leaves the effect of x^2 unresolved. No fixed linear combination of x and y equals x^2 for all states. The notebook's restricted fit consequently has a larger one-step error in y and a distorted fitted eigenvalue. The added observable supplies the missing nonlinear dependence.

A closed dictionary is the exception. For the map $x_{k+1} = x_k^2$, following x linearly requires x^2, following x^2 requires x^4, and so on,

$$Kx = x^2, \qquad Kx^2 = x^4, \qquad Kx^4 = x^8, \qquad \dots, \tag{7.9}$$

and no finite set of powers closes. A linear description exists, but it requires infinitely many observables. For general nonlinear dynamics, an exact representation containing the required observables can require an infinite-dimensional function space. A finite fitted dictionary then approximates the dynamics, and its quality depends on the choice of observables and on the states used for fitting.

The linear structure has a direct interpretation through eigenfunctions. An observable η with $K\eta = \mu\eta$ is only multiplied by the number μ at each step,

$$\eta(\mathbf{x}_k) = \mu^k \, \eta(\mathbf{x}_0). \tag{7.10}$$

In the three-observable example, x, x^2 and $y + \frac{c}{b-a^2}x^2$ are eigenfunctions with the eigenvalues a, a^2 and b, as can be checked by applying the matrix above. In this example the nonzero eigenfunction magnitudes decay geometrically, so their logarithms are linear in k. Other Koopman eigenvalues can describe growth or oscillation. Any observable in the span of eigenfunctions is forecast without approximation as a sum of such geometric sequences. The eigenvalues are the time scales and frequencies of the system, and the linear models of Section 6 are the special case in which the eigenfunctions are linear functions of the window.

The same construction applied to delay vectors reproduces the linear models of Section 6. DMD on the dictionary of delayed values $(s_k, \dots, s_{k-p+1})$ returns a matrix whose first row agrees with the least-squares AR weights to numerical precision and whose remaining rows shift the window. AR is the Koopman approximation on the dictionary of raw delayed values. Its failure on the Lorenz record shows that this dictionary is far from closed for chaotic dynamics, and the remedy is to enlarge it with nonlinear observables of the window.

7.3 Nonlinear vector autoregression

Products of delayed measurements provide a concrete finite dictionary of nonlinear observables. They let the forecast depend on interactions between past values while retaining a least-squares solution for the coefficients. Nonlinear vector autoregression applies this construction to the windows already defined for the benchmark.

Definition 7.2 (NVAR)**.** Nonlinear vector autoregression [23] uses the features

$$\boldsymbol{\phi}_k = (1, \; s_k, \dots, s_{k-L+1}, \; s_{k-i}s_{k-j} \;\; (0 \le i \le j \le L-1)) \tag{7.11}$$

and the forecast $\hat{s}_{k+\tau} = \mathbf{w}_\tau^\top \boldsymbol{\phi}_k$, with readout weights $\mathbf{w}_\tau$ fitted for each horizon.

The feature vector contains the constant, the L linear terms and the $L(L+1)/2$ distinct products of two window values, $1 + L + L(L+1)/2$ entries in total. For $L = 2$ the forecast is

$$\hat{s}_{k+\tau} = w_0 + w_1 s_k + w_2 s_{k-1} + w_3 s_k^2 + w_4 s_k s_{k-1} + w_5 s_{k-1}^2, \tag{7.12}$$

AR(2) plus three quadratic terms. Through the product term $w_4 s_k s_{k-1}$, the effect of s_k on the forecast depends on s_{k-1}, which is the interaction that a linear model lacks. Figure 26 shows the structure of the features for $L = 3$. The forecast is linear in $\mathbf{w}_\tau$, so it is fitted by a linear least-squares problem.

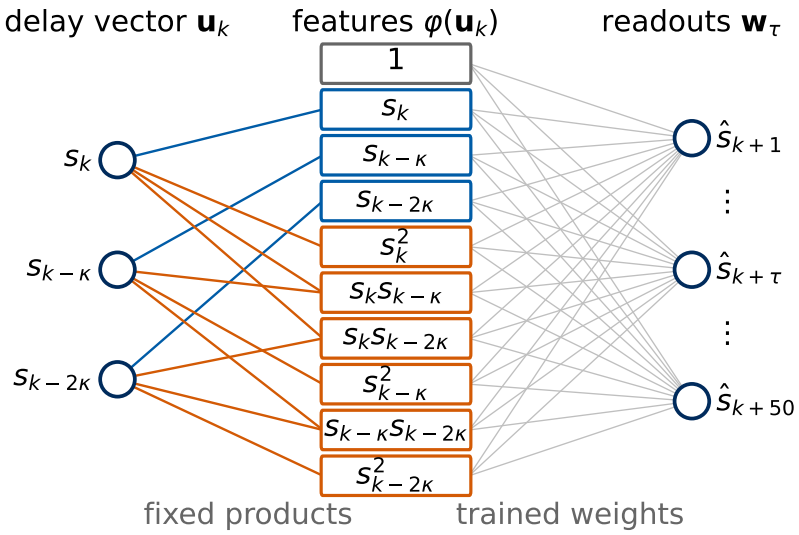


Figure 26: NVAR features for a window of three values. Each value feeds its linear feature and the products that contain it. Only the readout weights are fitted.

The products of window values are strongly correlated with each other, since neighbouring values of a smooth signal are similar. The matrix $X^\top X$ of equation (6.2) is then nearly singular, and the least-squares solution can contain large weights of opposite signs that cancel on the fitting data and amplify noise elsewhere. Ridge regression [24] adds a penalty on the size of the weights. With the intercept b separated from the non-constant features $\mathbf{v}_k$ and left unpenalised, the loss is

$$\mathcal{L}(\mathbf{w}, b) = \sum_k \left(s_{k+\tau} - b - \mathbf{w}^\top \mathbf{v}_k\right)^2 + \lambda \|\mathbf{w}\|^2. \tag{7.13}$$

Setting the derivative with respect to b to zero gives $b = \bar{y} - \bar{\mathbf{v}}^\top \mathbf{w}$, with the means $\bar{y}$ of the targets and $\bar{\mathbf{v}}$ of the features over the fitting data. Substituting this and setting the gradient with respect to $\mathbf{w}$ to zero gives

$$\left(V_c^\top V_c + \lambda \mathbb{1}\right) \mathbf{w} = V_c^\top \mathbf{y}_c, \tag{7.14}$$

where V_c holds the centred feature vectors as rows and $\mathbf{y}_c$ the centred targets. For $\lambda > 0$ the matrix is positive definite, so the solution is unique. Ridge improves conditioning, but a very small λ can leave the system ill-conditioned: its spectral condition number is $(\sigma_{\max}^2 + \lambda)/(\sigma_{\min}^2 + \lambda)$, where the σ_i are singular values of V_c (including zeros for missing column rank). As $\lambda \to \infty$ the weights vanish and the forecast tends to the mean of the fitting targets, and as $\lambda \to 0$ ordinary least squares is recovered when V_c has full column rank. The penalty and the window length are selected together on the validation block. The window need not consist of consecutive values. Taking every κ-th value, with the stride κ suggested by the delay embedding of Section 4, covers a longer stretch of the past with the same number of features.

Figure 27 shows one-step NVAR forecasts on the test blocks of the laser and of the three Mackey–Glass signals. For the Mackey–Glass signals the forecast and the target coincide within the line width. For the laser the points of the scatter plot stay near the diagonal also at the highest pulses, where the linear models bend away. Figure 28 follows the laser forecast to longer horizons, with one ridge weight vector $\mathbf{w}_\tau$ per horizon. The forecast keeps the timing of the pulses at five and ten samples ahead, while its accuracy for the heights decreases slowly, and the capacity of the Mackey–Glass forecasts stays close to one for the periodic signal and decreases slowly for the chaotic ones.

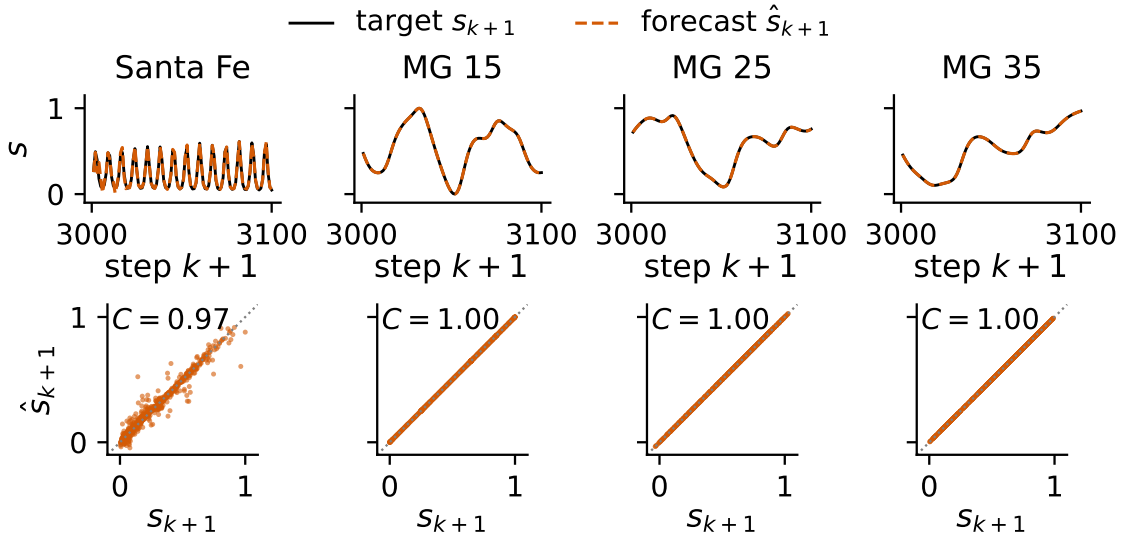


Figure 27: One-step NVAR forecasts on the test blocks of the laser and of the Mackey–Glass signals with $\tau_{\mathrm{MG}} = 15$, 25 and 35. Top, target and forecast against the step. Bottom, forecast against target with the capacity C. The window length, the stride and the penalty were chosen on the validation block in the NVAR notebook, whose settings differ slightly from those of the comparison of Table 3.

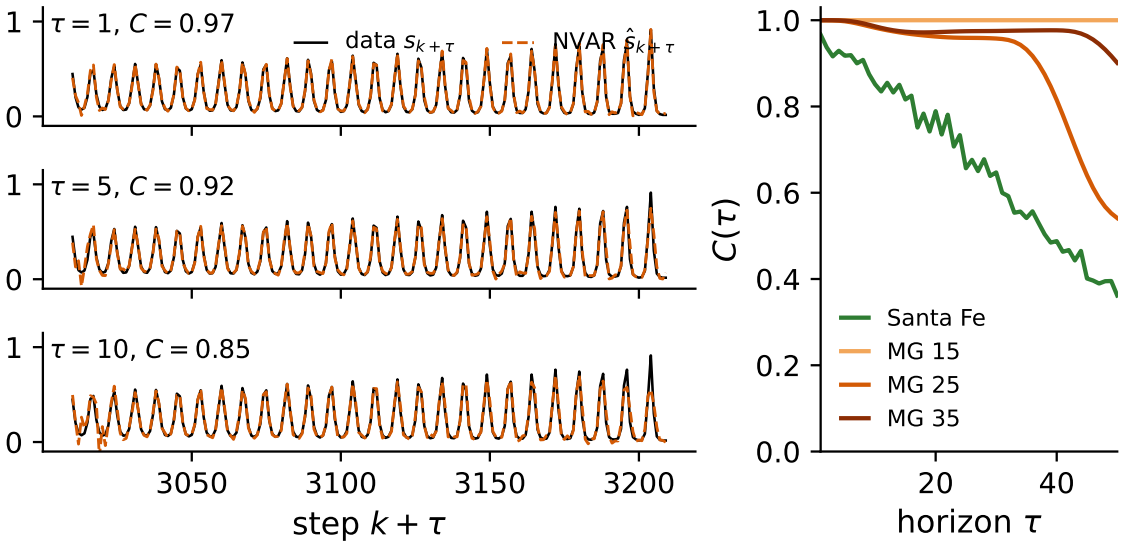


Figure 28: NVAR forecasts several steps ahead. Left, the laser record and the forecasts at the horizons $\tau = 1$, 5 and 10, each with its own readout. Right, the capacity $C(\tau)$ against the horizon for the laser and the Mackey–Glass signals.

On the laser, the validation block selects a window of 32 values, which gives $32 + 528 = 560$ non-constant features. The quadratic terms double the total capacity of AR, and the gain is largest at the longer horizons, where AR has lost most of its accuracy (Table 3). In the third row of Figure 22 the points remain near the diagonal also at the high pulses. A product such as $s_k s_{k-1}$ is large only when two successive samples are both large, which is how the model recognises a high pulse and anticipates the collapse that follows it. On the chaotic Mackey–Glass signals NVAR improves on AR in the same way (Table 7).

The heuristic of Section 7.1 fails where its premise fails. On the Lorenz coordinate NVAR reaches the same total capacity as AR (Table 7). The Lorenz field is quadratic in the full state (x, y, z), but the record measures only x, and the hidden coordinates are complicated functions of the window. A quadratic function of the state is then a function of the window of much higher complexity, which quadratic monomials of the window do not represent, and the advantage of NVAR is lost, as anticipated in the discussion of the one-step map.

8 Neural networks with learned features and memory

8.1 The multilayer perceptron on a window

NVAR prescribes its dictionary. When the hidden variables are complicated functions of the window, as for the Lorenz coordinate, a prescribed dictionary must grow rapidly to contain the needed observables, and a model that learns its features from the data becomes preferable. The networks are trained under the protocol of Section 5. Each reads a window of $L = 64$ values and returns all 50 horizons at once, $\hat{\mathbf{y}}_k = (\hat{s}_{k+1}, \ldots, \hat{s}_{k+50})$, a direct forecast with a shared set of features. The loss is the mean squared error over the 50 outputs and the fitting origins. The parameters are updated by the Adam optimiser on minibatches of 128 windows, the gradient is rescaled to norm at most 1 before every update (gradient clipping, explained in Section 8.2), and training stops when the validation loss has not improved for 20 epochs, with the parameters of the best epoch retained.

The multilayer perceptron (MLP) with one hidden layer of 64 units computes

$$\hat{\mathbf{y}}_k = W_2 \tanh(W_1 \mathbf{u}_k + \mathbf{b}_1) + \mathbf{b}_2, \tag{8.1}$$

with $W_1 \in \mathbb{R}^{64\times 64}$, $\mathbf{b}_1 \in \mathbb{R}^{64}$, $W_2 \in \mathbb{R}^{50\times 64}$ and $\mathbf{b}_2 \in \mathbb{R}^{50}$, in total 7410 parameters. The hidden activations $\tanh(W_1 \mathbf{u}_k + \mathbf{b}_1)$ play the role of the NVAR features, but they are learned. Each hidden unit computes one weighted combination of the window and bends it through the tanh, and the training chooses the combinations that help the forecast. On the laser the MLP matches NVAR (Table 3).

The memory of the MLP is its window, and the window matters more than the embedding dimension suggests. Shortening it from 64 to 8 values and then to 2 lowers the total capacity markedly (Table 6), although the laser passes the false-neighbour ratio criterion at three delay coordinates. That empirical result does not prove an exact reconstruction, even without noise, and intensity retains the sign ambiguity described earlier. With the 8-bit rounding of the laser record, a plausible explanation is that a longer history averages the measurement errors and separates states that a short window confuses. The price of a long window is a large first weight matrix, which grows with L, and the MLP treats every position of the window as a separate input with its own weights, although the dynamics that produced the values is the same at every step.

8.2 Recurrent networks

A fixed-size summary of the past can avoid assigning separate parameters to every position in a long window. The dynamical system that produced the record processes its past one step at a time with a fixed law, and it carries forward only its state. A recurrent neural network imitates this structure. It reads the record one value at a time and keeps a hidden state $\mathbf{h} \in \mathbb{R}^{n_h}$, a vector of n_h numbers that is updated with each new value by the same learned rule. The Elman network [25], the simplest such rule, starts from $\mathbf{h} = 0$, reads the window from the oldest value to the newest, and computes

$$\mathbf{h}_j = \tanh\left(W_h \mathbf{h}_{j-1} + W_x s_j + \mathbf{b}\right), \qquad j = k - L + 1, \dots, k, \tag{8.2a}$$

$$\hat{\mathbf{y}}_k = W_y \mathbf{h}_k + \mathbf{b}_y. \tag{8.2b}$$

Here $W_h \in \mathbb{R}^{n_h \times n_h}$ is the recurrent matrix, $W_x \in \mathbb{R}^{n_h}$ the input weights, $\mathbf{b}$ a bias, and $(W_y, \mathbf{b}_y)$ the readout applied to the last state.

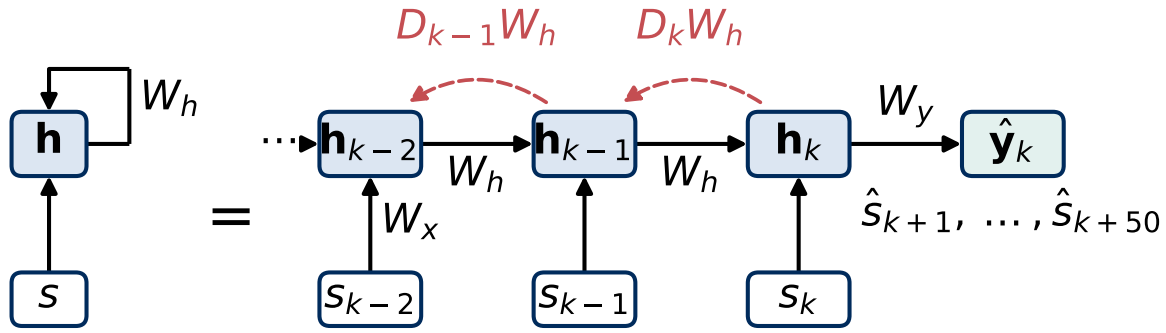


Figure 29: A recurrent network. Left, a cell with its recurrent connection. Right, the same cell unrolled in time with shared weights W_h, W_x and the final readout W_y. Only the last hidden state supplies the forecasts for all horizons. The dashed red arrows indicate the derivative factors used in training through time, developed below. Bias terms are omitted from the drawing.

Figure 29 shows the network unrolled over the window, one box per value of the record. Equation (8.2a) has two terms inside the tanh, and each has a role. The term $W_h \mathbf{h}_{j-1}$ carries forward what the network has retained from all earlier values, and the term $W_x s_j$ writes the new value into the state. The tanh mixes the two nonlinearly and keeps every component of the state between -1 and 1, so the state cannot grow without bound however long the record. After the last value the state $\mathbf{h}_k$ is a fixed-length summary of the whole window, and the readout (8.2b) is a linear map from this summary to the forecasts. The training decides what the summary contains, since the only demand on $\mathbf{h}_k$ is that the readout can produce good forecasts from it. Equation (8.2a) is itself a discrete-time dynamical system driven by the record, and its state is a learned counterpart of the delay vector of Section 4. The delay vector stores selected past values themselves, while the hidden state stores whatever functions of the past the forecast needs.

A network with a single unit shows how the state retains past input. With the update $h_j = \tanh(0.5 h_{j-1} + s_j)$, the initial state $h_0 = 0$ and the inputs $s_1 = 1$, $s_2 = 0$,

$$h_1 = \tanh 1 \approx 0.762, \qquad h_2 = \tanh(0.5 \cdot 0.762) \approx 0.363. \tag{8.3}$$

The newest input is zero, yet the state still reflects the earlier input, attenuated by the recurrent weight and by the saturation of tanh. Each further step multiplies the trace of

the old input by roughly the recurrent weight, so with a weight of 0.5 an input is practically forgotten after ten steps.

The same weights act at every step, so the number of parameters does not depend on the window length. For $n_h = 32$ the cell (8.2a) has $32 \cdot 32 + 32 + 32 = 1088$ parameters and the readout $50 \cdot 32 + 50 = 1650$, in total 2738, fewer than half the parameters of the MLP.

Training through time

The parameters are trained by gradient descent, and the gradient must be propagated through all steps of the recursion. Unrolling equation (8.2a) over the window yields the deep feed-forward network of Figure 29, whose layers share their weights, and the chain rule applied backwards along it is called backpropagation through time. The derivative of one state with respect to the previous one is

$$\frac{\partial \mathbf{h}_j}{\partial \mathbf{h}_{j-1}} = D_j W_h, \qquad D_j = \operatorname{diag}(1 - \mathbf{h}_j^2), \tag{8.4}$$

since $\tanh'(a) = 1 - \tanh^2 a$. The influence of the state m steps before the end on the last state is the product

$$\frac{\partial \mathbf{h}_k}{\partial \mathbf{h}_{k-m}} = D_k W_h \, D_{k-1} W_h \cdots D_{k-m+1} W_h, \tag{8.5}$$

and every gradient with respect to an early input or an early use of the weights contains such a product. The entries of each D_j lie in $(0, 1]$, so $\|\partial \mathbf{h}_k / \partial \mathbf{h}_{k-m}\|_2 \le \|W_h\|_2^m$.

In practice the gradient is not assembled from these products one by one. It is computed by a backward recursion that visits every step once. Let $\mathcal{L}$ be the loss of one window and write $\bar{\mathbf{h}}_j = \partial \mathcal{L} / \partial \mathbf{h}_j$ for the sensitivity of the loss to the hidden state at step j, including all the ways in which $\mathbf{h}_j$ influences later states. The last state enters only the readout (8.2b), so $\bar{\mathbf{h}}_k = W_y^\top \, \partial \mathcal{L} / \partial \hat{\mathbf{y}}_k$. Every earlier state enters only the next update, and the chain rule through equation (8.2a) gives

$$\bar{\mathbf{h}}_{j-1} = W_h^\top D_j \, \bar{\mathbf{h}}_j, \qquad j = k, k-1, \ldots, k-L+2. \tag{8.6}$$

The sensitivity flows backwards through the same network that the state flows forwards through, with the transposed matrix. Each step contributes to the gradient of the shared weights through the quantity $D_j \bar{\mathbf{h}}_j$, the sensitivity to the argument of the tanh at step j,

$$\frac{\partial \mathcal{L}}{\partial W_h} = \sum_j D_j \bar{\mathbf{h}}_j \, \mathbf{h}_{j-1}^\top, \qquad \frac{\partial \mathcal{L}}{\partial W_x} = \sum_j D_j \bar{\mathbf{h}}_j \, s_j, \qquad \frac{\partial \mathcal{L}}{\partial \mathbf{b}} = \sum_j D_j \bar{\mathbf{h}}_j, \tag{8.7}$$

summed over all steps of the window, because the same weights act at every step. Iterating equation (8.6) from the end of the window produces the products (8.5), so the contribution of step $k - m$ to the sums (8.7) carries the factor $\partial \mathbf{h}_k / \partial \mathbf{h}_{k-m}$. The cost of the backward pass equals that of the forward pass, but all hidden states of the forward pass must be stored, since equations (8.6) and (8.7) use them.

Equation (8.5) has the structure of the tangent propagator M_t of Section 3, a product of Jacobians along a trajectory, here the trajectory of the hidden state. Its growth rate is a finite-time Lyapunov exponent of the hidden dynamics. If the hidden dynamics contracts, the product decays exponentially in m and the distant past has no influence on learning, the vanishing-gradient problem [26]. If it expands, the product grows exponentially and the gradient explodes [27]. The same property that a forecaster needs, a state that remembers the past without amplifying it, is therefore also what makes the network trainable.

The growth rate, the slope of $\ln \|\partial \mathbf{h}_k / \partial \mathbf{h}_{k-m}\|_2$ against m, is measured for untrained networks with $n_h = 32$ whose recurrent matrix is a random orthogonal matrix scaled to spectral radius ϱ, so that all its singular values equal ϱ and the bound reads $m \ln \varrho$ (Table 4 and Figure 30). For $\varrho = 0.5$ the norm decays at about the rate of the bound. For $\varrho = 1$ the bound permits no decay, yet the norm still decays, because every unit driven away from zero has $\tanh' < 1$. For $\varrho = 1.5$ the bound allows strong growth, but the norm grows only slightly, because large states saturate the units and make the factors D_j small. The saturation of tanh thus protects against explosion only partly and makes vanishing gradients the more common failure.

Table 4: Rate per step at which $\ln \|\partial \mathbf{h}_k / \partial \mathbf{h}_{k-m}\|_2$ changes with m in recurrent networks with $n_h = 32$, compared with the bound $\ln \varrho$ for a scaled orthogonal recurrent matrix. The last row is the network trained on the laser with clipping.

recurrent matrix	measured rate	bound $\ln \varrho$
orthogonal, $\varrho = 0.5$	-0.755	-0.693
orthogonal, $\varrho = 1$	-0.146	0
orthogonal, $\varrho = 1.5$	+0.030	0.405
trained, spectral radius 1.40	-0.048	–

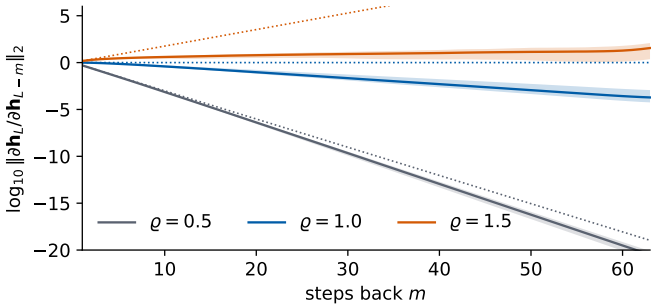


Figure 30: $\log_{10} \|\partial \mathbf{h}_k / \partial \mathbf{h}_{k-m}\|_2$ against the number m of steps back for untrained recurrent networks with $\varrho = 0.5$, 1 and 1.5. Shaded bands show the spread over random windows, and dotted lines the bound $m \log_{10} \varrho$.

Gradient clipping limits the damage of exploding gradients. Whenever the norm of the full gradient $\mathbf{G}$ exceeds a threshold θ_c, it is rescaled as $\mathbf{G} \to \mathbf{G}\, \theta_c / \|\mathbf{G}\|$, which keeps the direction and bounds the step. An experiment initialises the recurrent matrix at twice an orthogonal matrix, in the expanding regime, and trains three networks with and three without clipping at $\theta_c = 1$. Without clipping the gradient norms during training are more than a hundred times larger, the parameter updates overshoot, and the trained networks forecast far worse (Table 6). With the standard initialisation and clipping, the trained recurrent matrix has a spectral radius above one, and the Jacobian product first grows and then decays slowly (Table 4). Training places the hidden dynamics near the boundary between contraction and expansion, where information from early in the window can still reach the output. The trained RNN exceeds the MLP on the laser with fewer than half of its parameters (Table 3).

8.3 Long short-term memory

The difficulty of the Elman network lies in the structure of equation (8.5). Whatever the network retains has to pass through the same matrix W_h and the same saturating tanh at every step, so retained information and its gradient are multiplied by a similar factor over

and over, and a product of many similar factors either vanishes or explodes. The long short-term memory (LSTM) cell [28, 29] removes this structure. It adds a second vector, the memory $\mathbf{c}_j$, which is changed from step to step only by componentwise multiplication with numbers between 0 and 1 and by addition, and never passes through a weight matrix. Which components are kept, which are overwritten and which are shown to the rest of the network is decided at every step by gates, vectors with entries between 0 and 1 computed from the current input and the previous hidden state.

With the concatenated input $[\mathbf{h}_{j-1}, s_j]$, the logistic function $\sigma(a) = 1/(1+e^{-a})$ with values in $(0,1)$, and the componentwise product $\odot$, one step of the cell is

$$\mathbf{f}_j = \sigma\big(W_f[\mathbf{h}_{j-1}, s_j] + \mathbf{b}_f\big), \qquad \text{forget gate} \tag{8.8a}$$
$$\mathbf{i}_j = \sigma\big(W_i[\mathbf{h}_{j-1}, s_j] + \mathbf{b}_i\big), \qquad \text{input gate} \tag{8.8b}$$
$$\mathbf{g}_j = \tanh\big(W_g[\mathbf{h}_{j-1}, s_j] + \mathbf{b}_g\big), \qquad \text{candidate memory} \tag{8.8c}$$
$$\mathbf{c}_j = \mathbf{f}_j \odot \mathbf{c}_{j-1} + \mathbf{i}_j \odot \mathbf{g}_j, \qquad \text{cell memory update} \tag{8.8d}$$
$$\mathbf{o}_j = \sigma\big(W_o[\mathbf{h}_{j-1}, s_j] + \mathbf{b}_o\big), \qquad \text{output gate} \tag{8.8e}$$
$$\mathbf{h}_j = \mathbf{o}_j \odot \tanh(\mathbf{c}_j). \qquad \text{hidden-state output} \tag{8.8f}$$

The forecast is read from the last hidden state $\mathbf{h}_k$ by the readout (8.2b), as for the Elman network.

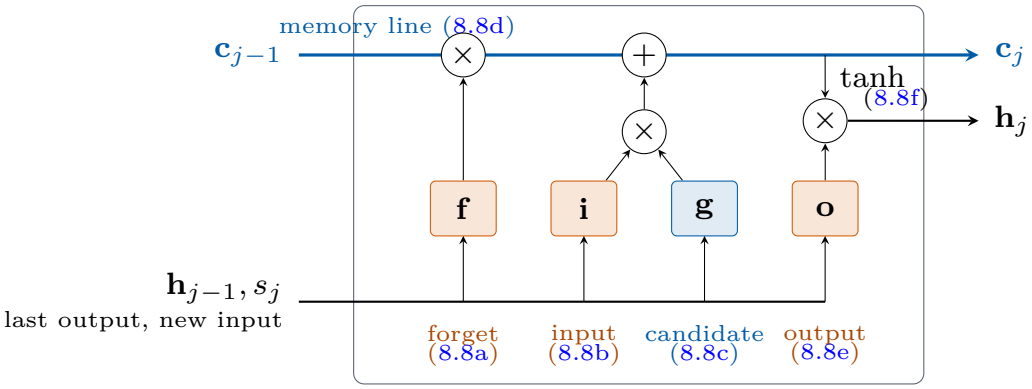


Figure 31: One step of the LSTM cell with the lines of equation (8.8). The memory $\mathbf{c}$ runs along the top line, where the forget gate (8.8a) multiplies it and the input gate (8.8b) adds a fraction of the candidate (8.8c), which together form the update (8.8d). The output gate (8.8e) decides how much of $\tanh(\mathbf{c}_j)$ appears in the hidden state (8.8f), which is passed to the next step together with the next value of the record.

The flow of information through one step can be followed in Figure 31 from the bottom to the top. The previous hidden state $\mathbf{h}_{j-1}$ and the new value s_j enter four affine maps with separate weights. Three of them pass through the logistic function and become the gates (8.8a), (8.8b) and (8.8e), each a vector of n_h numbers between 0 and 1, one per memory component. The fourth passes through the tanh and becomes the candidate (8.8c), the new content proposed for the memory, with entries between -1 and 1. The gates carry no content of their own. They only decide how strongly a content flows along a path.

On the memory line two operations take place, and equation (8.8d) contains both. The first term, $\mathbf{f}_j \odot \mathbf{c}_{j-1}$, is the forgetting. A component with forget value near 1 keeps its old memory, and one with a value near 0 erases it. The second term, $\mathbf{i}_j \odot \mathbf{g}_j$, is the writing. The input gate (8.8b) chooses the components into which the candidate (8.8c) is written and how strongly. Because the two terms are computed from the same current input, the cell

can recognise an event in the record, such as the collapse of a laser pulse, and respond by clearing some components and writing new content into them in the same step, while other components carry their content past the event unchanged.

The memory itself is not used directly for the forecast. The output gate (8.8e) decides which components of the memory are relevant at this step, and equation (8.8f) exposes them, squashed by tanh, as the hidden state $\mathbf{h}_j$. This hidden state has two roles. It is passed to the next step, where it enters all four affine maps and so controls the gates of the next step, and after the last value of the window it enters the readout. A component can therefore be stored in $\mathbf{c}$ for many steps while being hidden from the gates and the forecast, and be revealed when it becomes useful.

A single memory component shows the equations at work. Let the memory hold $c_{j-1} = 0.8$, and let the current input produce the forget value $f_j = 0.9$, the input value $i_j = 0.2$, the candidate $g_j = -0.5$ and the output value $o_j = 0.7$. Equations (8.8d) and (8.8f) give

$$c_j = 0.9 \cdot 0.8 + 0.2 \cdot (-0.5) = 0.62, \qquad h_j = 0.7 \tanh 0.62 \approx 0.386. \tag{8.9}$$

Most of the old content is kept and a small amount of new content of the opposite sign is added. Each of the four affine maps acts on $n_h + 1$ inputs, so the cell has $4(n_h(n_h + 1) + n_h) = 4352$ parameters for $n_h = 32$, and the forecaster with its readout has 6002.

The LSTM inside a forecaster

The cell (8.8) is one building block. Figure 32 shows where it sits in the network that produces the forecasts of the laser record. The input is the window of the last $L = 64$ values of the record, rescaled to $[0, 1]$. The values enter one at a time, the oldest first, into copies of the same cell with the same weights. Each copy receives one scalar s_j together with the two vectors $\mathbf{h}_{j-1}$ and $\mathbf{c}_{j-1}$ of 32 components each from the previous step, applies equations (8.8a) to (8.8f), and passes the updated $\mathbf{h}_j$ and $\mathbf{c}_j$ on. Both vectors start at zero before the oldest value. After the newest value s_k the hidden state $\mathbf{h}_k$ summarises the window, and a single linear layer, the readout (8.2b), maps its 32 components to the 50 forecasts $\hat{s}_{k+1}, \ldots, \hat{s}_{k+50}$. The memory $\mathbf{c}_k$ is not used by the readout. Its only role is to carry information along the window.

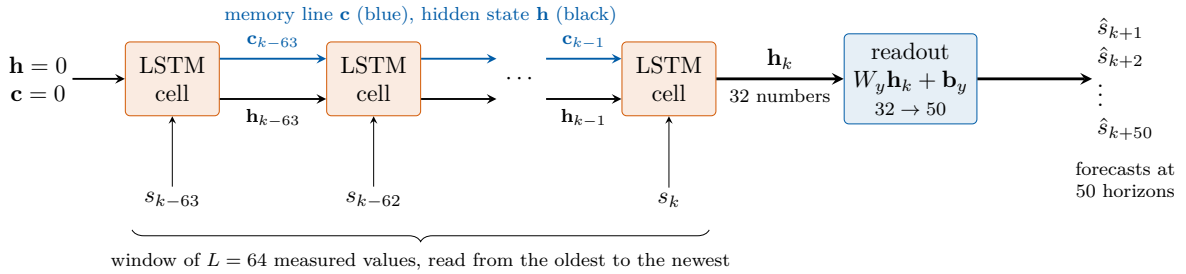


Figure 32: The LSTM forecaster of the laser record. The same cell of Figure 31, with the same weights, reads the window one value at a time and passes its memory $\mathbf{c}$ and hidden state $\mathbf{h}$ to the next step. The last hidden state $\mathbf{h}_k$ is mapped by a linear readout to the forecasts at all 50 horizons. In training, the forecasts are compared with the measured values $s_{k+1}, \ldots, s_{k+50}$, and the gradient flows back from the readout through all 64 copies of the cell.

Following one forecast through this network shows how the pieces cooperate. While the cell reads the early part of the window, its gates decide which features of the passing pulses are written into the memory, such as the height of recent maxima, which a forecast of the

next collapse could use. When it reaches the newest values, the output gate exposes the components of the memory that matter for what comes next, and the readout combines them into a forecast for each horizon. In training, the mean squared error of the 50 forecasts, averaged over the fitting windows, is differentiated with respect to all weights. The gradient enters through the readout, reaches $\mathbf{h}_k$, and is carried back along the window by a backward recursion, given below in equation (8.11), through every copy of the cell. Since the copies share their weights, each weight matrix collects contributions from all 64 steps. The Elman forecaster of Figure 29 has the same structure with a single state vector in place of the pair $(\mathbf{h}, \mathbf{c})$, and the GRU forecaster of the next subsection likewise.

The additive form of (8.8d) is what solves the problem of the Elman network. Holding the gates fixed, $\partial \mathbf{c}_j / \partial \mathbf{c}_{j-1} = \operatorname{diag}(\mathbf{f}_j)$, so along the memory line the gradient is multiplied only by forget values, with no weight matrix and no saturating derivative. A constant forget value f retains the fraction $f^m = e^{-m/\tau_{\text{mem}}}$ after m steps, with the memory time

$$\tau_{\text{mem}} = -\frac{1}{\ln f}, \tag{8.10}$$

9.5 steps for $f = 0.9$ and 99.5 steps for $f = 0.99$. The cell keeps information and its gradient alive over long times by holding its forget values near one, and it can choose this separately for each component and change it from step to step.

The backward recursion of the LSTM makes this explicit. With $\bar{\mathbf{c}}_j = \partial \mathcal{L} / \partial \mathbf{c}_j$ and $\bar{\mathbf{h}}_j$ as for the Elman network, the memory $\mathbf{c}_j$ influences the loss in two ways, through the hidden state (8.8f) of the same step and through the memory (8.8d) of the next step, so

$$\bar{\mathbf{c}}_j = \bar{\mathbf{h}}_j \odot \mathbf{o}_j \odot \left(1 - \tanh^2 \mathbf{c}_j\right) + \mathbf{f}_{j+1} \odot \bar{\mathbf{c}}_{j+1}. \tag{8.11}$$

The second term is the backward image of the memory line. It passes the sensitivity from step $j+1$ to step j multiplied only by the forget gate, while the first term adds what the hidden state at step j requires. From $\bar{\mathbf{c}}_j$ the chain rule through equation (8.8d) gives the sensitivities of the gates and of the candidate, $\bar{\mathbf{f}}_j = \bar{\mathbf{c}}_j \odot \mathbf{c}_{j-1}$, $\bar{\mathbf{i}}_j = \bar{\mathbf{c}}_j \odot \mathbf{g}_j$ and $\bar{\mathbf{g}}_j = \bar{\mathbf{c}}_j \odot \mathbf{i}_j$, and equation (8.8f) gives $\bar{\mathbf{o}}_j = \bar{\mathbf{h}}_j \odot \tanh \mathbf{c}_j$. Each of these is multiplied by the derivative of its activation function, $\sigma' = \sigma(1-\sigma)$ for the gates and $1 - \tanh^2$ for the candidate, which yields the sensitivity $\boldsymbol{\delta}_j^{(f)}$, $\boldsymbol{\delta}_j^{(i)}$, $\boldsymbol{\delta}_j^{(g)}$, $\boldsymbol{\delta}_j^{(o)}$ to the argument of each of the four affine maps. The weights then receive, as in equation (8.7),

$$\frac{\partial \mathcal{L}}{\partial W_f} = \sum_j \boldsymbol{\delta}_j^{(f)} \left[\mathbf{h}_{j-1}, s_j\right]^\top \tag{8.12}$$

and the same expressions with i, g and o. Finally $\bar{\mathbf{h}}_{j-1}$ collects the parts of $W_f^\top \boldsymbol{\delta}_j^{(f)} + W_i^\top \boldsymbol{\delta}_j^{(i)} + W_g^\top \boldsymbol{\delta}_j^{(g)} + W_o^\top \boldsymbol{\delta}_j^{(o)}$ that act on $\mathbf{h}_{j-1}$. These paths through the gates contain weight matrices and saturating derivatives and can vanish as in the Elman network, but the path (8.11) along the memory line does not, and it is enough for the sensitivity of a distant step to reach the gradient.

The initial value of the forget bias $\mathbf{b}_f$ decides whether this possibility is used from the start of training. A test with untrained networks driven by windows of 200 random values compares the sensitivity of the last state to the oldest and to the newest input (Table 5). In the Elman network and in an LSTM with $\mathbf{b}_f = 0$, whose forget values are near $1/2$, the oldest input has practically no influence. With $\mathbf{b}_f = 3$ the forget values are near 0.95, and the oldest input is as influential as the newest. A positive initial forget bias is therefore standard practice.

Trained on the laser, the LSTM with $n_h = 32$ has the slowest decay of capacity with the horizon and the highest total capacity of all models, and the spread over initialisations is

Table 5: Ratio of the sensitivity of the last state to the oldest and to the newest input, for untrained networks with $n_h = 32$ driven by windows of 200 random values.

network	sensitivity ratio
Elman	$10^{-22.4}$
LSTM, $\mathbf{b}_f = 0$	$10^{-29.6}$
LSTM, $\mathbf{b}_f = 3$	$10^{0.1}$

small (Table 3). Its capacity grows with the number of hidden units (Table 6). The trained network nevertheless makes little use of long memory. For each unit the forget value is averaged over all steps of the test windows, and the memory time $-1/\ln \bar{f}$ of this mean $\bar{f}$ has a median over the units of 2.5 steps on the laser and between three and four steps on the Mackey–Glass signals. These values describe only the direct retention of the memory line. Information can also persist through the hidden state and the gates, so they do not measure the full memory of the network. The comparison with the Elman network suggests that the advantage of the LSTM on these records lies less in long memory than in the gated, additive update, which makes the training better conditioned.

Table 6: Total capacity C_{tot} on the laser test block when one setting of a network is varied. For the clipping experiment the recurrent matrix of the Elman network is initialised at twice an orthogonal matrix, and the values are means over three initialisations, with the median gradient norm during training.

network	setting	C_{tot}	median gradient norm
MLP	window $L = 64$	35.3	
MLP	window $L = 8$	23.1	
MLP	window $L = 2$	18.1	
LSTM	$n_h = 4$	28.2	
LSTM	$n_h = 64$	45.6	
Elman, expanding initialisation	clipping at $\theta_{\mathrm{c}} = 1$	28.6	0.14
Elman, expanding initialisation	no clipping	8.3	19.23

8.4 Gated recurrent unit

The LSTM carries two vectors from step to step and uses four affine maps. The gated recurrent unit (GRU) [30] keeps the gated, additive update and simplifies the rest. It has a single state vector $\mathbf{h}_j$, which serves both as memory and as output, and two gates. In the convention implemented in the comparison notebook, the update gate retains the old state and the reset gate multiplies its recurrent linear transformation. One step is

$$\boldsymbol{\zeta}_j = \sigma\big(W_\zeta[\mathbf{h}_{j-1}, s_j] + \mathbf{b}_\zeta\big), \qquad \text{update gate} \tag{8.13a}$$

$$\mathbf{r}_j = \sigma\big(W_r[\mathbf{h}_{j-1}, s_j] + \mathbf{b}_r\big), \qquad \text{reset gate} \tag{8.13b}$$

$$\tilde{\mathbf{h}}_j = \tanh\big(W_x s_j + \mathbf{b} + \mathbf{r}_j \odot (W_h \mathbf{h}_{j-1})\big), \qquad \text{candidate state} \tag{8.13c}$$

$$\mathbf{h}_j = \boldsymbol{\zeta}_j \odot \mathbf{h}_{j-1} + (1 - \boldsymbol{\zeta}_j) \odot \tilde{\mathbf{h}}_j. \qquad \text{hidden-state update} \tag{8.13d}$$

The flow of information in Figure 33 parallels that of the LSTM. The update gate (8.13a) plays the roles of both the forget and the input gate of the LSTM, tied together. Equation (8.13d) is a componentwise interpolation between the old state and the candidate, so a component

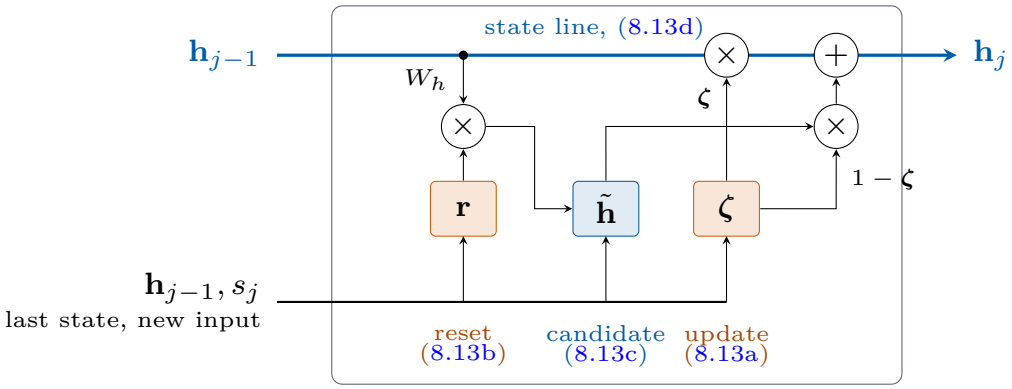


Figure 33: One step of the gated recurrent unit with the lines of equation (8.13). The state $\mathbf{h}$ runs along the top line. The update gate (8.13a) keeps the fraction $\boldsymbol{\zeta}$ of the old state and adds the fraction $1-\boldsymbol{\zeta}$ of the candidate (8.13c), as in (8.13d). The reset gate (8.13b) scales $W_h\mathbf{h}_{j-1}$ before it enters the candidate, which also receives the new value s_j from the input line.

with ζ near 1 is carried forward unchanged and one with ζ near 0 is replaced. Some texts use the complementary update gate. The original GRU applies the reset gate before the matrix, $W_h(\mathbf{r}_j \odot \mathbf{h}_{j-1})$, which is a slightly different unit. In the LSTM the forget and input gates are independent, and the memory can keep its content and add new content at the same time. In the GRU whatever is written displaces an equal fraction of what was there. The reset gate (8.13b) acts inside the candidate (8.13c). It decides how much of the old state is used to compose the new content. With $\mathbf{r}_j$ near 0 the candidate is computed from the new value alone, as if the history had been reset, which lets the unit start a new description of the record after an abrupt change. The GRU has no output gate, and the whole state is visible to the gates and to the readout at every step.

The interpolation (8.13d) gives the GRU the same protected path as the LSTM memory line. In the backward recursion,

$$\bar{\mathbf{h}}_{j-1} = \boldsymbol{\zeta}_j \odot \bar{\mathbf{h}}_j + (\text{terms through } \boldsymbol{\zeta}_j,\ \mathbf{r}_j \text{ and } \tilde{\mathbf{h}}_j), \tag{8.14}$$

where the bracket collects the paths through the gates (8.13a) and (8.13b) and the candidate (8.13c), each with a weight matrix and the derivative of its activation, as for the LSTM. The first term contains no weight matrix, so a component with an update gate near one passes both its content and its gradient through many steps. The gradients of the weights are again sums over the steps of the window of the sensitivity of each affine map times its input, as in equation (8.7). In all three recurrent architectures these recursions are carried out by automatic differentiation and are never coded explicitly, but their form explains which architectures can learn long dependencies. With three affine maps in place of four, the GRU cell has three quarters of the parameters of an LSTM cell of the same width. On the benchmark signals the GRU and the LSTM perform alike (Table 7), and neither is consistently ahead of the other in this comparison.

8.5 Comparison of all forecasting models

A common fitting and testing protocol lets us compare changes in model architecture on the same forecasting problem. Table 7 collects the resulting total capacities. Figure 34 shows the capacity as a function of the horizon. The figures also contain an echo state network (ESN), a recurrent network whose recurrent and input matrices are fixed at random values and whose

linear readout of the hidden state is the only fitted part, obtained by the ridge regression of Section 7.3. The desired state is determined by the input history after the influence of an arbitrary starting state decays, a condition called the echo state property. We omit the initial states from the readout fit to allow that transient to decay, the washout. In the comparison notebook, fitting begins at sample 63, so the preceding 63 states serve as washout. This finite exclusion needs checking against the reservoir's rate of forgetting. Like NVAR, the ESN supplies a fixed nonlinear dictionary with a linear readout. Fitting the readout leaves the stability of its recursive free run to be checked separately.

Table 7: Total capacity C_{tot} (maximum 50) on the test blocks of the benchmark signals. MG denotes Mackey–Glass with the delay τ_{MG} indicated. The MA model was not fitted to the Lorenz signal. ESN denotes an echo-state network, a recurrent network with fixed random recurrent weights in which only the linear readout is trained [20].

model	Lorenz x	MG 15	MG 25	MG 35	laser
persistence	8.4	19.5	13.3	11.8	5.5
AR	15.4	49.8	33.1	38.0	16.5
MA	–	7.1	7.2	7.4	4.7
NVAR	15.4	50.0	45.5	48.6	34.7
MLP	29.4	50.0	48.4	49.5	35.3
RNN	43.8	50.0	49.4	49.5	42.5
GRU	45.1	50.0	49.8	49.9	43.0
ESN	39.2	50.0	48.3	49.8	34.7
LSTM	43.8	50.0	49.9	49.9	44.3

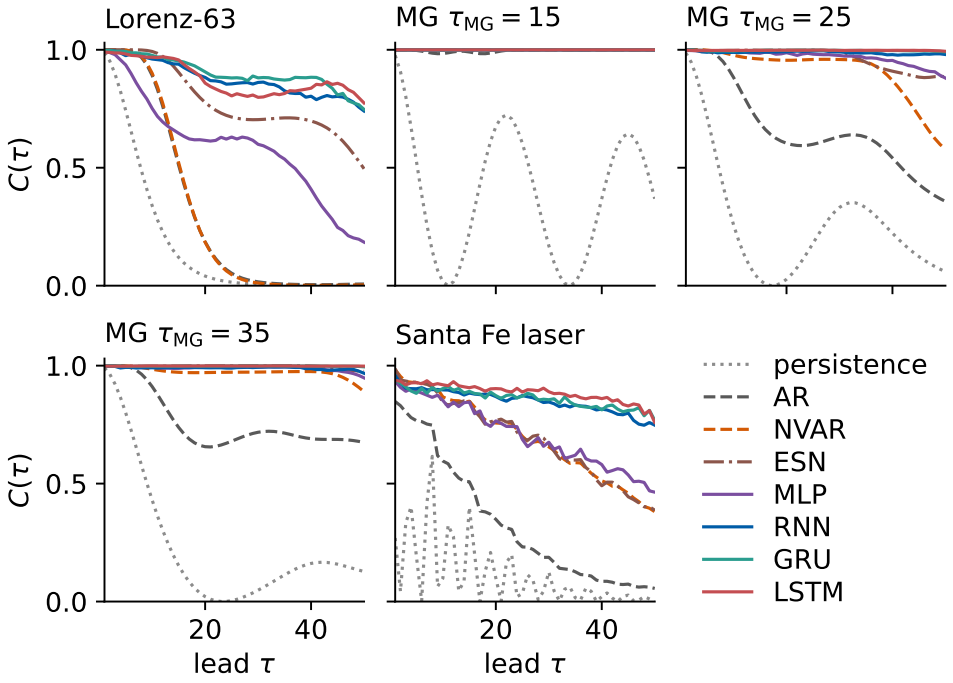


Figure 34: Capacity $C(\tau)$ against the horizon for all models on the five benchmark signals.

For the periodic Mackey–Glass signal, a linear model is already competitive, and every fitted model except MA reaches or nearly reaches the maximum of 50, while persistence stays far below it. The recurrence argument of Section 6 explains why a signal dominated by a few oscillatory components can be well approximated this way. For the chaotic signals the forecast

map is nonlinear, so the models separate according to how much of it they can represent. AR is lowest, NVAR and the MLP, with fixed or learned nonlinear features of a fixed window, are higher, and the recurrent networks, which in addition build a learned state, are highest. The separation is largest on the Lorenz coordinate, where the hidden variables are complicated functions of the window and the quadratic dictionary of NVAR does not apply. There the recurrent networks reach about 44 while AR and NVAR stay at 15.4.

The direct forecasts are evaluated on true past values, the inputs on which the models were trained. A free run tests a different property. From the origin $k_0 = 2999$ every model predicts one step ahead, appends the prediction to its window and repeats for 200 steps (Figure 35). The number of steps during which the error stays below a fifth of the training range is given in the last column of Table 3. The LSTM stays close for 110 steps, compared with 47 for the echo-state network, the next longest. The AR free run relaxes to a small regular oscillation, as expected for a linear map whose leading eigenvalues lie close to the unit circle. The NVAR free run follows the pulses for a while and then leaves the range of the data altogether, because its quadratic feedback is unstable away from the data. A model trained only on true inputs, a procedure called teacher forcing, meets its own errors for the first time in a free run, and nothing in its training controls its response to them. This mismatch between training and use is called exposure bias. The free-run ranking comes from one origin and one initialisation, and chaos amplifies small differences between models, so it is far less robust than the ranking of the direct forecasts.

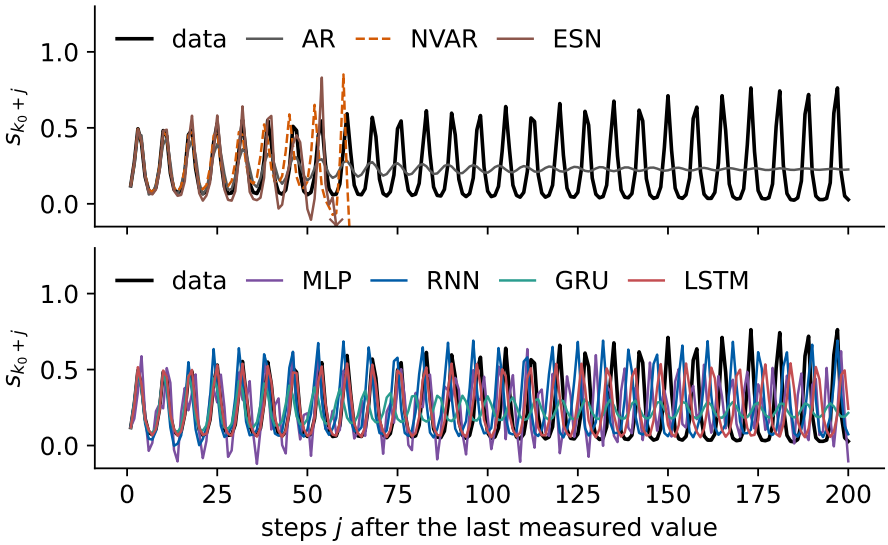


Figure 35: Free runs on the laser from the origin $k_0 = 2999$, the last sample before the test block. Top, AR, NVAR and ESN. Bottom, the networks. The data are shown in black, and crosses mark where a curve leaves the plotted range.

The window-based neural free run uses the already trained 50-output network. At each step it keeps only output 1, appends that prediction, drops the oldest window value and evaluates the shifted window again. Each RNN or LSTM call starts from zero hidden state (and zero LSTM cell state), then rereads the whole window; the hidden state is not carried between these calls. The ESN instead continues its reservoir state from the forecast origin. These choices specify the recursive experiment and distinguish it from a continuously stateful recurrent forecaster.

9 Forecasting measured records from nature

The benchmark signals of the preceding sections were chosen because their origin is known. The Lorenz and Mackey–Glass signals are solutions of known equations, and the laser record comes from a system whose law has few variables and is known in principle. Most records that a scientist meets are of a different kind. A weather station records the daily temperature, oceanographers record monthly indices of the sea surface temperature such as the El Niño index of the tropical Pacific, solar observatories count sunspots, the Mauna Loa observatory records the carbon dioxide content of the atmosphere, hydrologists record the discharge of rivers, physiologists the electrical activity of the heart and the brain, and ecologists the numbers of animals, as in the century-long fur records of the Hudson's Bay Company for the Canadian lynx. Several of these records have a long history in the subject. Yule introduced the autoregressive model in 1927 to describe the sunspot number [31], and the sunspot and lynx records have served as test cases for forecasting methods ever since.

Such records differ from the benchmarks in four respects. No law of motion is established for the quantity that is measured, or the law involves far more variables than are recorded, so the relevant state is unknown. The record is affected by influences that are external to the system as far as the record is concerned, such as the weather systems that pass over a temperature station. The statistics of the record change in time, through a seasonal cycle, a trend or changes of the instruments. Finally, a record contains only one realisation of the process, which cannot be repeated with other initial conditions.

The procedure of this chapter applies to such records without change, and its diagnostics now carry the main weight. Before any model is fitted, the record is described, the parts of its variation that are understood, such as the seasonal cycle, are separated from the rest, and the autocorrelation, the power spectrum and the embedding criteria of Section 4 are used to judge whether a low-dimensional deterministic description is plausible. Simple reference forecasts that need no model of the system then set the standard that a learned model must beat. We apply the procedure to two records at opposite ends of the range, the daily temperature in central England, which is dominated by unresolved weather, and the monthly sunspot number, which is dominated by a nearly regular cycle.

9.1 Temperature in central England

Temperature varies through seasonal forcing, weather fluctuations and long-term changes. A forecast must improve on the seasonal pattern before its skill can be attributed to information in the recent observations. The Central England record provides a measured example in which climatology and persistence have distinct roles.

The data and the question

The Central England Temperature series of the Met Office Hadley Centre is one of the longest instrumental records [32, 33]. It gives the daily mean temperature in °C, representative of a triangular region between Lancashire, London and Bristol, from 1 January 1772 to the present, without missing values (Table 8). The mean over the last thirty years is more than a degree higher than over the first thirty, the warming of the climate seen in a single record. The forecasting experiments use the last thirty years, from 1 January 1996, and ask how well the temperature of the coming days can be forecast from the record of the preceding days.

The answer is expressed through the autocorrelation (2.36) and two stochastic reference mod-

els. The simplest stochastic model with memory is the AR(1) process $s_k = \phi s_{k-1} + \xi_k$, with independent innovations ξ_k of zero mean and variance σ^2. Each value retains the fraction ϕ of the previous one and receives a new random kick. For $|\phi| < 1$ the process has a stationary solution, whose variance follows from $\mathrm{var}(s) = \phi^2 \mathrm{var}(s) + \sigma^2$ as $\sigma^2/(1-\phi^2)$, and its autocorrelation $\rho(m) = \phi^m$ decays geometrically. The random walk $s_k = s_{k-1} + \xi_k$, the case $\phi = 1$, keeps every past kick and has a variance that grows linearly with time. A record whose autocorrelation decays geometrically is consistent with the AR(1) process, and its future beyond a few correlation times is then as uncertain as a value drawn at random from the record.

Table 8: The Central England Temperature record. Statistics of the climatology and the anomaly refer to the forecasting period from 1996.

quantity	value
full record	1 January 1772 to 26 October 2026, 93 071 days
mean temperature 1772–1801 and 1996–2025	9.13 °C and 10.45 °C
forecasting period	from 1 January 1996, 11 257 days
test block	27 August 2020 to 26 October 2026, 2232 origins
climatology, mean and annual amplitude	10.25 °C, 6.30 °C
variance of the temperature explained by the climatology	74%
standard deviation of temperature and anomaly	5.19 °C and 2.64 °C
autocorrelation of the anomaly at 1, 2, 5, 10, 20 days	0.784, 0.560, 0.257, 0.087, 0.034
mutual information of the anomaly at $\kappa = 0$, first minimum	2.040 nats, 0.011 nats at $\kappa = 12$ days
ratio test of false neighbours below 5%	$m = 4$
combined fraction of false neighbours at $m = 4$ and $m = 8$	0.169 and 0.25

Climatology, anomaly and two reference forecasts

Daily temperature is dominated by the seasonal cycle, which is understood and needs no forecasting model. The climatology is the expected temperature for the day of the year, modelled as a mean with an annual and a semi-annual harmonic fitted by least squares on the fitting block,

$$c(t) = a_0 + \sum_{n=1}^{2} \Big[a_n \cos \frac{2\pi n t}{365.25} + b_n \sin \frac{2\pi n t}{365.25} \Big], \tag{9.1}$$

with t in days. The anomaly $a_k = T_k - c(t_k)$ is the departure of the temperature T_k from the climatology, the part that weather forecasting concerns. The climatology explains three quarters of the variance of the daily temperature, and the anomaly that remains has a short memory. Its autocorrelation falls below $1/e$ after four days and is negligible after ten (Table 8).

Meteorology judges forecasts against two references that require no model of the atmosphere, and their errors follow from the autocorrelation. Assume a stationary anomaly with zero mean, variance σ_a^2 and autocorrelation $\rho(\tau)$. The climatology forecast $\hat{T}_{k+\tau} = c(t_{k+\tau})$ has the error $a_{k+\tau}$, so its mean squared error is σ_a^2 at every horizon. The persistence forecast $\hat{T}_{k+\tau} = T_k$ has, over a few days in which the climatology hardly changes, the error $a_{k+\tau} - a_k$ and

$$\mathrm{MSE}_{\mathrm{pers}} \approx \langle (a_{k+\tau} - a_k)^2 \rangle = 2\sigma_a^2 - 2\langle a_k a_{k+\tau} \rangle = 2\sigma_a^2 (1 - \rho(\tau)). \tag{9.2}$$

Persistence is better than climatology while $\rho(\tau) > 1/2$. It uses the present weather and ignores the season, and climatology does the reverse. The two are combined optimally by

damped persistence.

Damped persistence. *Among forecasts of the form* $\hat{a}_{k+\tau} = b\,a_k$*, the mean squared error* $\sigma_a^2(1 - 2b\rho(\tau) + b^2)$ *is minimal at* $b = \rho(\tau)$*. The forecast* $\hat{T}_{k+\tau} = c(t_{k+\tau}) + \rho(\tau)a_k$ *has* MSE $= \sigma_a^2(1 - \rho(\tau)^2)$*, which is never larger than the error of climatology or of persistence of the anomaly.*

Derivation. Expanding the square gives $\langle (a_{k+\tau} - b a_k)^2 \rangle = \sigma_a^2 - 2b\sigma_a^2\rho + b^2\sigma_a^2$, whose derivative in b vanishes at $b = \rho$. Inserting $b = \rho$ gives $\sigma_a^2(1-\rho^2) \le \sigma_a^2$, and $2(1-\rho)-(1-\rho^2) = (1-\rho)^2 \ge 0$ shows that it does not exceed the persistence error. □

Damped persistence starts from the present anomaly and lets it relax towards the climatology as the horizon grows, at the rate set by the autocorrelation. For an AR(1) anomaly, $\rho(\tau) = \phi^\tau$ and damped persistence is the conditional mean of $a_{k+\tau}$ given the past, the best forecast in the mean-square sense. Persistence and damped persistence of the anomaly differ by the factor $\rho(\tau)$ and have the same capacity, since a correlation is unchanged by scaling, while their errors differ. On the temperature record the root-mean-square error (RMSE) is therefore reported alongside C.

Diagnostics and forecasts

The diagnostics of Section 4 give a picture opposite to that of the laser (Table 8). The ratio test of false nearest neighbours falls below 5% at four coordinates, as for a low-dimensional attractor, but the combined fraction is high and grows with m, the behaviour found for independent Gaussian values in Table 2. The record of a single location is inconsistent with a low-dimensional deterministic reconstruction at these settings. This does not prove that the atmosphere is stochastic. The temperature at one place depends on weather systems extending over thousands of kilometres that the local record does not contain, and on this record their effect acts as unresolved degrees of freedom, the epistemic randomness of Section 1.

The forecasts use the protocol of Section 5 with 60% of the days for fitting, 20% for validation and 20% for testing, and horizons from 1 to 20 days, so that C_{tot} has the maximum 20. Table 9 first shows a trap. On the raw temperature every forecast scores highly, and climatology, which uses no measured value at all, scores highest, with a capacity of about 0.74 at every horizon, the fraction of the variance explained by the season. These values measure the seasonal cycle and say nothing about forecasting the weather. A model fitted to the raw temperature must first learn the season, and its score mixes the two.

Table 9: Total capacity C_{tot} (maximum 20) of the forecasts on the temperature test block, horizons 1 to 20 days. The raw temperature was forecast by climatology, persistence and AR only.

target	climatology	persistence	AR	MA	ARMA	NVAR	MLP	RNN	LSTM
temperature	14.887	11.537	12.737	–	–	–	–	–	–
anomaly	–	1.280	1.244	1.121	1.237	1.298	1.235	1.256	1.238

On the anomaly all models score nearly the same, and the scores are small. The capacity one day ahead is close to $\rho(1)^2 = 0.784^2 = 0.615$ for every model, and $C(\tau)$ follows $\rho(\tau)^2$ to zero within about ten days. No model extracts predictable structure beyond the short autocorrelation of the record. The errors tell the same story (Table 10). Persistence is good for one day and poor afterwards, climatology has the same error at every horizon, and

damped persistence, which combines them, is as good as the best learned model at every horizon. The nonlinear models and the networks therefore add almost nothing to the optimal linear combination of the two references. NVAR was selected for this comparison because it had the largest anomaly capacity on the test block, which makes the comparison exploratory.

Table 10: Root-mean-square error in °C of temperature forecasts on the test block. The NVAR forecast is the anomaly forecast added to the climatology, and damped persistence uses $\rho(\tau)$ estimated on the fitting block.

forecast	1 day	5 days	20 days
persistence	1.79	3.36	4.19
climatology	2.8	2.8	2.8
damped persistence	1.69	2.68	2.79
NVAR on the anomaly	1.67	2.67	2.77

The contrast with the laser does not mean that the methods failed. Their success on the laser depended on the existence of a low-dimensional deterministic law observed in the record, which the diagnostics detected there and do not detect here. Weather services forecast the temperature successfully for about a week, but they do so with a model of the whole atmosphere and with measurements from the whole globe.

9.2 The sunspot number

The sunspot record supplies a second measured test with a much longer characteristic cycle. Its activity rises and falls fairly regularly, while cycle amplitude and timing vary. Forecasting from a window can use the phase of this cycle, and the longer horizons test the limits of that information.

Counting sunspots

The visible surface of the Sun carries dark patches, sunspots, where strong magnetic fields suppress the flow of heat from the interior, so that the gas there is cooler and emits less light than its surroundings. Their number rises and falls with a period of about eleven years, the solar cycle. At the maximum of a cycle the Sun shows hundreds of spots and produces frequent flares and eruptions of magnetised plasma, which disturb radio communication, satellite navigation and power grids on the Earth. At the minimum the disc can be free of spots for weeks. The cycle is generated by the dynamo of the Sun, the motion of the electrically conducting plasma that amplifies and reverses the magnetic field, and no model of the dynamo forecasts the height of the next cycle reliably.

Since 1848 solar activity has been measured by a count introduced by Rudolf Wolf, who also reconstructed it back to 1749 from the drawings and notes of earlier observers. An observer counts the groups of spots g and the individual spots f and forms

$$R = 10\,g + f. \tag{9.3}$$

The factor 10 gives a group about the weight of ten single spots, so that R follows the overall activity and depends less on how many small spots the telescope resolves. A single spot also counts as a group, so three groups with three, five and one spots give $R = 10 \cdot 3 + 9 = 39$. The counts depend on the instrument and the observer, and the international sunspot number of

the World Data Center SILSO at the Royal Observatory of Belgium combines the observations of a worldwide network of stations with a scaling factor for each station. The series was recalibrated in 2015, and its version 2.0 removes several inconsistencies of the older series [34].

One sample of the record used here is the monthly mean of the daily values, with $\Delta t = 1$ month (Table 11). The quality of the record changes in time. For its oldest part the station statistics are unknown, while in recent years hundreds of observations enter each monthly value. The values are never negative, and their distribution is skewed towards large values, because months near the minima are the most frequent and very high values occur only at the peaks of strong cycles.

Figure 36 shows the measured signal at three time scales. The full record reveals changes in cycle amplitudes over centuries. The last forty years separate the shapes of successive cycles, and the last five years resolve the month-to-month variation within the most recent rise and peak. These are the monthly observations without smoothing.

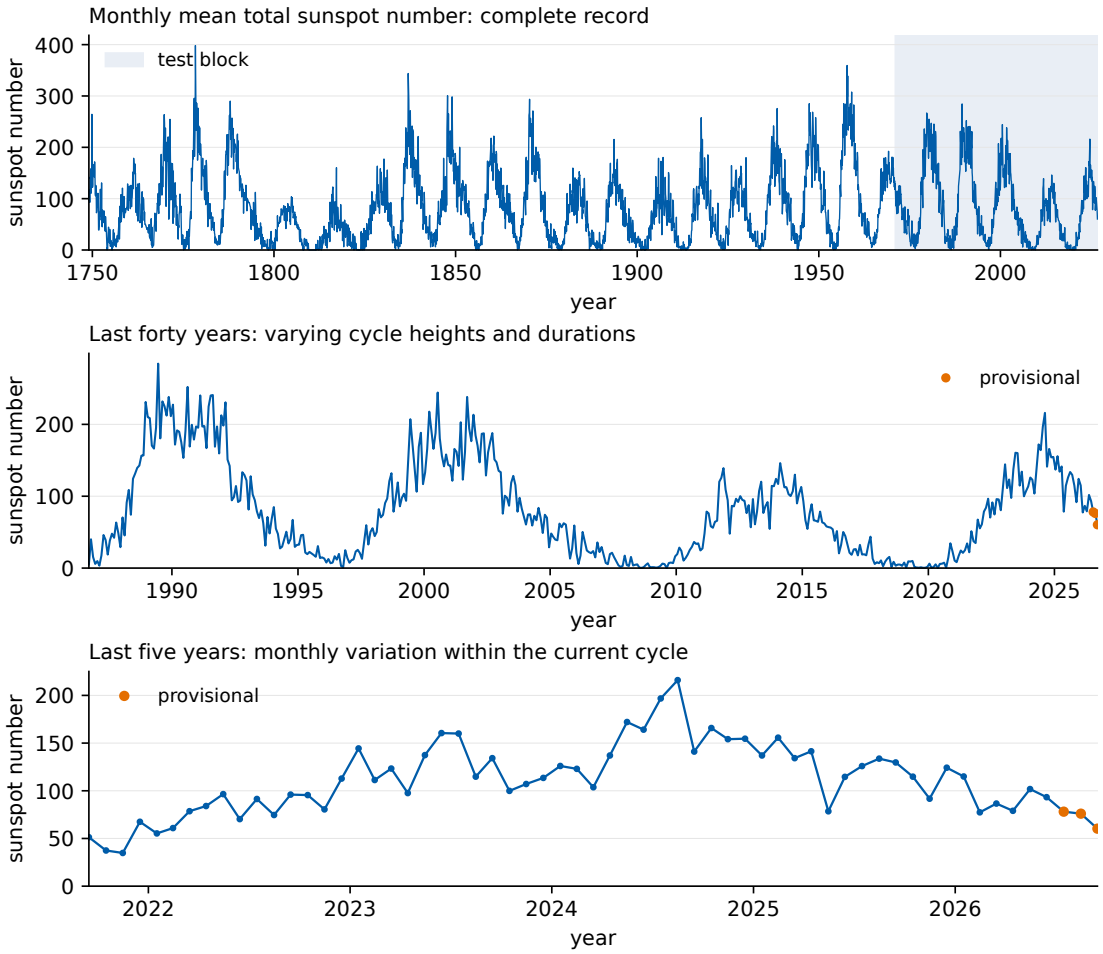


Figure 36: Monthly SILSO sunspot number in the complete record, the last forty years and the last five years. Shading marks the test block used for forecasting, and orange points mark the three provisional months in the stored data snapshot. All panels use the same record as the forecasting notebook.

Table 11: The monthly sunspot number (SILSO version 2.0).

quantity	value
record	January 1749 to September 2026, 3333 months, about 25 cycles
months without station statistics	828 oldest months
observations per monthly value in the last year	736 to 1371
mean, standard deviation, skewness	82.2, 67.5, 0.91
maximum	398.2 (1778)
months without any spot	67
autocorrelation at 1 month, 1 year, 20 months	0.918, 0.724, 0.495
autocorrelation minimum and second maximum	-0.443 at 5.25 years, 0.579 at 10.67 years
largest periodogram peaks in the 8–15 year band	11.11, 10.68, 9.92, 11.57 years
variance in periods between 8 and 15 years	64%
first minimum of the mutual information	$\kappa = 34$ months
ratio test of false neighbours below 5%	$m = 4$
combined fraction of false neighbours, $m = 4$ to 8	0.064 to 0.093
test block	1971 to 2026, five cycles

A nearly periodic record

A strictly periodic signal of period P has an autocorrelation that oscillates with the same period and never decays, since the signal one period later is the same signal. The sunspot record is different. Its autocorrelation stays high over the first year, turns negative at half a cycle and returns to a second maximum after one cycle, but this maximum is well below one (Table 11). The oscillation shows the cycle, and the decay of its maxima shows that the cycles differ in length and height. The phase of the cycle is gradually forgotten, so the record is nearly periodic over a few cycles and irregular over many.

The power spectrum makes the same point in frequency. A record of about 278 years resolves periods near eleven years only to a few months, and within this resolution the periodogram shows a cluster of peaks between ten and twelve years instead of a single line. The cycle has no single sharp period, and periods between 8 and 15 years carry about two thirds of the variance. A broadened peak of this kind is the spectral signature of an oscillation whose phase drifts, in contrast to the single line of a clock.

The delay embedding of Section 4 selects a delay of about a quarter of the cycle. For an oscillation a quarter period is the natural choice, since $\sin(\omega t)$ and $\sin(\omega t - \pi/2) = -\cos(\omega t)$ are as different as two values of one oscillation can be, and together they fix its phase. Four coordinates satisfy the ratio test, so the embedding window spans $(4 - 1) \cdot 34 + 1 = 103$ months, about four fifths of a cycle. The combined fraction of false neighbours lies between the few percent of the Lorenz record and the high values of independent noise and of the temperature anomaly. The sunspot record therefore has a strong deterministic component, the cycle, together with irregularities that four delay coordinates do not resolve.

Forecasts of the sunspot number

The forecasts follow the protocol of Section 5 with 60% of the months for fitting, 20% for validation and 20% for testing, horizons from 1 to 20 months and windows of at most 20 months. The test block covers the five cycles since 1971.

Table 12: Sunspot forecast capacity $C(\tau) = r(\tau)^2$ at representative horizons in months and total capacity $C_{\mathrm{tot}} = \sum_{\tau=1}^{20} C(\tau)$. All eight models use the same 648 test origins. Settings are selected before testing, with input spans at most twenty months. The primary NVAR uses five values spaced four months apart, so its inputs are strided. Values come from the executed sunspot notebook.

model	$C(1)$	$C(5)$	$C(10)$	$C(20)$	C_{tot}
persistence	0.872	0.763	0.632	0.268	11.740
AR	0.893	0.812	0.734	0.582	14.461
MA	0.849	0.304	0.000	0.000	3.032
ARMA	0.887	0.781	0.633	0.255	11.811
NVAR	0.882	0.801	0.739	0.609	14.636
MLP	0.891	0.829	0.770	0.663	15.364
RNN	0.880	0.823	0.778	0.648	15.179
LSTM	0.880	0.823	0.768	0.677	15.287

Figure 37 shows the complete horizon dependence behind the table. Persistence retains high correlation at short horizons, while AR, NVAR and the neural networks retain more of the signal's variation as the horizon grows. MA returns to a constant forecast after its finite innovation window, giving zero capacity under the scoring convention. Small differences among the stronger models on this single chronological split do not establish a ranking across independent runs.

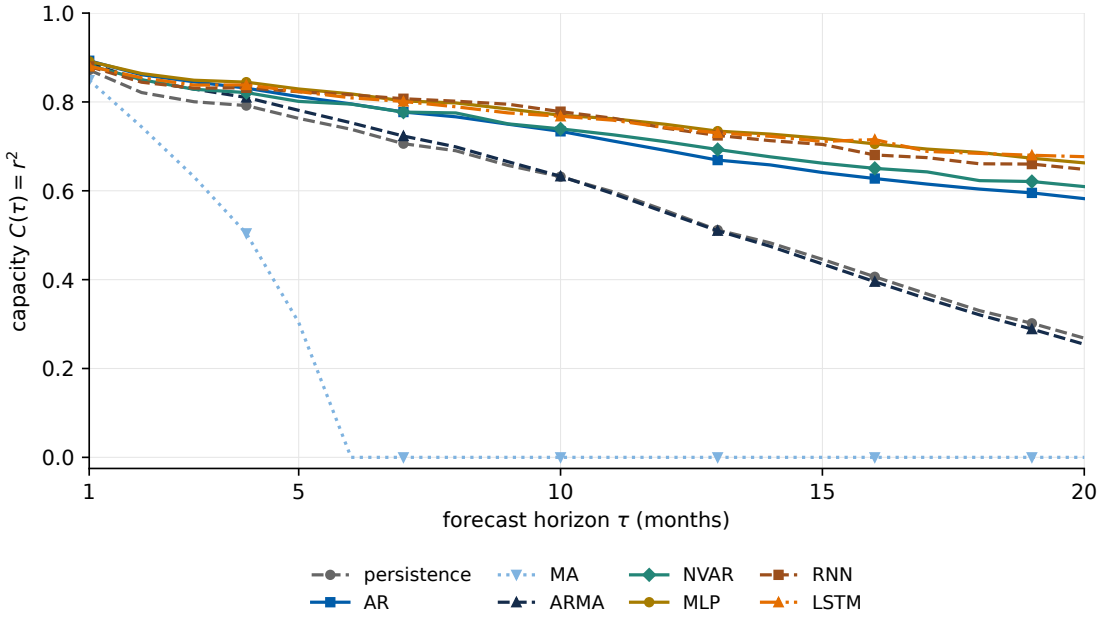


Figure 37: Sunspot forecast capacities for all eight models at horizons from one to twenty months, evaluated on the same 648 test origins. The curves and Table 12 use the same stored predictions and score arrays. Capacity measures squared correlation and does not by itself measure amplitude accuracy.

One month ahead all eight models have squared correlations between 0.85 and 0.9, which is what the autocorrelation at one month allows, since the activity changes little within a month. The models separate at longer horizons (Table 12). Twenty months ahead, the squared correlations are about a quarter for persistence, three fifths for AR and NVAR, and

two thirds for the networks. These fractions describe the variance explained after optimal affine recalibration of each forecast. The last value alone does not say whether the activity is rising or falling, while a window of 20 months shows the slope and the curvature, and a model that reads them continues the cycle. The ARMA(1,1) model selected on the validation block adds almost nothing to persistence, and the MA(5) forecast returns to the mean after five months, so its capacity is zero from $\tau = 6$ on.

The errors depend on the phase. For the best model, the MLP, the mean absolute error at 20 months is almost half as large again when the cycle rises during the year before the target month than when it falls, 38.6 against 26.4. The rise of a new cycle and the height it will reach are the hardest part of the forecast, which is the same difficulty that the dynamo models face. A few forecasts are negative, all near the minima. Squaring a forecast of $\sqrt{R}$ gives a nonnegative result. A forecast z of $\ln(R+1)$ transforms back to $e^z - 1$, which is nonnegative only if $z \geq 0$; that route also needs an output constraint or clipping. Transforming the target changes the loss and the conditional quantity being estimated.

Delay diagnostics can suggest forecast inputs, but their settings must be chosen without the test record. Repeating MI and FNN on the fitting block alone selects $\kappa = 23$ months and $m = 4$, a span of 70 months. The full-record values $\kappa = 34$ and $m = 4$ describe the record and are not used to select the inputs. With these four values as input, NVAR raises its total capacity from 14.636 to 15.345 and comes close to the networks, while AR changes from 14.461 to 14.306, both with the ridge penalty chosen on the validation block. The diagnostics thus propose a window, and only the forecast on the validation block decides whether it helps.

10 Neural ordinary differential equations

Forecast maps predict measured values at selected horizons. When the experiment records the full state, we can instead fit a continuous law whose integration supplies trajectories from different initial conditions. This changes both the training problem and the physical conclusions that a learned model can support.

10.1 Learning the vector field

Section 2 described a deterministic system by its law of motion $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$. When the vector field $\mathbf{F}$ is known, everything about the dynamics follows from it. Solving the equation, analytically where possible and numerically otherwise, gives the trajectory from any initial state, forward and backward in time and at any time, including the times between the samples of an experiment. The fixed points are the zeros of $\mathbf{F}$, their stability follows from the eigenvalues of its Jacobian, the divergence tells whether volumes in phase space shrink, and the Lyapunov exponents and the attractor follow from long integrations. This is the knowledge that Laplace's ideal in Section 1 describes, and it answers both the predictive and the explanatory question of the chapter.

In many problems the law is not known, while the state can be measured. A pendulum with an unknown friction can be filmed, so that its angle and angular velocity are recorded at many times, a chemical reaction can be followed by measuring the concentrations of all its species, and a simulation can store its full state at every step. The data are then samples of trajectories in phase space, and the vector field that generated them is unknown. We learn $\mathbf{F}$ from such data by modelling it as a neural network $\mathbf{F}_{\boldsymbol{\Theta}}$ with parameters $\boldsymbol{\Theta}$, and then to use the learned field in place of the unknown one.

Definition 10.1 (Neural ODE)**.** A neural ordinary differential equation is a dynamical system

$$\dot{\mathbf{x}} = \mathbf{F}_{\boldsymbol{\Theta}}(\mathbf{x}) \tag{10.1}$$

whose vector field is a neural network with parameters $\boldsymbol{\Theta}$ [35]. Its prediction from an initial state $\mathbf{x}_0$ at time t is the solution

$$\hat{\mathbf{x}}(t) = \mathbf{x}_0 + \int_0^t \mathbf{F}_{\boldsymbol{\Theta}}\big(\hat{\mathbf{x}}(t')\big)\, dt', \tag{10.2}$$

computed by a numerical solver.

The network receives a state and returns a velocity, a vector of the same dimension. A neural ODE makes one structural assumption, that the measured variables form an autonomous first-order system, so that the measured state determines its own velocity. Everything else about the law is learned. The difference from the forecasting models of the previous sections is the object that is learned. A forecast map G_τ maps a window to a value at one fixed horizon and is tied to the sampling interval of the training data. A one-step forecast combined with the window shift defines a discrete dynamical system, whose fixed points and Jacobian eigenvalues can be studied, as can those of a learned full-state map. They describe the evolution over one sampling interval, and the instantaneous vector field follows from them only indirectly. A direct forecast at one horizon does not by itself specify the dynamics between samples. A learned vector field provides instantaneous velocities and divergence, and a solver evaluates its trajectories at arbitrary times.

In this training protocol the network is compared with observed states through the integrated solution (10.2). The target data contain no full derivative vectors. For the pendulum example, both angle and angular velocity are observed state components, while the angular acceleration is absent. The network predicts the pair $(\dot{\theta}, \dot{\omega})$ and learns it through trajectory agreement. Training therefore requires a numerical solver inside the loss, and the properties of that solver matter for what is learned.

Figure 38 shows the available data and the object to be inferred. The observation points occupy only part of the phase plane. The network supplies a derivative vector at any queried state, and integrating these vectors predicts motion from an initial condition. Comparing that motion with a held-out trajectory tests the learned field where data are available. Arrows outside the observed region remain extrapolations.

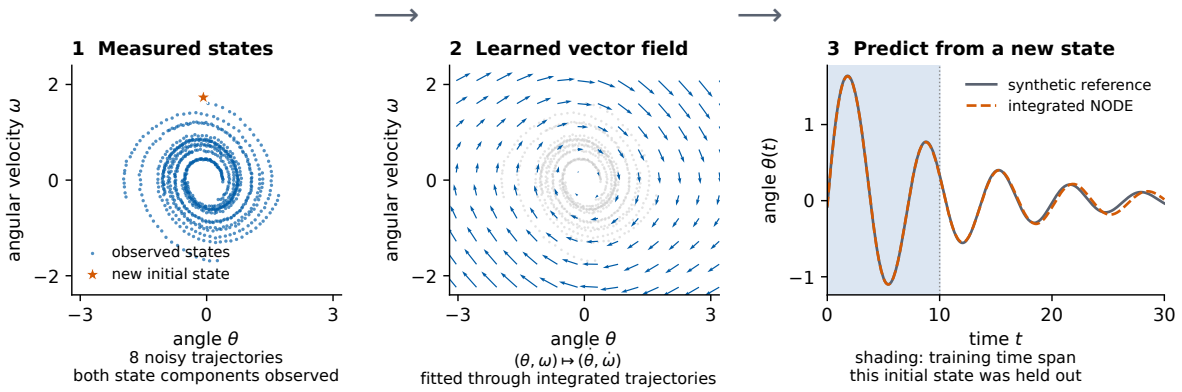


Figure 38: Learning a pendulum vector field from measured trajectories. Left, eight noisy training trajectories provide both angle and angular velocity every 0.1 time units over $0 \le t \le 10$. The star marks a held-out initial state. Centre, arrows are the trained network's derivative vectors $(\dot\theta, \dot\omega)$, with the observed states faded behind them. The fit uses integrated trajectory segments without derivative targets. Right, integration from the starred state predicts the angle to $t = 30$. Shading indicates the time span of the training trajectories. The starred trajectory is held out throughout, and the synthetic reference uses the generating equation only for validation.

10.2 Numerical integration and residual networks

A neural vector field becomes a trajectory only after numerical integration. The simplest solver follows the arrow of that field over a short step. The Taylor expansion of the solution around the time $t_n = n\Delta t$ reads

$$\mathbf{x}(t_n + \Delta t) = \mathbf{x}(t_n) + \Delta t\, \dot{\mathbf{x}}(t_n) + \tfrac{1}{2}\Delta t^2\, \ddot{\mathbf{x}}(t_n) + O(\Delta t^3), \tag{10.3}$$

and $\dot{\mathbf{x}}(t_n) = \mathbf{F}(\mathbf{x}(t_n))$ is known at the current state. Dropping the terms of second and higher order gives the explicit Euler step

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \Delta t\, \mathbf{F}(\mathbf{x}_n). \tag{10.4}$$

The integration starts at the initial state $\mathbf{x}_0$, evaluates the vector field there, moves a distance $\Delta t\,|\mathbf{F}(\mathbf{x}_0)|$ along the arrow to $\mathbf{x}_1$, evaluates the field again at the new point, and repeats. The numerical trajectory is a polygon of straight segments, each tangent to the true trajectory at its starting point. Each step makes an error of order Δt^2, the first neglected Taylor term, called the local error. Reaching a fixed time T takes $T/\Delta t$ steps, and the local errors add up to a global error of order Δt. Euler's method is therefore of first order, and halving the step halves the error.

For the harmonic oscillator the error has a systematic direction: each Euler step moves outside the circular orbit. With $\ddot x = -x$ with $v = \dot x$, one Euler step maps (x, v) to $(x + \Delta t\, v,\ v - \Delta t\, x)$, and

$$(x + \Delta t\, v)^2 + (v - \Delta t\, x)^2 = (1 + \Delta t^2)(x^2 + v^2), \tag{10.5}$$

because the cross terms cancel. Every step multiplies the radius of the phase-space circle by $\sqrt{1 + \Delta t^2}$, so the numerical solution spirals outwards and, after the $2\pi/\Delta t$ steps of one period, has the radius $(1 + \Delta t^2)^{\pi/\Delta t}$ instead of 1. With 13 steps per period the radius nearly quadruples, and with $\Delta t = 0.01$ it grows by about 3% (Figure 39). The numerical flow expands phase-space areas, although the exact flow conserves them by Liouville's theorem. A

neural field trained through Euler steps would learn to compensate this artificial expansion, and the learned law would then describe the solver as much as the system.

Higher accuracy comes from evaluating the field several times within a step. The classical fourth-order Runge–Kutta method (RK4) computes four slopes,

$$\begin{aligned} \mathbf{k}_1 &= \mathbf{F}(\mathbf{x}_n), & \mathbf{k}_2 &= \mathbf{F}(\mathbf{x}_n + \tfrac{1}{2}\Delta t\,\mathbf{k}_1), \\ \mathbf{k}_3 &= \mathbf{F}(\mathbf{x}_n + \tfrac{1}{2}\Delta t\,\mathbf{k}_2), & \mathbf{k}_4 &= \mathbf{F}(\mathbf{x}_n + \Delta t\,\mathbf{k}_3), \end{aligned} \qquad \mathbf{x}_{n+1} = \mathbf{x}_n + \frac{\Delta t}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4). \tag{10.6}$$

The first slope is taken at the start of the step, the second and third at trial points in the middle, and the fourth at a trial point at the end. Their weighted average, with weights $1, 2, 2, 1$ and two different trial slopes at the midpoint, approximates the mean slope over the step. The weights are chosen so that the step agrees with the Taylor expansion up to the term Δt^4, so the local error is of order Δt^5 and the global error of order Δt^4. Halving the step reduces the error sixteen-fold, at the cost of four evaluations of the field per step. Library solvers go one stage further and adapt the step size, estimating the local error by comparing two formulas of different order and shrinking the step where the solution changes quickly. For conservative systems, symplectic integrators preserve phase-space volume at every step. No solver removes the growth of errors on a chaotic attractor, which belongs to the equation and not to the method. The neural ODEs of this chapter use RK4 with a fixed step.

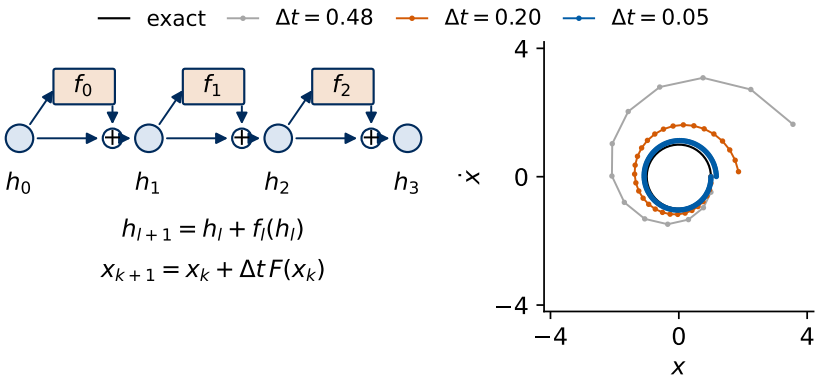


Figure 39: Left, three residual blocks $\mathbf{h}_{l+1} = \mathbf{h}_l + \mathbf{f}_l(\mathbf{h}_l)$ and the corresponding Euler step. Right, Euler solutions of the harmonic oscillator over one period for three step sizes, with the exact circle.

The Euler step (10.4) also connects neural ODEs to the networks of earlier lectures. A residual network [36] computes $\mathbf{h}_{l+1} = \mathbf{h}_l + \mathbf{f}_l(\mathbf{h}_l)$, where each block adds a learned correction to its input. With $\mathbf{f}_l = \Delta t\,\mathbf{F}_{\boldsymbol{\Theta}}$ shared across layers, a residual network is a sequence of Euler steps, with the layer index in the role of time [37, 38]. Letting the number of layers grow as $T/\Delta t$ turns the network into the solution of $\dot{\mathbf{x}} = \mathbf{F}_{\boldsymbol{\Theta}}(\mathbf{x})$ over the time T. A neural ODE is in this sense a residual network of continuous depth, and its training is the training of a very deep network with shared weights.

10.3 The training protocol

The data are measured states $\mathbf{y}_k$ at the times t_k, on one or several trajectories, with measurement noise. From a measured state $\mathbf{y}_0$ the model predicts the states at the later observation times, and the parameters minimise the squared deviation averaged over the K observation

times and the d state components,

$$\mathcal{L}(\boldsymbol{\Theta}) = \frac{1}{Kd} \sum_{k=1}^{K} \left\| \hat{\mathbf{x}}(t_k) - \mathbf{y}_k \right\|^2, \qquad \hat{\mathbf{x}}(t_0) = \mathbf{y}_0. \tag{10.7}$$

For a long chaotic trajectory this loss is unsuitable. By Section 3 any small error of the field makes the predicted trajectory diverge from the data within a few Lyapunov times, after which the loss depends erratically on $\boldsymbol{\Theta}$ and its gradient carries no useful information. The training trajectories are therefore cut into short segments of S sampling intervals. A minibatch contains B segments with measured initial states $\mathbf{y}_{k_b}$, $b = 1, \dots, B$. The model is started at each measured initial state, integrated over its segment, and compared with the measured states of that segment,

$$\mathcal{L}(\boldsymbol{\Theta}) = \frac{1}{BSd} \sum_{b=1}^{B} \sum_{j=1}^{S} \left\| \hat{\mathbf{x}}_b(t_{k_b+j}) - \mathbf{y}_{k_b+j} \right\|^2, \qquad \hat{\mathbf{x}}_b(t_{k_b}) = \mathbf{y}_{k_b}. \tag{10.8}$$

Here d is the number of measured state components. The squared errors are averaged over the components as well as over the segments and the observation times. We call this procedure multiple shooting. Classical multiple shooting also treats the initial states of the segments as unknowns with continuity constraints, which this simplified version omits [39].

The complete protocol consists of the following steps. The measured trajectories are divided in time into fitting, validation and test parts, as in Section 5. For every update, B segments are drawn at random from the fitting part. Each is integrated with RK4 from its measured initial state through the network $\mathbf{F}_{\boldsymbol{\Theta}}$, which is evaluated four times per step. The loss (10.8) compares the integrated states with the measured ones, its gradient with respect to $\boldsymbol{\Theta}$ is computed as described next, and the Adam optimiser updates $\boldsymbol{\Theta}$. Settings such as the segment length S are chosen on the validation part, and the trained field is finally tested by free runs from many initial states of the test part, which are integrated far beyond the length of a segment.

10.4 Gradients through the solver

Training a vector field requires knowing how a change in its weights changes the integrated trajectory and its loss. The gradient $d\mathcal{L}/d\boldsymbol{\Theta}$ can be obtained in two ways. In the first, the solver with a fixed step is a composition of arithmetic operations and network evaluations, and reverse-mode automatic differentiation through all of them gives the exact gradient of the discretised loss. Its structure is clearest for the Euler step (10.4) with the field $\mathbf{F}_{\boldsymbol{\Theta}}$. The step $\mathbf{x}_{n+1} = \mathbf{x}_n + \Delta t\, \mathbf{F}_{\boldsymbol{\Theta}}(\mathbf{x}_n)$ is a recurrent update of the kind met in Section 8.2, with the state $\mathbf{x}_n$ and without an input. Writing $\bar{\mathbf{x}}_n = \partial \mathcal{L}/\partial \mathbf{x}_n$ and J_n for the Jacobian of $\mathbf{F}_{\boldsymbol{\Theta}}$ at $\mathbf{x}_n$, the chain rule gives the backward recursion

$$\bar{\mathbf{x}}_n = \left(\mathbb{1} + \Delta t\, J_n^{\top}\right) \bar{\mathbf{x}}_{n+1} + \frac{\partial \mathcal{L}}{\partial \mathbf{x}_n}\Big|_{\text{direct}}, \qquad \frac{\partial \mathcal{L}}{\partial \boldsymbol{\Theta}} = \sum_n \Delta t \left(\frac{\partial \mathbf{F}_{\boldsymbol{\Theta}}}{\partial \boldsymbol{\Theta}}\Big|_{\mathbf{x}_n} \right)^{\top} \bar{\mathbf{x}}_{n+1}, \tag{10.9}$$

where the direct term is the derivative of the loss with respect to $\mathbf{x}_n$ when $\mathbf{x}_n$ is an observation time and zero otherwise. The sensitivity is carried backwards from the end of the segment by the transposed Jacobian of one step, as in equation (8.6), and every step contributes to the gradient of the shared parameters, as in equation (8.7). For RK4 the same chain rule runs through the four evaluations of each step. The memory needed to store the intermediate states grows with the number of steps.

A model with one parameter shows what this computes. For the decay law $\dot{x} = -ax$ from $x_0 = 1$, one Euler step of size 0.1 predicts $\hat{x}_1 = 1 - 0.1a$, and a measured value $y_1 = 0.92$ gives

the loss and its derivative

$$\mathcal{L}(a) = (0.08 - 0.1a)^2, \qquad \frac{d\mathcal{L}}{da} = -0.2\,(0.08 - 0.1a) = -0.006 \quad \text{at } a = 0.5. \tag{10.10}$$

The derivative is negative, so gradient descent increases a and produces the faster decay that the data require.

The second way, the adjoint method, is the limit $\Delta t \to 0$ of the recursion (10.9). Dividing $\bar{\mathbf{x}}_n - \bar{\mathbf{x}}_{n+1} = \Delta t\, J_n^\top \bar{\mathbf{x}}_{n+1}$ by Δt gives a differential equation for the sensitivity, which is integrated backwards in time and needs no stored trajectory.

Adjoint method for the gradient. *For a loss $\mathcal{L}(\mathbf{x}(T))$ of the final state, define the adjoint $\mathbf{a}(t)$ by*

$$\dot{\mathbf{a}} = -J(t)^\top \mathbf{a}, \qquad \mathbf{a}(T) = \frac{\partial \mathcal{L}}{\partial \mathbf{x}(T)}, \qquad J(t) = \frac{\partial \mathbf{F}_{\boldsymbol{\Theta}}}{\partial \mathbf{x}}\Big|_{\mathbf{x}(t)}. \tag{10.11}$$

Then

$$\frac{d\mathcal{L}}{d\boldsymbol{\Theta}} = \int_0^T \mathbf{a}(t)^\top \frac{\partial \mathbf{F}_{\boldsymbol{\Theta}}}{\partial \boldsymbol{\Theta}}\Big|_{\mathbf{x}(t)}\, dt. \tag{10.12}$$

Derivation. A change $\delta\boldsymbol{\Theta}$ changes the trajectory by $\delta\mathbf{x}(t)$, which obeys the linearised equation $\delta\dot{\mathbf{x}} = J\,\delta\mathbf{x} + (\partial \mathbf{F}_{\boldsymbol{\Theta}}/\partial\boldsymbol{\Theta})\,\delta\boldsymbol{\Theta}$ with $\delta\mathbf{x}(0) = 0$, since the initial state is measured. By the product rule,

$$\frac{d}{dt}(\mathbf{a}^\top \delta\mathbf{x}) = \dot{\mathbf{a}}^\top \delta\mathbf{x} + \mathbf{a}^\top \delta\dot{\mathbf{x}} = -\mathbf{a}^\top J\,\delta\mathbf{x} + \mathbf{a}^\top J\,\delta\mathbf{x} + \mathbf{a}^\top \frac{\partial \mathbf{F}_{\boldsymbol{\Theta}}}{\partial \boldsymbol{\Theta}}\delta\boldsymbol{\Theta} = \mathbf{a}^\top \frac{\partial \mathbf{F}_{\boldsymbol{\Theta}}}{\partial \boldsymbol{\Theta}}\delta\boldsymbol{\Theta}. \tag{10.13}$$

The adjoint equation is chosen so that the terms with J cancel. Integrating from 0 to T with $\delta\mathbf{x}(0) = 0$ gives $\mathbf{a}(T)^\top \delta\mathbf{x}(T) = \int_0^T \mathbf{a}^\top (\partial \mathbf{F}_{\boldsymbol{\Theta}}/\partial\boldsymbol{\Theta})\, dt\, \delta\boldsymbol{\Theta}$, and the left side equals $\delta\mathcal{L}$ by the chain rule and the definition of $\mathbf{a}(T)$. □

The adjoint equation is the tangent equation (3.1) of Section 3 with the transposed Jacobian and run backwards. Errors propagate forwards with J, sensitivities backwards with $J^\top$. The adjoint is integrated from T to 0 together with the state, which is recomputed by running the neural ODE backwards, and with the running integral of equation (10.12). With observations at several times, $\mathbf{a}$ receives a jump $\partial\mathcal{L}/\partial\mathbf{x}(t_k)$ at each observation time. The products $J^\top\mathbf{a}$ and $(\partial \mathbf{F}_{\boldsymbol{\Theta}}/\partial\boldsymbol{\Theta})^\top\mathbf{a}$ cost about one backward pass through the network, and the memory does not depend on the number of steps. The same equations are the costate equations of optimal control. The two gradients are not identical. The adjoint is the gradient of the continuous problem and backpropagation the gradient of its discretisation, so they differ by the discretisation error. On a damped pendulum with an untrained network the difference is of order 10^{-8} with 50 RK4 steps and shrinks by about $4^4 = 256$ with each division of the step by four, the convergence of a fourth-order solver. Running a dissipative flow backwards expands volumes, so the backward reconstruction of the state in the adjoint method can lose accuracy for strongly contracting systems. For the small networks of this chapter the memory of backpropagation is affordable, and all models are trained by backpropagation through RK4.

10.5 A damped pendulum

The damped pendulum tests whether trajectory fitting recovers a field whose energy loss and fixed-point stability can be checked independently. It is the damped pendulum of Section 2.3, equations (2.10) with $\gamma = 0.2$. Its vector field, its fixed points and their stability are known from equations (2.15) to (2.18), the hanging position is a stable spiral and the upright position

a saddle, and its energy decreases at the rate (2.12). Each of these properties can be checked on the learned field.

The model has access to measured states only. Both state variables, the angle θ and the angular velocity ω, are recorded every 0.1 time units on eight trajectories started from different initial states, each over $0 \le t \le 10$, with Gaussian measurement noise of standard deviation 0.01 added to every value. The initial energies are such that every trajectory swings without passing over the top, so the data cover the region around the hanging position and never visit the upright position. Equation (2.10) is used only to generate these data and to judge the result. The model knows neither the form of the gravitational term, nor that friction is present, nor that $\dot\theta = \omega$. It knows only that the state is two-dimensional and that its velocity is a function of the state. The settings of the training are listed in Table 13. The final component-averaged loss is about 2.1×10^{-4}. This is the level that noise alone produces. Even the exact field starts each segment from a noisy measured state and is compared with noisy targets, and over a short segment the two noise variances add to about $2\sigma^2 = 2 \times 10^{-4}$ per component.

Table 13: Data and training of the two neural ODEs. Both fields are multilayer perceptrons with tanh units, trained by Adam through RK4 with the multiple-shooting loss (10.8). Wall-clock times were measured on a shared machine and support comparisons of orders of magnitude only.

	damped pendulum	Lorenz system
measured	θ, ω	x, y, z, standardised per component
sampling	0.1, eight trajectories over $0 \le t \le 10$	0.02, 8000 samples on one trajectory after a transient of 50 time units
noise	Gaussian, standard deviation 0.01	none
split	all for fitting	4000 fitting, 1000 validation, 3000 test samples
network	$2 \to 64 \to 64 \to 2$, 4482 parameters	$3 \to 64 \to 64 \to 3$, output $\times 10$, 4611 parameters
segments	64 segments of $S = 10$ intervals per mini-batch	$S = 10$, chosen on the validation part from $S = 5$, 10, 20 (median t_{div} 2.80, 3.30, 2.48 Lyapunov times)
optimisation	3000 Adam steps, 52 s in this run, final loss 2.1×10^{-4}	4000 Adam steps, 128 segments per mini-batch

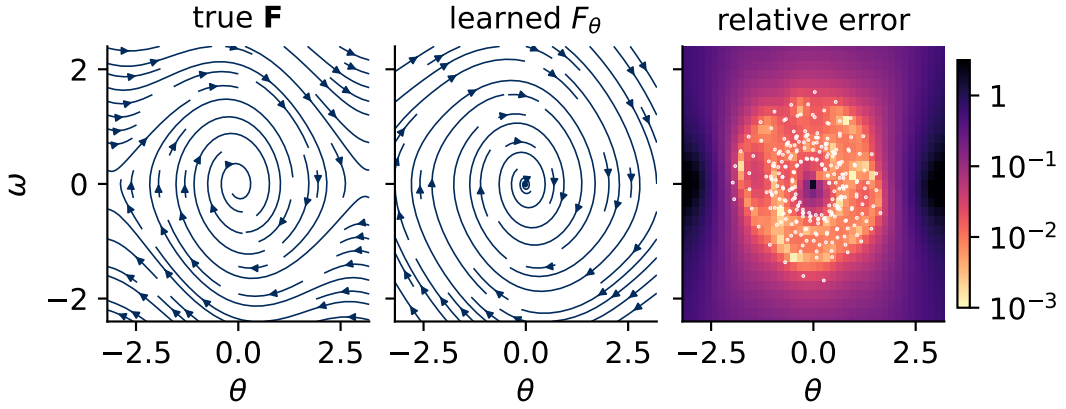


Figure 40: True vector field (left), learned field $\mathbf{F}_\Theta$ (middle) and relative error $|\mathbf{F}_\Theta - \mathbf{F}|/|\mathbf{F}|$ (right) in the phase plane of the damped pendulum. White dots are the measured states.

Once trained, the network can be evaluated at any point of the phase plane, and Figure 40

compares the learned field with the true one. Close to the measured states the learned arrows agree with the true ones to about one percent, and farther away the relative error grows to tens of percent. Near the saddles at $\theta = \pm\pi$, which no trajectory visited, the learned streamlines differ visibly from the true ones. The network has learned the law where the data constrain it and has extrapolated arbitrarily elsewhere. The learned field is used in the same way as a known one. Integrated from three new initial states up to $t = 30$, three times the length of the training trajectories, it forecasts the angle with a maximal error of 8.2×10^{-2}, because these trajectories spiral into the hanging position, where the data are densest. The field has not been validated for trajectories passing over the top, where the training data provide little constraint.

10.6 The Lorenz system with the full state measured

The second test returns to the Lorenz system (2.28), now with all three coordinates recorded.

The stored data are one long trajectory. The Lorenz equations are integrated from an arbitrary initial state, the first 50 time units are discarded so that the trajectory has settled on the attractor, and the next 8000 states are stored at the interval $\Delta t = 0.02$,

$$\mathbf{x}_k = (x(k\Delta t),\, y(k\Delta t),\, z(k\Delta t)), \qquad k = 0, \dots, 7999. \tag{10.14}$$

The record is divided in time into the fitting part, $k < 4000$, the validation part, $4000 \le k < 5000$, and the test part, $k \ge 5000$. The three coordinates have different ranges, so each is standardised with the mean $\bar{x}_i$ and the standard deviation s_i of the fitting part,

$$q_{k,i} = \frac{x_{k,i} - \bar{x}_i}{s_i}, \qquad i = 1, 2, 3, \tag{10.15}$$

and every model works with the standardised state $\mathbf{q}_k$.

The network represents the vector field of the standardised state. It receives a state $\mathbf{q} \in \mathbb{R}^3$ and returns a velocity $\dot{\mathbf{q}} \in \mathbb{R}^3$,

$$\dot{\mathbf{q}} = \mathbf{F}_{\boldsymbol{\Theta}}(\mathbf{q}) = 10 \cdot \mathrm{MLP}_{\boldsymbol{\Theta}}(\mathbf{q}), \tag{10.16}$$

where the MLP has two hidden layers of 64 tanh units, 4611 parameters in total, and the fixed factor 10 matches the typical size of the standardised velocities, so that the network outputs are of order one. The network never sees a time series as input. It sees one state and returns one velocity, and the time dependence enters only through the solver.

The loss is the multiple-shooting loss (10.8) with the standardised states $\mathbf{q}_k$ in place of $\mathbf{y}_k$, $d = 3$ components and $B = 128$ segments drawn at random from the fitting part, each integrated with one RK4 step of size Δt per sample. The trained quantities are the weights and biases $\boldsymbol{\Theta}$ of the MLP and nothing else. The gradient of the loss is computed by backpropagation through the RK4 steps, and 4000 Adam updates are made. The segment length S is the only setting chosen on the validation part (Table 13). Short segments emphasise the one-step error, while long segments constrain the forecast over longer times at the price of a rougher loss, and the intermediate value $S = 10$ is best.

After training, the learned equation is used without any further data. A free-run forecast starts from one measured state $\mathbf{q}_{k_0}$ of the test part and integrates equation (10.16) forward with RK4 for 500 steps,

$$\hat{\mathbf{q}}(t_{k_0}) = \mathbf{q}_{k_0}, \qquad \hat{\mathbf{q}}(t_{k_0+n}) = \varphi^{\boldsymbol{\Theta}}_{n\Delta t}(\mathbf{q}_{k_0}), \qquad n = 1, \dots, 500, \tag{10.17}$$

where $\varphi_t^{\boldsymbol{\Theta}}$ is the flow map of the learned field. Only the initial state is measured. Every later state is produced by the model from its own previous state, so errors accumulate as in any free run. The forecast is compared with the measured trajectory $\mathbf{q}_{k_0+n}$, and the divergence time t_{div} is the first lead at which the root-mean-square error of the three standardised components exceeds 0.5. Repeating this from many origins of the test part gives the median divergence time. With the Lyapunov time of 55 samples, a forecast that stays valid for about three Lyapunov times stays valid for about 165 samples.

A free run of the learned system over 400 time units, 20 000 steps from the first test state and transformed back by inverting equation (10.15), stays bounded and fills both lobes (Figure 41). Its long-time statistics, the mean of z and the standard deviation of x, agree with those of the data to better than one percent (Table 15), although individual trajectories diverge after about three Lyapunov times. The model reproduces the climate of the system, its statistics over long times, while the weather, the individual trajectory, is predictable only over the horizon set by chaos, the distinction made in Section 3.

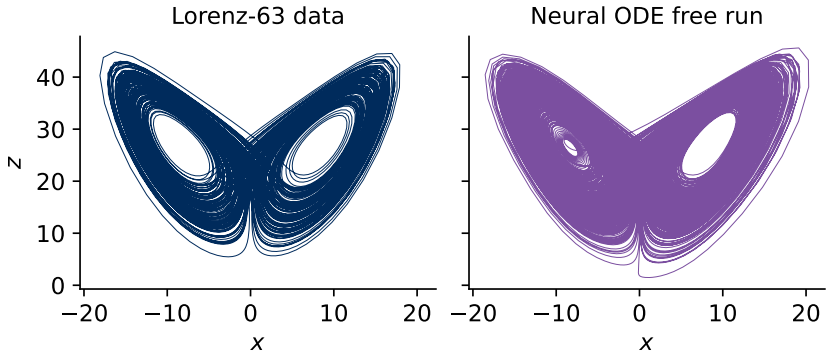


Figure 41: Lorenz data (left) and a free run of 400 time units of the learned neural ODE (right), projected on the (x, z) plane.

The result is compared with discrete-time models fitted to the same data under the same protocol, AR and NVAR fitted to the increment $\mathbf{q}_{k+1} - \mathbf{q}_k$ of the standardised state $\mathbf{q}$ on a window of p states, and an MLP map $\mathbf{q}_{k+1} = \mathbf{q}_k + G_{\boldsymbol{\Theta}}(\mathbf{q}_k)$, the discrete-time counterpart of the neural ODE.

Table 14: Free-run forecasts of the full Lorenz state. t_{div} is the median divergence time over the test origins, and $C(n)$ the capacity averaged over the three components at a lead of n Lyapunov times.

model	parameters	t_{div} [Lyapunov times]	$C(1)$	$C(2)$	$C(3)$	$C(5)$
AR, $p = 8$	75	0.24	0.104	0.018	0.015	0.177
NVAR, $p = 3$	165	7.69	1.000	1.000	1.000	0.997
MLP map, width 128	17411	2.71	0.986	0.875	0.566	0.118
neural ODE, $S = 10$	4611	2.88	0.996	0.877	0.689	0.094

NVAR follows the trajectory more than twice as long as the networks (Table 14), the reverse of the ranking on the scalar Lorenz record in Table 7. The explanation is the argument of Section 7. With the full state measured, the one-step increment of a quadratic vector field is close to a quadratic polynomial of the state, which NVAR contains in its dictionary and fits by a linear least-squares problem with a unique solution. The networks must build quadratic functions from tanh units by stochastic optimisation from 4000 samples. When only x is measured, the hidden coordinates are complicated functions of the window and the quadratic

dictionary of the window no longer contains the one-step map. The suitability of a model depends on what is measured, as much as on the system. AR is linear and cannot represent the switching between the lobes. The neural ODE and the MLP map have similar forecast performance within the spread over test origins. Separately, the neural ODE uses about a quarter of the parameters.

10.7 Physical diagnostics of the learned field

The value of a learned vector field lies in what can be done with it after training. The methods of Sections 2 and 3 apply to $\mathbf{F}_{\boldsymbol{\Theta}}$ as they apply to a known law, because the network is a differentiable function whose Jacobian is available by automatic differentiation. Its zeros are found by Newton's method, their stability from the eigenvalues of the Jacobian, the dissipation from the divergence, and the chaos from the Lyapunov exponents of the learned flow. Table 15 compares these quantities with those of the Lorenz field.

Table 15: Properties of the learned neural ODE and of the Lorenz field. C_+ and C_- are the two fixed points of steady convection rolls turning in opposite senses, equation (2.30). At each of them the Jacobian has one real eigenvalue and a pair of complex conjugate eigenvalues, and the table lists this complex pair, whose real part decides whether the point attracts or repels nearby states. The fixed points of the learned field are found by Newton's method started from the true ones, and the divergence is averaged over the test states.

quantity	Lorenz field	learned field
eigenvalues at the origin	-22.83, -2.67, 11.83	-13.53, -3.27, 11.03
complex eigenvalues at C_-	$0.094 \pm 10.195\,i$	$0.15 \pm 10.20\,i$
complex eigenvalues at C_+	$0.094 \pm 10.195\,i$	$-0.33 \pm 9.72\,i$
mean divergence	-13.67	-13.88
largest Lyapunov exponent	0.906 ± 0.007	0.899 ± 0.010
mean of z in a long free run	23.42 (data)	23.57
standard deviation of x	7.89 (data)	7.88

The comparison separates two kinds of property. The quantities averaged over the attractor, the divergence, the largest Lyapunov exponent and the long-time statistics, are learned well, because every training sample contributes to them. Local properties in regions that the data do not visit are learned poorly. The origin is a saddle in both fields, with the correct number of unstable directions, but the strongly contracting eigenvalue is too small in magnitude by almost a factor of two. The steady rolls illustrate the danger. The learned C_- is an unstable spiral close to the true one, but the learned C_+ is a stable spiral, while the true C_+ is weakly unstable. The real part of the complex pair is about one percent of its imaginary part and decides the stability, and the data, which never stay near the fixed points, constrain it poorly. A conclusion drawn from the learned field about the stability of the steady rolls would be wrong. A learned field is a hypothesis about the law, reliable where the data are informative and to be tested elsewhere. The next section asks how prior knowledge of part of the law can reduce this uncertainty.

11 Hybrid models and symbolic regression for a partly known law

The learned Lorenz field reproduces trajectories while assigning the wrong stability to one steady roll. This failure shows how little some local properties are constrained by the observed motion. A known part of the equation can reduce the freedom left to the network and make the remaining unknown term easier to identify.

11.1 Known and unknown parts of a law

The neural ODE of the previous section learned every component of the field, and the pendulum experiment showed the price. The learned field was accurate near the data and arbitrary elsewhere, and the learned Lorenz field had a fixed point of the wrong stability. A network that must learn the whole vector field spends the information in the data on all of it, including the parts that physics already fixes, and it has nothing to guide it where the data are absent.

In science the law is rarely unknown as a whole. Usually a part is established beyond doubt and a part is uncertain. The equations of a climate model contain the conservation of mass, momentum and energy, which are certain, and terms for the effect of clouds and of turbulence smaller than the grid, which are not. A model of a chemical reaction network knows which species react, while the rate laws of some reactions are uncertain. A mechanical system obeys Newton's equations with known gravitational and elastic forces, while the friction depends on surfaces, lubricants and the surrounding air and is usually known only roughly. In each case the natural model keeps the certain part as it is and learns only the uncertain part from the data. We keep the certain part, learn the uncertain part, and then ask a second question. A network learned for the uncertain part is a function without a formula, and the scientific aim is often a formula, a law that can be compared with theory, interpreted and used outside the range of the data.

The pendulum with unknown friction is the example. A pendulum in air obeys

$$\ddot{\theta} = -\sin\theta + R(\dot{\theta}), \tag{11.1}$$

in units where the small-oscillation frequency is one. Gravity is known, and the angular deceleration R caused by friction depends on the bearing, the bob and the air. Three laws are common, and each corresponds to a different physical mechanism. Viscous friction $-\gamma\dot{\theta}$, proportional to the velocity, describes a lubricated bearing or slow motion through a fluid. Quadratic drag $-c\,\dot{\theta}|\dot{\theta}|$ describes a bob moving fast through air, where the force comes from the momentum transferred to the air pushed aside. Dry friction $-\mu\,\mathrm{sign}(\dot{\theta})$, of constant magnitude, describes two solid surfaces sliding on each other and is smoothed as $-\mu\tanh(\dot{\theta}/\varepsilon)$ with $\varepsilon = 0.05$. Each law opposes the motion, so the energy (2.11), with $\omega = \dot{\theta}$, obeys $\dot{E} = \dot{\theta}\,R(\dot{\theta}) \le 0$, which generalises equation (2.12). Identifying which law acts identifies the mechanism of dissipation.

11.2 The hybrid model

The hybrid model keeps the gravitational term and lets a small network $U_{\boldsymbol{\Theta}}(\theta, \dot{\theta})$ of the angle and the angular velocity represent the friction,

$$\ddot{\theta} = -\sin\theta + U_{\boldsymbol{\Theta}}(\theta, \dot{\theta}), \tag{11.2}$$

a construction called a universal differential equation [40]. The network receives both variables because the form of the friction is not assumed, and whether it depends on the angle is left to the data. It is trained as the neural ODE of Section 10. Equation (11.2), written as a first-order system for $(\theta, \dot\theta)$, is integrated by RK4 from measured states, compared with the measured states that follow, and the gradient of the loss is propagated through the solver to the parameters of $U_{\boldsymbol\Theta}$. The known term $-\sin\theta$ enters the integration but has no parameters.

The data are a single measured swing of 201 states $(\theta, \dot\theta)$ over $0 \le t \le 20$, recorded every $\Delta t = 0.1$ and starting at rest at $\theta = 2$, with measurement noise of standard deviation 0.02. They are generated with quadratic drag, $c = 0.15$, which the model does not know. The division of labour changes what the data must provide. The network has to learn a function that is small compared with gravity, the friction, and it learns it as a correction inside the equation, so that every measured state constrains it through the whole subsequent motion. On the measured states the learned friction deviates from the true one by about 8% in relative root-mean-square terms, although the friction is a small fraction of the total force.

11.3 A sparse formula for the learned friction

A trained network gives the friction as a function without a formula. To obtain a formula, the learned function is expanded in a library of candidate terms $\mathcal{B}_j$,

$$U_{\boldsymbol\Theta} \approx \sum_{j=1}^{8} \alpha_j \mathcal{B}_j, \qquad \{\mathcal{B}_j\} = \{1,\ \sin\theta,\ \theta^2,\ \theta\dot\theta,\ \dot\theta,\ \dot\theta|\dot\theta|,\ \tanh(\dot\theta/\varepsilon),\ \dot\theta^3\}, \tag{11.3}$$

evaluated on the measured states. The library is the hypothesis space. It contains the three friction laws, terms that no friction law has, and terms in θ, because the network receives the angle and the regression must itself discover that the friction depends on $\dot\theta$ alone.

Ordinary least squares with all eight columns fails in an instructive way. On a single decaying swing several velocity terms are strongly correlated, since $\dot\theta$, $\dot\theta|\dot\theta|$ and $\dot\theta^3$ all change sign together and grow together, and the fit distributes the friction over all of them with coefficients of both signs that largely cancel. The fit reproduces the data, but the formula has no physical meaning. The remedy rests on the observation that physical laws are usually simple, with few terms. Sequentially thresholded least squares (STLSQ) [41] enforces this. It solves the regression, sets every coefficient with magnitude below a threshold to zero, solves again with the remaining columns, and repeats until the set of active terms no longer changes. A small ridge penalty prevents correlated columns from cancelling. STLSQ retains one term, $-0.1497\,\dot\theta|\dot\theta|$, within 0.2% of the hidden coefficient. A single term survives over a wide range of thresholds, from about a quarter to almost half of the root-mean-square of $U_{\boldsymbol\Theta}$, so the result does not hinge on a fine choice. The same procedure recovers the viscous and the dry law with one term each. The pendulum experiment thus identifies the mechanism of dissipation, drag by the air, and measures its strength.

The hybrid model is what makes the regression possible on noisy data. The original sparse-identification method regresses numerical derivatives of the data directly on the library [41]. Here the acceleration would have to be estimated from the measured angular velocity by the central difference $(\dot\theta_{k+1} - \dot\theta_{k-1})/(2\Delta t)$. Dividing by $2\Delta t$ amplifies the measurement noise to the standard deviation $\sigma/(\sqrt{2}\,\Delta t) \approx 0.141$, larger than the root-mean-square friction 0.100, so the estimated acceleration is dominated by noise and, with the same threshold, no term survives. The hybrid model never differentiates the data. The solver integrates the model, and the comparison of whole segments averages the measurement noise.

11.4 Extrapolation with prior knowledge

The value of each level of prior knowledge appears in extrapolation. The four models are integrated to $t = 100$, five times the measured window, and compared with the true motion over $20 < t \leq 100$ (Table 16). Gravity alone, without friction, keeps swinging and is soon completely wrong. The neural ODE of the whole field is better but has to extrapolate both gravity and friction into amplitudes and velocities that the data never reached. The hybrid model extrapolates only the friction and is several times more accurate. The recovered formula is better still by almost an order of magnitude, because its functional form matches the generator and remains correct for any velocity.

Table 16: Root-mean-square error of the angle over $20 < t \leq 100$, beyond the measured window $0 \leq t \leq 20$, for models with increasing prior knowledge.

model	what is learned	RMS angle error
gravity alone	nothing	1.479
neural ODE	the whole vector field	0.176
hybrid model (11.2)	the friction as a network	0.047
recovered equation	the friction as a formula	0.006

Every piece of correct prior knowledge removes something that would otherwise have to be learned from the data, and what is not learned cannot be learned wrongly. The recovery succeeded because the library contained the true term. Had the drag been proportional to $|\dot{\theta}|^{1.8}$, STLSQ would have returned the closest combination of the available terms, which would fit the swing well and still be a wrong law. A formula selected from a library is a hypothesis restricted by that library, and a fit to one trajectory does not establish its validity outside the measured range. Measurements at other amplitudes, or a physical argument for the form of the law, are needed to confirm it.

12 Physics-informed neural networks

A physical equation can also constrain a fitted solution directly. The hybrid model placed a network inside a numerical differential-equation solver. A physics-informed neural network represents the solution itself and trains it using the equation, observations and the conditions that select the desired solution.

12.1 Learning a solution when the law is known

Suppose we observe an oscillator at twelve times during the beginning of its motion and want its position later. Many smooth curves can pass through the observations and then follow different paths. Fitting the positions alone gives no test of which continuation is physically plausible. If the equation $m\ddot{x} + c\dot{x} + kx = 0$ is known, however, every candidate curve also makes a prediction for velocity and acceleration. We can check whether those three quantities satisfy the force balance, including at times when no position was measured.

We fit the same network to the same twelve positions twice in Figure 42. The second fit also tests the equation throughout $0 \leq t \leq 10$ and follows the damped motion beyond the observation window. The law constrains the curve at those later times during training, even though the positions are unobserved.

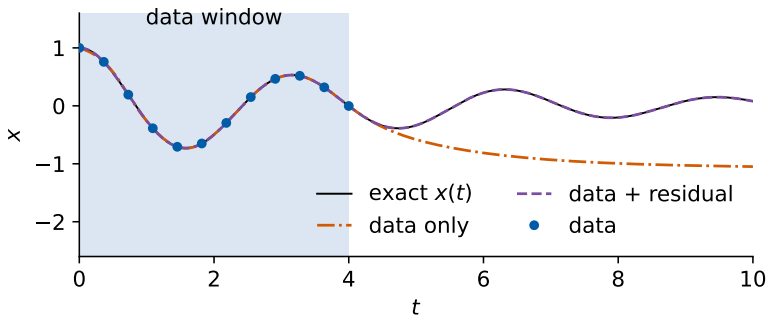


Figure 42: Damped oscillator from 12 samples in the shaded window. The network trained on the data alone (dash-dotted) departs from the exact solution beyond the window, while the network trained with the residual of the law (dashed) follows it over the whole interval.

This use of observations together with a known law is called data assimilation. Two related tasks use the same construction. The forward problem computes a solution from the law and initial conditions without trajectory measurements. The inverse problem infers an unknown physical parameter from observations. The law is known, possibly up to a few parameters, and helps determine the solution. In Section 10, the measured solution was used to learn the law. The heat equation with a few thermometers in a rod, or a flow inferred from dye concentration [42], poses the same question in space and time.

Consider the forward problem first. The mass, damping and spring constant of an oscillator are given, together with its initial position and velocity. We seek a function of time whose acceleration, velocity and position obey the equation of motion. A physics-informed neural network (PINN) represents this unknown solution by a network [43, 44]. For every choice of weights $\boldsymbol{\Theta}$, the network $x_{\boldsymbol{\Theta}}(t)$ is one candidate curve. Training changes the whole curve until it approximately satisfies the equation and the initial conditions.

The input is a time t, and the output is a position. In the neural ODE of Section 10, the network instead receives a state and returns its velocity. A numerical solver then constructs a trajectory. The hybrid model of Section 11 also uses a solver, with a network supplying one unknown term of the velocity. A PINN evaluates the trajectory directly. Once it is trained for one initial-value problem, a position at a new time inside its training interval requires one network evaluation. A different initial condition generally requires a new fit for the PINN used here, whereas a learned vector field can be integrated from different initial states.

12.2 Equation residual and automatic differentiation

The equation of motion tests a candidate curve at times when no position was measured. We evaluate the curve and its first two derivatives and substitute them into the known law. The result is the residual,

$$r_{\boldsymbol{\Theta}}(t) = \frac{m\ddot{x}_{\boldsymbol{\Theta}}(t) + c\dot{x}_{\boldsymbol{\Theta}}(t) + kx_{\boldsymbol{\Theta}}(t)}{k}. \tag{12.1}$$

The numerator is the imbalance of the three terms in the force equation. Dividing by the positive spring constant k gives the residual the unit of position. The exact solution has $r(t) = 0$ at every time, so the equation supplies a target value of zero even where the position is unknown. The times t_i at which this test is applied are called collocation points. They are chosen by us and need not coincide with measurement times. In the oscillator notebook they are 200 equally spaced times in $[0, 10]$.

The derivatives are derivatives of the candidate curve. For one hidden layer,

$$x_{\boldsymbol{\Theta}}(t) = \sum_j a_j \tanh(w_j t + b_j) + c_0, \tag{12.2}$$

$$\dot{x}_{\boldsymbol{\Theta}}(t) = \sum_j a_j w_j [1 - \tanh^2(w_j t + b_j)], \tag{12.3}$$

$$\ddot{x}_{\boldsymbol{\Theta}}(t) = -2 \sum_j a_j w_j^2 \tanh(w_j t + b_j)[1 - \tanh^2(w_j t + b_j)]. \tag{12.4}$$

These expressions follow by applying the chain rule twice. Automatic differentiation applies the same rule through every layer, evaluating derivatives of the network up to floating-point error, without estimating derivatives from noisy measurements. Smooth tanh units supply the second derivative required here. A piecewise linear ReLU network has zero second derivative between its bends and is unsuitable for this direct construction of the oscillator residual.

Two differentiations, both drawn in Figure 43, have distinct jobs. Differentiation with respect to t gives velocity and acceleration for the residual. Differentiation of the resulting loss with respect to $\boldsymbol{\Theta}$ tells the optimiser how to change the curve. When the network receives the rescaled time $2t/T - 1$, the chain rule supplies the factors $2/T$ and $(2/T)^2$ in the first and second derivatives.

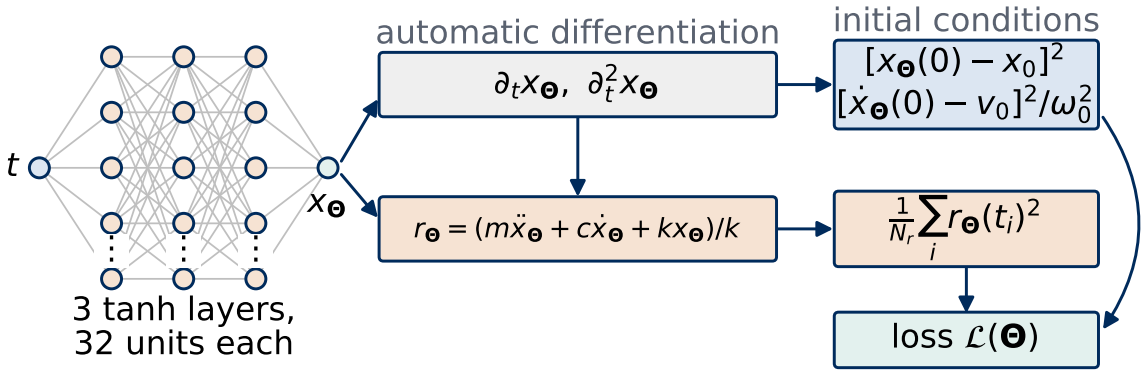


Figure 43: The forward oscillator PINN. Time enters the network, which returns a candidate position. Its time derivatives give the equation residual and the initial velocity. Their penalties form the loss, whose gradient updates the network weights.

12.3 The equation and the information that selects its solution

The equation alone has many solutions. For an initial-value problem, the initial position and velocity select one of them. Measurements provide another kind of information: the candidate curve should pass close to the observed positions. We keep these contributions separate in the loss so that their roles are visible.

Definition 12.1 (PINN loss)**.** For the oscillator, write

$$\mathcal{L}_r = \frac{1}{N_r} \sum_{i=1}^{N_r} r_{\boldsymbol{\Theta}}(t_i)^2, \qquad \mathcal{L}_{\mathrm{IC}} = [x_{\boldsymbol{\Theta}}(0) - x_0]^2 + \frac{[\dot{x}_{\boldsymbol{\Theta}}(0) - v_0]^2}{\omega_0^2}, \tag{12.5}$$

$$\mathcal{L}_d = \frac{1}{N_d} \sum_{j=1}^{N_d} [x_{\boldsymbol{\Theta}}(t_j) - y_j]^2, \qquad \omega_0 = \sqrt{k/m}. \tag{12.6}$$

The training objective is

$$\mathcal{L}(\boldsymbol{\Theta}) = w_r\mathcal{L}_r + w_{\mathrm{IC}}\mathcal{L}_{\mathrm{IC}} + w_d\mathcal{L}_d, \tag{12.7}$$

with nonnegative weights. $\mathcal{L}_r$ measures violation of the law at N_r collocation points, $\mathcal{L}_{\mathrm{IC}}$ departure from the given initial conditions, and $\mathcal{L}_d$ disagreement with N_d measured positions y_j. The data term is omitted when no measurements are supplied. The factor ω_0^{-2} gives the initial-velocity penalty the same units as the other terms.

For forward solving we use $w_r = w_{\mathrm{IC}} = 1$ and no data term. For the sparse-data experiment we use $w_r = w_d = 1$ and omits the initial-condition penalty: the measured positions constrain which motion is reconstructed. The data-only comparison keeps just $\mathcal{L}_d$. Thus the available information determines which terms are present. Because $\mathcal{L}_r$ is a mean, adding collocation points samples the law more densely without multiplying its weight relative to the initial conditions. The weights still matter, since equally scaled loss values can produce very different parameter gradients [45].

Two candidate curves that fail different tests

Take the dimensionless oscillator $\ddot{x} + 0.4\dot{x} + 4x = 0$ with $x(0) = 1$ and $\dot{x}(0) = 0$. The curve $x(t) = 0$ has zero residual everywhere, but its initial-position penalty is 1. The curve $x(t) = \cos(2t)$ satisfies both initial conditions, but substitution into equation (12.1) gives

$$r(t) = \frac{-4\cos(2t) - 0.8\sin(2t) + 4\cos(2t)}{4} = -0.2\sin(2t). \tag{12.8}$$

It fails the equation because its oscillations never lose amplitude to damping. At $t = \pi/4$, a single collocation point contributes $r^2 = 0.04$ before averaging. The correct damped motion must satisfy both tests. No measured target position at $\pi/4$ was needed to reject the undamped candidate.

One training step evaluates the current curve and its time derivatives at the collocation points, evaluates the required initial conditions and data predictions, forms equation (12.7), and differentiates this scalar with respect to all weights. The notebook first uses Adam and then L-BFGS, an optimiser that uses information from successive gradients to refine the fit. All collocation times contribute to the same update, so the network is trained on the whole interval at once, without stepping forward in time.

A function with zero residual *everywhere* and correct initial conditions is the solution of the initial-value problem, by the uniqueness theorem of Section 2. Zero residual at a finite set of points is a weaker statement, and a small training loss alone does not establish accuracy. The residual must also be checked at times unused in training, together with the initial-condition errors and, when available, comparison with a reference solution. Values outside the collocation interval are extrapolations of the network and receive no such constraint from the training loss.

12.4 Reconstruction from sparse observations

Sparse oscillator measurements test whether the differential equation constrains the continuation of a fitted curve. We use $m = 1$, $c = 0.4$, $k = 4$, $x_0 = 1$, $v_0 = 0$ on $0 \le t \le 10$, about three periods, and a network with three hidden layers of 32 units, 2209 parameters. Twelve noise-free values of the solution at equally spaced times in $0 \le t \le 4$ are given, and the same architecture and initial weights are trained once on the data alone and once with

the residual added at 200 collocation points on the whole interval (Figure 42). Neither fit receives an explicit initial-condition penalty. Within the data window both networks fit well, the one with the residual even better (Table 17). Beyond the data window the difference is qualitative. The network trained on data alone drifts away, with errors as large as the solution itself, while the network trained with the residual follows the remaining periods. The interval $4 < t \leq 10$ lies inside the collocation domain: the equation constrains the continuation even though positions were measured only at earlier times.

In the separate forward-solving benchmark, with initial conditions and no trajectory data, classical integration is much cheaper (Table 17). RK4 reaches a smaller error with a few hundred evaluations of the right-hand side in microseconds, while the PINN needs millions of evaluations of the network and its derivatives during a training of tens of seconds. These wall-clock times were measured on a shared machine and compare orders of magnitude only. Hardware, software and system load affect the recorded values. Learning a function by optimisation is a substantial cost when an ordinary differential-equation solver already supplies the required trajectory.

Table 17: Physics-informed networks for the oscillator and the heat equation. Errors are maximal absolute errors of the position, and parameter estimates are means ± standard deviations over five noise realisations.

experiment	method	result
oscillator, 12 samples in $0 \leq t \leq 4$	data only, inside / beyond the window	2.6×10^{-2} / 1.2×10^{0}
	data and residual, inside / beyond	3.6×10^{-5} / 3.0×10^{-4}
oscillator, forward problem	RK4, step 0.1, 400 evaluations	5.0×10^{-5}
	PINN, about 2.6×10^{6} evaluations, 26 s	1.2×10^{-3}
oscillator, damping $c = 0.4$	PINN	0.401 ± 0.006
	least-squares fit of the solution	0.400 ± 0.004
heat equation, $D = 0.1$	PINN	0.0998 ± 0.0006

12.5 Inferring a physical parameter and a field

Suppose the spring constant and mass are known but the damping c is not. Both the curve and this one physical coefficient can be fitted to measured positions. In the residual, replace c by e^{α} and train α together with $\boldsymbol{\Theta}$; the exponential keeps the damping positive. The data term asks the curve to match the observations, and the residual term asks that curve and coefficient to satisfy the same equation. A curve with the wrong decay rate can fit a few observations, but it must also account for their evolution through the damping term. Identification succeeds only when the observations constrain the coefficient: the motion $x(t) = 0$, for example, gives no information about c.

We use 25 positions with Gaussian noise of standard deviation 0.02, with $\mathcal{L} = \mathcal{L}_r + \mathcal{L}_d$ and no initial-condition penalty. Across five noise realisations the recovered c is close to the true value 0.4 (Table 17). The reference method fits the analytical oscillator solution by least squares, allowing c, x_0 and v_0 to vary. Both methods achieve similar accuracy in this example. The PINN learns the trajectory and the coefficient together; the reference fit evaluates a solution for each trial coefficient.

A partial differential equation adds spatial coordinates to the input. For heat diffusion in a

rod, the network $u_{\boldsymbol{\Theta}}(x,t)$ receives a position and a time and returns the temperature there. The known law and its residual are

$$\partial_t u = D\,\partial_x^2 u, \qquad r_{\boldsymbol{\Theta}}(x,t) = \partial_t u_{\boldsymbol{\Theta}}(x,t) - D\,\partial_x^2 u_{\boldsymbol{\Theta}}(x,t). \tag{12.9}$$

Automatic differentiation now takes one derivative with respect to time and two with respect to position. Collocation points fill the interior of the space–time domain. Separate penalties impose the initial temperature profile at $t = 0$ and the temperatures $u(0,t) = u(1,t) = 0$ at the ends of the rod. These conditions select a heat flow from the many fields that satisfy the diffusion equation.

We first solve this forward problem with $D = 0.1$, then make $D = e^{\alpha}$ trainable and add measurements from four thermometers at 20 times, with noise of standard deviation 0.01. The network reconstructs the temperature between the sensors while estimating the diffusivity. Reconstructing the field from a law and incomplete measurements is the data-assimilation part of this inverse problem. The mean estimate differs from the true value by less than one percent (Table 17); a least-squares fit through a Crank–Nicolson solver gives comparable estimates and is faster here. These examples demonstrate how residuals combine a law with incomplete measurements. They do not establish a general computational advantage over a conventional solver.

Prior knowledge decides what is learned

The prior knowledge determines the object that is learned. With only a record, a forecast map of the window is learned. With the full state measured, the vector field is learned. With part of the law known, the unknown part is learned inside the equation and can be converted into a formula. With the whole law known, the solution is learned from the residual, and unknown parameters are estimated with it.

13 The laser record revisited

The chapter began with two questions about the laser record, a forecast and a law. The predictive question has a quantitative answer. The dependence of successive observations motivates forecasting from a window, and Section 4 gives sufficient conditions for such a window to retain state information, and the best model of Table 7, the LSTM, reaches $C = 0.92$ at a horizon of ten samples, more than one pulse period, and follows a free run for 110 samples. Both results come from one record, one test block and the settings described, and the free run from a single origin.

The explanatory question remains partly open. The diagnostics select a delay of 2 samples and 3 coordinates at the thresholds of Section 4, so a low-dimensional deterministic description of the pulses is consistent with the record. No vector field was learned for the laser, either in delay coordinates or in the physical variables, and the parameters of equations (1.3) were not estimated. The methods of Sections 10 and 11 assume a measured state, while the detector records only the intensity. Fitting equations (1.3) with the intensity as the measurement function would advance the explanatory question.

Symbols

Several letters carry more than one meaning, each in its own context, following the usual conventions of the fields involved. The table lists them with the sections in which each

meaning applies.

symbol	meanings
$\mathbb{1}$, $\mathbb{1}_n$	identity matrix, with explicit size n when needed
s_k	measured record at sample k (throughout)
$\mathbf{u}_k$, L	delay vector or dense window of L measurements in forecasting
$\mathbf{x}$, $\mathbf{F}$, h	state, law of motion $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$, measurement function (throughout); $\mathbf{F}_{\boldsymbol{\Theta}}$ learned vector field (Section 10)
$\mathbf{h}_k$	hidden state of a recurrent network (Section 8)
κ	delay of the delay embedding (from Section 4 on)
g	coupling constant of the laser; gravitational acceleration; observable g (Section 7); candidate $\mathbf{g}_j$ of the LSTM (bold); number of sunspot groups
μ	mean of a stationary process; eigenvalue of a Jacobian or of the Koopman operator; dry-friction coefficient
m	number of delay coordinates (Section 4); lag of the autocorrelation; steps back in a recurrent network; mass
X	amplitude of the convection rolls (Lorenz); design matrix of least squares; Fourier coefficient X_j
S, S_k	segment length of multiple shooting (Section 10); partial sums of Lyapunov exponents (Section 3)
A, $\mathcal{A}$	attractor; matrix of a linear map; $\mathcal{A}$ area of initial states of the pendulum
ω	angular velocity of the pendulum; angular frequency of an oscillation, ω_0 of small pendulum oscillations
α_j	coefficients of the library terms in symbolic regression (Section 11)
R	friction law of the pendulum (Section 11); sunspot number (Section 9.2)
η	eigenfunction of the Koopman operator (Section 7); η_k measurement noise (Section 2.6)
k	sample index (throughout); spring constant of the oscillator (Section 12)
T, δT, ΔT	temperature in the convection model, its deviation and the imposed difference (Section 2); daily temperature T_k (Section 9.1); final time of an integration (Section 10)
ξ, ζ	horizontal and vertical coordinates of the convecting layer (Section 2); ξ_k innovations of an MA model (Section 6); update gate $\boldsymbol{\zeta}$ of the GRU (bold)
Λ	wavenumber of the convection rolls, $\Lambda^2 = k_x^2 + k_z^2$ (Section 2)
a, $\mathbf{a}$	aspect ratio of the rolls (Section 2); temperature anomaly a_k (Section 9.1); adjoint $\mathbf{a}(t)$ (Section 10)
R_k, Q_k	triangular and orthogonal factors of the QR algorithm for Lyapunov exponents (Section 3)
χ, ν	thermal diffusivity and viscosity in the Lorenz model (Section 2)
σ	Prandtl number of the Lorenz model (Section 2); singular values σ_i of the error growth (Section 3); standard deviation of noise or innovations; σ_s standard deviation of the record; logistic function $\sigma(a)$ in the LSTM (Section 8)
ρ	heating parameter of the Lorenz model (Section 2); autocorrelation $\rho(m)$, $\rho(\tau)$ (from Section 2.6 on)
ϱ	spectral radius of a recurrent matrix (Section 8)
γ	decay rates $\gamma_c, \gamma_\perp, \gamma_\parallel$ of the laser (Section 1); autocovariance $\gamma(m)$ (Section 2.6); damping of the pendulum
λ	Lyapunov exponents λ_i (Section 3); ridge penalty (Sections 6–7)
Δ, Δt	error tolerance of the prediction horizon (Section 3); sampling interval
r_k, Δ_k	nearest-neighbour distance and difference in the added coordinate (Section 4); Pearson correlation $r(\tau)$ (Section 5); reset gate $\mathbf{r}_j$ of the GRU (bold); residual $r_{\boldsymbol{\Theta}}$ of a PINN (Section 12)
Υ, δT, ρ_m	stream function, temperature deviation and mass density of the convecting fluid (Lorenz derivation in Section 2)

Continued on the next page

symbol	meanings (continued)
$\mathrm{MI}(\kappa)$	mutual information at the delay κ (Section 4)
$\boldsymbol{\zeta}_j$	update gate of the GRU (Section 8)
$\mathcal{B}_j$	candidate terms of the library for symbolic regression (Section 11)
$\boldsymbol{\Theta}$	parameters of a neural network (throughout)
θ	angle of the pendulum; θ_c threshold of gradient clipping
φ_t	flow map, the state after a time t (Section 2)
ϕ_i, ϕ	weights of an autoregressive model; coefficient of the AR(1) process
$\boldsymbol{\phi}_k$	feature vector of NVAR (Section 7)
E	cavity field of the laser (Section 1); energy of the pendulum (Section 2.3)
$\mathcal{E}$, A	delay map and attractor, $\mathcal{E}(A)$ the reconstructed attractor (Section 4)
c	offset of an AR model; climatology $c(t)$ (Section 9.1); damping or drag coefficient; LSTM memory $\mathbf{c}_k$ (bold)
D	population inversion of the laser; dimension of an attractor (D_B, D_KY); diagonal matrices D_k of a recurrent network
τ	forecast horizon in samples, corresponding to physical time $\tau\Delta t$ (throughout); delay τ_MG of the Mackey–Glass equation; memory time τ_mem
$C(\tau)$, C_tot	capacity at horizon τ and its sum over the horizons
K, K_t	Koopman operator (Section 7); number of observation times in a loss
R, g, f	sunspot number, number of groups and number of individual spots (Section 9.2)

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